LaTeX M1 NOTE II

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Github: *https://github.com/Baudelaireee/Notebook*

Contents

1 Some group and linear algebra

This section can be seen as a breif review of linear agebra and group theory, it mainly focus on something related to lie theory and representation theory.

1.1 Semi-direct product

Definition 1.1. Let G and H be groups, and let $\varphi : H \to \mathrm{Aut}(G)$ be a group homomorphism. The **semi-direct product** $G \rtimes_\varphi H$ is the set $G \times H$ with the multiplication defined by

$$(g_1, h_1)(g_2, h_2) = (g_1\varphi(h_1)(g_2), h_1h_2)$$

for all $g_1, g_2 \in G$ and $h_1, h_2 \in H$.

Remark 1.1. If φ is trival, i.e. $\varphi(h) = \mathrm{id}_G$ for all $h \in H$, then the semi-direct product $G \rtimes_\varphi H$ is just the **direct product** $G \times H$.

Conversely, if $G \rtimes H$ is abelian, then φ must be trivial by caculating $(1, h)(g, 1) = (g, 1)(1, h)$, since it will be $\varphi(h)(g) = g$ for any h and g.

Lemma 1.1. Let $N \lhd G$ be a normal subgroup, and $H < G$ be a subgroup, then $\langle H \cup N\rangle = NH$.

Proof. □

Definition 1.2. A group G is an inner semi-direct product of N and H if $N \lhd G$, $H < G$ and $G = NH$ with $N \cap H = \{e\}$.

Remark 1.2. The definition implies that the group has semi-direct product structure, and the automorphism comes from the inner automorphism:

$$G \cong N \rtimes_\varphi H,$$

where $\varphi : H \to \mathrm{Aut}(N)$ is defined by $\varphi(h)(n) = hnh^{-1}$, it is well-defined since N is normal. To prove the isomorphism, we set $f : N \rtimes H \to G$ by $f(n, h) = nh$, then it is surjective since $NH = G$, and the it is indeed a morphism

$$\begin{aligned}
f((n_1, h_1)(n_2, h_2)) &= f(n_1\varphi(h_1)(n_2), h_1h_2) \\
&= n_1h_1n_2h_1^{-1}h_1h_2 \\
&= n_1h_1n_2h_2 \\
&= f(n_1, h_1)f(n_2, h_2)
\end{aligned}$$

So it is enough to check the injectivity: if $f(n, h) = e$, then $n = h^{-1} \in N \cap H$, thus $n = e$ and $h = e$.

Proposition 1.2. Let $G \rtimes H$ be a semi-direct product with respect to $\varphi : H \to \mathrm{Aut}(G)$, then G can be idetified as a normal subgroup of $G \rtimes H$ while H can be identified as a subgroup of $G \rtimes H$, and we have a splitting exact sequence

$$0 \to G \to G \rtimes H \to H \to 0$$

Proof. We consider the inclusion $i : G \to G \rtimes H$ defined by $i(g) = (g, 1)$ and the projection $\pi : G \rtimes H \to H$ defined by $\pi(g, h) = h$, then we can verify that i and π are homomorphisms, and $\ker \pi = i(G) \cong G$, so it shows that G can be identified as a normal subgroup of $G \rtimes H$ and the sequence is exact. Finally, to prove that the sequence splits, we define a section $s : H \to G \rtimes H$ by $s(h) = (1, h)$, then it is easy to check that s is a injective hormomorphism such that $\pi \circ s = id_H$, it shows that the sequence splits, and H can be identified as $s(H)$. $\square$

1.2 Minimal polynomial

Lemma 1.3. Let $f : V \to V$ be a linear endomorphism with V a finite-dimensional vector space over k, and $\mathcal{V} = \{v_1, \ldots, v_n\}$ be a basis of V such that A is the matrix of f with respect to $\mathcal{V}$, then the minimal polynomial of the matrix A is exactly the minimal polynomial of the endomorphism f.

Proof. We consider the twist map $\varphi : k^n \to V$ defined by $\varphi(e_i) = v_i$, where e_i is the standard basis of k^n, then we have the following commutative diagram:

$$\begin{tikzcd} k^n \arrow[r, "\varphi"] \arrow[d, "A"'] & V \arrow[d, "f"] \\ k^n \arrow[r, "\varphi"] & V \end{tikzcd}$$

thus for any polynomial $p \in k[X]$, we have $p(f) \circ \varphi = \varphi \circ p(A)$, hence f will have the same minimal polynomial as A. $\square$

Remark 1.3. We can consider two representations of $k[X]$-module by

$$\rho : k[X] \to \mathrm{End}(V), \quad X \mapsto f$$

and

$$\rho_A : k[X] \to \mathrm{End}(k^n), \quad X \mapsto A$$

then φ gives an isomorphism of two representations.

1.3 Decomposition of endomorphisms

Theorem 1.1 (Real-Polar decomposition)**.**
For any real endomorphism $A \in \mathrm{GL}_n(\mathbb{R})$, there exists a unique orthogonal endomorphism $O \in O(n)$ and a unique symmetric positive definite endomorphism $P \in S_n^{++}(\mathbb{R})$ such that

$$A = OP$$

Proof. Here is the outline of the proof:
(1) A **lemma** about the existence of square root are needed here: If $A \in S_n^{+}(\mathbb{R})$, then there exists a (unique) $B \in S_n^{+}(\mathbb{R})$ such that $B^2 = A$.
(2) Prove that A^tA is (semi-definite) positive, thus by the lemma there exist a (unique) squar root $P = \sqrt{A^tA}$, and we prove that P is positive definite.
(3) Define $O = AP^{-1}$, and prove that O is orthogonal.
(4) Prove the uniqueness of the decompoistion.

Here is the details:
(1) By real-spectral theorem, there exists $O \in O(n)$ such that

$$A = ODO^T, \quad D = \mathrm{dig}(\lambda_1, ..., \lambda_n)$$

where $\lambda_i \geq 0$ are the eigenvalues of A, and we set $\Sigma = \mathrm{dig}(\sqrt{\lambda_1}, ..., \sqrt{\lambda_n})$ such that

$$A = (O\Sigma O^T)^2$$

which shows that $B = O\Sigma O^T$ is a (unique) square root of A and it is positive since $\sqrt{\lambda_i} \geq 0$.

(2) $A^T A$ is clearly symmetric, and it is positive since

$$x^T(A^T A)x = \|Ax\| \geq 0$$

for any $x \in \mathbb{R}^n$ under the common inner product, and A is invertible such that $\|Ax\| = 0$ if and only if $x = 0$, thus $A^T A$ is positive definite, so does its square root (the eignevalues is strictly positive).

(3) $P^{-1} = (P^{-1})^T$ since P is symmetric, so we have

$$O^T O = (P^{-1})^T A^T A P^{-1} = (P^{-1})^T P^2 P^{-1} = I$$

(4) Suppose that there eixsts another decomposition $A = O_0 P_0$, then we have

$$A^T A = P_0^T O_0^T O_0 P_0 = P_0^2$$

so P_0 is the positive square root of $A^T A$, if the uniqueness of (1) is proved, we can finish the proof, but here we give another proof of the uniqueness only under the existence of the square root: we have $O_0^{-1} O = P_0 P^{-1}$, so it is easy to check that $P_0 P^{-1}$ is **orthogonal**; notice that P_0 and $A^T A$ commute, and there exists a polynomial f such that

$$f(A^T A) = P$$

it can be done by using lagrange interpolation, thus P_0 and P commute as two symmetric matrix, then P and P_0 can be diagonalized simultaneously by one orthogonal H such that

$$P = HDH^T \quad P_0 = HD_0H^T$$

where $D = \mathrm{dig}(\lambda_1, ..., \lambda_n)$ and $D_0 = \mathrm{dig}(\mu_1, ..., \mu_n)$, thus

$$P_0 P^{-1} = HD_0^{-1}DH^T$$

it is **symmetric and positive definite** since $D_0^{-1}D$ is digonal with positive entries, thus $P_0 P^{-1} \in O(n) \cap S_n^{++}(\mathbb{R}) = \{I\}$, which implies that $P_0 = P$ and $O_0 = O$. $\square$

Theorem 1.2 (Complex-Polar decomposition)**.**
For any complex endomorphism $A \in \mathrm{GL}_n(\mathbb{C})$, there exists a unique unitary endomorphism $U \in U(n)$ and a unique hermitian positive definite endomorphism $P \in H_n^{++}(\mathbb{C})$ such that

$$A = UP$$

The proof is similar to the real case, we set $P = \sqrt{A^* A}$ and $U = AP^{-1}$, and the lemma is based on the complex-spectral theorem for normal opreator: The postive Hermitian endomorphism has a unique positive Hermitian square root.

Remark 1.4. In particular, if $n = 1$, we have the polar decompistion in $\mathrm{GL}_1(\mathbb{C}) = \mathbb{C}^*$ by

$$z = |z|e^{iArg(z)}$$

it is easy to see that $U(1) \cong \mathbb{S}^1$, it is the original ideal of the decomposition. Similarly, in the real case, we can embed $\mathbb{C}$ into $\mathrm{GL}_2(\mathbb{R})$ by

$$a + ib \mapsto \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$$

and we have the polar decomposition for the matrix form of complex numbers:

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix} = \begin{pmatrix} \sqrt{a^2+b^2} & 0 \\ 0 & \sqrt{a^2+b^2} \end{pmatrix} \begin{pmatrix} \frac{a}{\sqrt{a^2+b^2}} & -\frac{b}{\sqrt{a^2+b^2}} \\ \frac{b}{\sqrt{a^2+b^2}} & \frac{a}{\sqrt{a^2+b^2}} \end{pmatrix}$$

Another useful decomposition is QR decomoposition, which is also called as Gram-Schimidt decompoistion. Suppose that

1.4 Lattice

Definition 1.3. Let V be a $\mathbb{R}$-vector space of dimension n

Definition 1.4. Let V be a $\mathbb{R}$-vector space of dimension n, a parallelotope of V with respect to a basis $\{v_1, \ldots, v_n\}$ is the set of points

$$P = \{\sum_{i=1}^{n} c_i v_i \mid c_i \in [0, 1)\}$$

The volume of P corresponds to the an inner product $\langle \cdot, \cdot \rangle$ is defined by

$$\mathrm{Vol}(P) = \sqrt{\det G(v_1, \ldots, v_n)}$$

where $G(v_1, \ldots, v_n)$ is the Gram matrix with entries $G_{ij} = \langle v_i, v_j \rangle$.

The definition is resonable with the following reasons:

1. Let P be of dimension 3 with respect to the euclidean inner product, then the volume of P is given by the triple product

   $$\mathrm{Vol}(P) = |v_1 \cdot (v_2 \times v_3)| = |\det[v_1, v_2, v_3]|$$

   with respect to the standard basis.

2. Let $\{v_i\}_i$ be an orthogonal basis, then the Gram matrix $G = I_n$ exactly, hence the volume of a hypercube is 1, which same as what we do in plane geometry and solid geometry.

3.

with the convention above, it allows us to construct the invrariant measure on any real inner product space.

Proposition 1.4. For any real inner product space V of dimension n, there exists a natural isometry ($\mathbb{R}^n$ equipped with the standard inner product)

$$\Phi : \mathbb{R}^n \to V, \quad (x_1, \ldots, x_n) \mapsto \sum_{i=1}^{n} x_i e_i$$

where $\{e_1, \ldots, e_n\}$ is an orthonormal basis of V, then $\mu = \lambda \circ \Phi^{-1}$ defines a Haar measure on V such that for any parallelotope P we have $\mu(P) = \mathrm{Vol}(P)$ exactly.

Proof. μ is exactly a pushforward measure, to verify that it is a Haar measure, we need to check the translation invariance: for any $v \in V$ and any measurable set $S \subset V$, we have

$$\mu(v + S) = \lambda(\Phi^{-1}(v + S)) = \lambda(\Phi^{-1}(v) + \Phi^{-1}(S)) = \lambda(\Phi^{-1}(S)) = \mu(S)$$

where the second equality is due to the linearity of Φ and the third equality is due to the translation invariance of the Lebesgue measure λ. Now we consider the parallelotope P with respect to a basis $\{v_1, \ldots, v_n\}$ and set $\Phi(a_i) = v_i$, then we have

$$\mu(P) = \lambda(\Phi^{-1}(P)) = |\det A| = \sqrt{\det A^t A}$$

where A is the matrix with columns a_i, and isometry shows that $A^T A$ is exactly the Gram matrix $G(v_1, \ldots, v_n)$, which finishes the proof. $\square$

2 Representation of groups

The representation theory appears in almost all branches of mathematics, but the basic ideas and techniques are quite similar, among them the representation theory of finite groups is the most basic one to start with.

2.1 Category and Schur's lemma

Definition 2.1. Let G be a finite group and k be a field, $\mathbf{Rep}_k(G)$ is a category definied as following:

- The objects is of the form (V, ρ), where V is a k-vector space such that ρ a group homomorphism
$$\rho : G \to \mathrm{GL}(V)$$
such a objects is called a **k-representation of G**.
- A morphism from (V, ρ) to (W, τ) is a k-linear map $T \in \mathcal{L}(V, W)$ such that for any $g \in G$ we have the following commutative diagram:
$$\begin{array}{ccc} V & \xrightarrow{T} & W \\ {\scriptstyle \rho(g)}\downarrow & & \downarrow{\scriptstyle \tau(g)} \\ V & \xrightarrow{T} & W \end{array}$$
such a morphism is called an **intertwiner** or a **G-equivariant**. The isomorphisms in the category is just a linear isomorphism such that above diagram commutes. Usually, we will denote the set of morphisms from (V, ρ) to (W, τ) by $\mathrm{Hom}_G(V, W)$, and the set of endomorphisms of (V, ρ) by $\mathrm{End}_G(V)$.
- In fact under the composition of map, the morphisms is well-defined and associative, and the identity inside a representation (V, ρ) is the identity map $T(x) = x$ for any $x \in V$, thus it is indeed a category.
- A representation is called faithful if ρ is injective, and it is called trivial if $\rho(g) = id_V$ for any $g \in G$.
- If V is finite-dimensional over k, then a representation (V, ρ) is called finite-dimensional, and the dimension of V is called the dimension of the representation.

It is the basic definition of representation of groups, so firstly we need to describle more structures of the objects and morphisms such that we can know more about what the category exactly looks like.

Definition 2.2. Let $W \subset V$ as a subspace, we say that W is invariant under the representation (V, ρ) if for any $g \in G$ we have
$$\rho(W) \subset W$$

and we define $(W, \rho|_W)$ as a subrepresentation of (V, ρ). There always exists two trivial subrepresentations: $(\{0\}, \rho|_{\{0\}})$ and (V, ρ) itself.

Remark 2.1. The definition is exactly natural, in the sense of morphism, subrepresentation is just a representation can be embedded into a larger representation by a injective intertwiner, i.e. if (W, τ) a representation with $W \subset V$ as a subspace, then the inclusion $i : V \to W$ will be an intertiwner such that

$$\tau(w) = \tau(g)(i(w)) = i(\rho(w)) = \rho(w)$$

by the definition of τ, then we can recover the definition of invariant.

In linear algebra, we always consider the invariant subspace of a linear opreator, in the context of representation, the invariant subspace is the invariant under a collection of linear opeartors, so there are some analogue.

Lemma 2.1. Let (V, ρ) and W, π be two reprsentation of G, then

1. If $T : V \to W$ is an interwiner, then $\ker T$ is an invariant subspace of V and $\text{im}T$ is an invariant subspace of W.
2. If $T \in \text{End}_G(V)$, then the eigenspaces of T are invariant subspaces of V.

Proof. $\square$

As we study groups by classifying their subgroups, we can also study representation by considering their subrepresentations, so it motivates us to consider a special case, which is analogous to simple groups.

Definition 2.3. A representation (V, ρ) is called irreducible if it has no non-trivial subrepresentation, i.e. the only subrepresentations are $(\{0\}, \rho|_{\{0\}})$ and (V, ρ) itself.

Theorem 2.1 (Schur)**.**
Let (V, ρ) and (W, π) be two irreducible representations of G, and $T : V \to W$ be an interwiner, then T is either an isomorphism or the zero map.

Proof. By above lemma, $\ker T$ is an invarant subspace of V, so we can conclude that $\ker T$ is either $\{0\}$ or V since (V, ρ) is irreducible. If $\ker T = V$, then T is the zero map; If $\ker T = \{0\}$, then T is injective, notice that $\text{im}T$ is an invariant subspace of W and (W, π) is irreducible, in this case we have $\text{im}T = W$ since T is injective, thus T is an isomorphism. $\square$

Remark 2.2. If two representations are not isomorphic, then intwerwiner must be zero, i.e. $\text{Hom}_G(V, W) = \{0\}$, and we can define a equivalence relation on **the irreducible repesentations** of G as following:

$$(V, \rho) \sim (W, \pi) \iff \text{they are isomorphic}$$

and we use $\hat{G}$ to denote the set of equivalence classses.

Corollary 2.2. $\mathrm{End}_G(V)$ is a division k-algebra for any irreducible representation (V, ρ) of G, **in particular if** $k = \mathbb{C}$ **and** V **is finite-dimensional**, then $\mathrm{End}_G(V) \cong \mathbb{C}$ is the set of scalar multiplications.

Proof. $\mathrm{End}_G(V)$ contains zero map and isomorphisms by schur's lemma, thus it is a divison algebra over k, and the unique finite-dimensional division algebra over $\mathbb{C}$ is $\mathbb{C}$ itself, so we have $\mathrm{End}_G(V) \cong \mathbb{C}$. □

2.2 Group algebra and convolution

Definition 2.4. Let G be a group and k be a field, then the group alegbra $k[G]$ is a k-algebra defined as following:

$$k[G] := \{\sum_{g \in G} c_g e_g \mid c_g \in k \text{ and } c_g = 0 \text{ for almost all } g\}$$

where e_g is a formal component with respect to g, which satisifies

$$\forall g, h \in G, \quad e_g \times e_h = e_{gh}$$

and e_1 is the multiplicative identity of $k[G]$, the operation is defined as following:
- Addition:

$$\sum_{g \in G} a_g e_g + \sum_{g \in G} b_g e_g = \sum_{g \in G} (a_g + b_g) e_g$$

- Scalar multiplication:

$$c \cdot \sum_{g \in G} a_g e_g = \sum_{g \in G} (c a_g) e_g$$

- Multiplication:

$$(\sum_{g \in G} a_g e_g) \times (\sum_{h \in G} b_h e_h) = \sum_{g \in G} c_g e_g$$

with $c_g = \sum_{uv=g} a_u b_v$.

Remark 2.3. Generally, the definition can be extended to a monoid and a ring insetead of a group and a field. If G is a finite group, then $k[G]$ is a finite-dimensional k-vector space with basis $\{e_g\}_{g \in G}$. Moreover, the group alegbra is a commutative algebra if and only if G is an abelian group.

Proposition 2.3. There exists a equivalence of catgegories

$$\mathbf{Rep}_k(G) \cong {}_{k[G]}\mathbf{Mod}$$

Proof. Let (V, ρ) be a representation of G, the definition indues a group action of G on V by ρ, i.e. a map

$$G \times V \to V, \quad (g, v) \mapsto \rho(g)(v)$$

defines a group action, thus we can extend it to be an $k[G]$-module. Conversely, for an $k[G]$-module M, the definition of scalar multiplication induces a group action of $k[G]$ on M, and by restriction we can get a represenation of G.

For morphisms,... $\square$

Remark 2.4. Similar argument can be applied to the right $k[G]$-module, but we are used to use left $k[G]$-module.

Definition 2.1. A **(left) regular representation** of G is the representation of G on $k[G]$ defined as following:

$$\rho : G \to \mathrm{GL}(k[G]), \quad g \mapsto \rho(g)$$

where $\rho(g)$ is the left multiplication by e_g:

$$\rho(g)(\sum_{h \in G} c_h e_h) = \sum_{h \in G} c_h e_{gh}$$

Remark 2.5. By above equivalence of categories, a representation of G is (left) regular if and only if the corresponding $k[G]$-module is $k[G]$ itself, or equivalently a free $k[G]$-module of rank 1.

More generally, any group action can be extended linearly to be a reprsentation, we construct as following

1. Let $G \circlearrowleft X$ be a group action with a well-defined operation $g * x$, firstly we can extend the set X to be a vector space over k

$$k[X] := \{\sum_{x \in X} c_x e_x \mid c_x \in k \text{ and } c_x = 0 \text{ for almost all } x \in X\}$$

where e_x is a formal component with respect to x, and toghether forms a basis of $k[G]$, it is easy to check that it is a vector space.

2. The group action can be equivalently defined as a group homomorphism $\sigma : G \to S_X$ with $\sigma(g)(x) = g * x$, then we can extend the morphism to

$$\rho : G \mapsto \mathrm{GL}(k[X]), \quad g \mapsto \rho(g)$$

where $\rho(g)$ is the linear extension of $\sigma(g)$:

$$\rho(g)(\sum_{x \in X} c_x e_x) = \sum_{x \in X} c_x e_{g*x}$$

It is easy to check that ρ is a group homomorphism, and we consider a natural inclusion $\iota : X \to K[X]$ by $\iota(x) = e_x$, then we have the following commutative diagram:

$$\begin{CD} X @>{\sigma(g)}>> X \\ @V{\iota}VV @VV{\iota}V \\ k[X] @>{\rho(g)}>> k[X] \end{CD}$$

which shows that ρ is exactly the linear extension of σ as a representation of G.

3. The extension can be applied to morphism of G-sets. if G is a group acting on two sets X and Y with opreation $*$ and $\star$ respectively, then we consider a G-map $f : X \to Y$ which satisfies

$$\forall g \in G, x \in X, \quad f(g * x) = g \star f(x)$$

then we can extend f to be a linear map $\tilde{f} : k[X] \to k[Y]$ by

$$\tilde{f}(\sum_{x\in X} c_x e_x) = \sum_{x\in X} c_x e_{f(x)}$$

Let ρ_X and ρ_Y be reprsentations defined as 2, then we can check

$$\begin{aligned} \rho_Y(g) \circ \tilde{f}(\sum_{x\in X} c_x e_x) &= \rho_Y(g)(\sum_{x\in X} c_x e_{f(x)}) \\ &= \sum_{x\in X} c_x e_{g\star f(x)}) \\ &= \sum_{x\in X} c_x e_{f(g*x)}) \\ &= \tilde{f}(\sum_{x\in X} c_x e_{g*x}) \\ &= \tilde{f} \circ \rho_X(g)(\sum_{x\in X} c_x e_x) \end{aligned}$$

which shows that $\tilde{f}$ is an interwiner between ρ_X and ρ_Y.

The constuction shows that representation thoery is a powerful tool, as a core tool in group theory, group action can be contained in represetation theory, which can be concluded by the language of category as following:

Proposition 2.4. Let $L : G-\mathbf{Set} \to \mathbf{Rep}_k(G)$ be a functor defined by the construction above:
- On objects: $L(X) = (k[X], \rho)$ where ρ is the linear extension of the group action of G on X.
- On morphisms: $L(f) = \tilde{f}$ where $\tilde{f}$ is the linear extension of the G-map f.
Then L is a well-defined functor which staifies

(a) L is faithful but not full, and for any G-set X, the image $L(X)$ is called a **permutation representation** of G.

(b) L is a free functor.

$$\begin{tikzcd} X \arrow[r, "\eta_X"] \arrow[dr, "f"'] & U(k[X]) \arrow[d, "U(\widetilde{f})"] \\ & U(V) \end{tikzcd}$$

Now we turn our attention to the finite-dimensional representation of finite groups

Definition 2.5. Let G be a finite group and k be a field, we set the space of functions on G by $\mathcal{F}(G)$, and we define the operation such that it is a k-alegbra:
- Addition:

$$(f+g)(x) = f(x) + g(x)$$

- Scalar multiplication:

$$(c \cdot f)(x) = cf(x)$$

- Multiplication:

$$(fg)(x) = f(x) \cdot g(x)$$

where $f, g \in \mathcal{F}(G)$ and $c \in k$.

As a vector space, $\mathcal{F}(G)$ is finite-dimensional with basis $\{\delta_g\}_{g \in G}$ where δ_g is the delta function defined by

$$\delta_g(x) = \begin{cases} 1 & x = g \\ 0 & x \neq g \end{cases}$$

hence there exists a natural isomorphism $\mathcal{F}(G) \cong k[G]$ by sending δ_g to e_g.

However, as k-alegbra the group alegbra $k[G]$ is not isomorphic to $\mathcal{F}(G)$, since $k[G]$ can be non-commutative if G while $\mathcal{F}(G)$ is always commutative, but a different multiplication can be defined on $\mathcal{F}(G)$ such that we can get an isomorphism.

Lemma 2.5. For any $f_1, f_2 \in \mathcal{F}(G)$, we define the **convolution** of f_1 and f_2 by

$$(f_1 * f_2)(g) = \frac{1}{|G|} \sum_{t \in G} f_1(t) f_2(t^{-1}g)$$

it has the following properties:

(a) The convolution is associative.

(b) The convolution is commutative if and only if G is an abelian group.

(c) $\delta_g * \delta_h = \delta_{gh}$ for any $g, h \in G$.

(d) $(\mathcal{F}(G), +, *)$ is isomorphic to group algebra $k[G]$ as k-alegbra.

(e) $f(g_1^{-1} x g_2) = (\delta_{g_1} * f * \delta_{g_2^{-1}})(x)$

Proof. $\square$

Proposition 2.6. For any $g \in G$, we define the left translation on $\mathcal{F}(G)$ by

$$L_g : \mathcal{F}(G) \to \mathcal{F}(G), \quad L_g(f)(x) = f(g^{-1}x)$$

then L_g is an algebra automorphism of $(\mathcal{F}(G), +, *)$ and the map $g \mapsto L_g$ defines a representation $(\mathcal{F}(G), L)$, which is equivalent to the left regular representation of G.

Simiarly, the right translation on $\mathcal{F}(G)$ defined by

$$R_g : \mathcal{F}(G) \to \mathcal{F}(G), \quad R_g(f)(x) = f(xg)$$

and $g \mapsto R_g$ induces a representation $(\mathcal{F}(G), R)$ of G, it is equivalent to the right regular representation of G.

Proof. $\square$

Remark 2.6. This proposition explicitly constructs a left and right regular representation via the convolution on algebra of functions on a finite group G, more generally, if G is finite then there exists a equivalence of categories

$$\mathbf{Rep}_k(G) \cong {}_{\mathcal{F}(G)}\mathbf{Mod}$$

where $\mathcal{F}(G)$ is the **convolution algebra**.

Finally, we will write a representation of finite group G as a $\mathcal{F}(G)$-module, we set a representation $\rho : G \to \mathrm{GL}(V)$, we should extend it such that the following diagram commutes:

$$\begin{tikzcd} G \arrow[r, "\rho"] \arrow[d, hook, "g\mapsto\delta_g"'] & \mathrm{GL}(V) \arrow[d, hook] \\ \mathcal{F}(G) \arrow[r, "\widetilde{\rho}"'] & \mathrm{End}(V) \end{tikzcd}$$

we define the extension as following:

$$\widetilde{\rho}(f)(v) = \frac{1}{|G|} \sum_{g \in G} f(g) \cdot \rho(g)(v)$$

for any $f \in \mathcal{F}(G)$ and $v \in V$, it needs to check that $\widetilde{\rho}$ is a ring homomorphism, then it induces a $\mathcal{F}(G)$-module structure on V. Moreover, when $f = \delta_h$ for some $h \in G$, we have

$$\widetilde{\rho}(\delta_h) = \rho(h)$$

so exactly $\widetilde{\rho}$ gives a extension exactly.

2.3 Character and orthogonality relations

Definition 2.6. Suppose that (V, ρ) is a finite-dimensional representation of G, we define the character of (V, ρ) as a function $\chi_\rho : G \to k$ by

$$\chi_\rho(g) = \mathrm{tr}(\rho(g))$$

for any $g \in G$.

Notice that if $\dim V = 1$, then $\mathrm{GL}(V) \cong k^\times$, which shows that the character of the representation will be the representation itself: a group homomorphism from G to $k^\times$. Generally, the character of a representation is not necessary a group homomorphism.

Definition 2.7. Let G be a group and k be a field, we denote $\mathrm{Cl}(G) = G/\sim_c$ be the quotient of G by the conjugacy relation

$$x \sim y \iff \exists g \in G, y = gxg^{-1}$$

a function $f : \mathrm{Cl}(G) \to k$ is called a **class function**, or equivalently, a function $f : G \to k$ such that

$$f(gxg^{-1}) = f(x)$$

for any $g, x \in G$.

Remark 2.7. The algebraic properties of trace shows that the character of a representation is always a class function, thus it can be regarded as a function on $\mathrm{Cl}(G)$.

The conjugacy classes can be described by group action $G \circlearrowleft G$ with operation $g * x = gxg^{-1}$, then $\mathrm{Cl}(G)$ is exactly the set of orbits with

$$C_G(x) = \{g \in G \mid gxg^{-1} = x\}$$

the centralizer (stablizer) of x in G. When G is finite, $[G : C_G(x)]$ is exactly the size of orbit of x.

Proposition 2.7. Let (V, ρ) be a finite-dimensional representation of G, then
(a) $\chi_\rho(e) = \dim V$.
(b) $\chi_{\rho^*}(g) = \chi_\rho(g^{-1}) = \mathrm{tr}(\rho(g)^{-1})$ for any $g \in G$.
Let (W, π) be another finite-dimensional representation of G, then
(c) $\chi_{\rho\oplus\pi} = \chi_\rho + \chi_\pi$
(d) $\chi_{\rho\otimes\pi} = \chi_\rho \cdot \chi_\pi$

3 Toplogical Groups

3.1 Basic properties

Group and Topological space is two different objects, group is algebraic structure, topological space is geometric structure, but it is not strange to combine them together.

Definition 3.1. A **topological group** is a object $(G, \mathcal{T}, \cdot)$ such that

- $(G, \cdot)$ is a group
- $(G, \mathcal{T})$ is a topological space
- The group structure is compatible with topological structure, i.e. multiplication

  $$m : G \times G \to G, \quad (x, y) \mapsto x \cdot y$$

  and inverse

  $$i : G \to G, \quad x \mapsto x^{-1}$$

  are continuous maps.

With the basic definition, we can talk about the properties of topological groups, here is a useful lemma, it shows that the topological group is **homogeneous**, i.e. the local topology is same at any point.

Lemma 3.1. Let G be a topological group, then for any $g \in G$, we define **left translation** and **right translation** by

$$L_g : G \to G, \quad x \mapsto gx$$
$$R_g : G \to G, \quad x \mapsto xg$$

Then both L_g and R_g are homeomorphisms on G.

Proof. It is easy to verify that $L_{gh} = L_g \circ L_h$, and $R_{gh} = R_h \circ R_g$, then we have

$$L_g^{-1} = L_{g^{-1}}, \quad R_g^{-1} = R_{g^{-1}}$$

and we can check the continuity of L_g and R_g by composition. $\square$

It is an important in the following proofs of properties, we first talk about the basic properties of topological groups.

Proposition 3.2. Let G be a topological group, then

(1) each open subgroup $H < G$ is also closed.

(2) each closed subgroup $H < G$ with finite index is also open.

Proof. The subgroup H gives a partition of G by left cosets:

$$G = \bigsqcup_{g \in \mathcal{S}} gH$$

with $\mathcal{S} \subset G$ the set of representatives in G/H, and we fix the representatives for H be e, then

$$G - H = \bigsqcup_{g \in S - \{e\}} gH$$

If H is open, then by translation each gH is also open, thus the union of open sets $G - H$ is also open, hence H is closed, which finishes that proof of (1). For the statement (2), the finite index shows that $\mathcal{S}$ is finite, and we suppose that the index is n. If $n = 1$, then $H = G$ is open clearly; otherwise, we can deduce that $G - H$ is the finite union of closed sets gH with $g \in \mathcal{S} - \{e\}$, thus it is closed, it can be done similarly by translation, hence H is open sice $G - H$ is closed. □

Then it is the **separation** of topological groups.

Proposition 3.3. Let G be a topological groups, then the following statements are equivalent:

(1) G is Hausdorff (T2).

(2) The singleton $\{e\}$ is closed in G.

(3) G is T1.

Proof. (3) implies (2) is clear, since T1 space makes all singletons closed; (1) implies (3) is just the implication from strong to weak separation axiom.

(2) implies (1): G is Hausdorff if and only if the diagonal set

$$\Delta = \{(g, g) \in G \times G \mid g \in G\}$$

is closed under the product topology, and topolgical group induces a continous map $f(x, y) = xy^{-1}$ for any $(x, y) \in G \times G$, then $\Delta = f^{-1}(\{e\})$ is closed by continuity. □

Corollary 3.4. Let G be a topological group and $H < G$ be a subgroup, then the quotient space G/H is **Hausdorff** if and only if H is **closed** in G.

Another topological properties is about **connectness**:

Proposition 3.5. Let G be a topological group, then

(1) If G is connected, then any open set containing e generates G.

(2) The connected component G_0 of identity $e \in G$ is a **closed normal subgroup** of G, and each connected component of G is homeomorphic to G_0 by translation.

Proof. (1) We denote $\langle A \rangle$ be the subgroup generated by $A \subset G$, and we take A be an open set containing e, we will prove that $\langle A \rangle$ is open and closed, by the connectness of G, which implies that $\langle A \rangle = G$. It is clear that

$$\bigcup_{x \in \langle A \rangle} xA \subset \langle A \rangle$$

conversely, for any $a \in \langle A \rangle$, we have $a = ae$ with $e \in A$, so a is in the union such that we can replace the inculsion by equality. Notice that $xA = L_x(A)$ is open since A is open, so $\langle A \rangle$ is also open as the union of open sets; By the propoistion 3.2, $\langle A \rangle$ is also closed since it is open subgroup, hence we finish the proof of (1).

(2) Any connected component is closed in general topology, which can be proved by contradition to $G_0 \subsetneq \overline{G_0}$. To prove that G_0 is a subgroup, we take a continous map $f(x, y) = xy^{-1}$ on $G \times G$, then $f(G_0 \times G_0)$ is connected since the finite product of connected spaces is connected, and we notice that $a = f(a, e)$ for any $a \in G_0$, so $G_0 \subset f(G_0 \times G_0)$ and then we can take equality by the maximality of connected component, and it shows that G_0 is exactly a subgroup.

To prove the normality, we take the conjugation map for any $g \in G$:

$$\sigma_g : G \to G, \quad x \mapsto gxg^{-1}$$

it is homeomorphism by composition $\sigma_g = L_g \circ R_g^{-1}$. Here $\sigma_g(e) = e$ and the image is also connected, so $\sigma_g(G_0) \subset G_0$ by the maximality of connected component, so it must be normal. $\square$

Notice that if $H < G$ is a subgroup, then it induces a natural equivalence relation on G by left cosets:

$$x \sim y \iff x = yh, \text{ for some } h \in H$$

similarly we can define right cosets here, and then it gives a quotient space $G/\sim$ and we denote it by G/H, and it is natrually endowed with the quoitent topology with a (continous) quoitent map:

$$\pi : G \to G/H, \quad g \mapsto gH$$

In particular, if H is a normal subgroup, then G/H is exactly a topological group by UPQ:

$$\begin{tikzcd} G \times G \arrow[r, "f"] \arrow[d, "\pi \times \pi"'] & G \arrow[d, "\pi"] \\ G/N \times G/N \arrow[r, "f_{G/N}"] & G/N \end{tikzcd}$$

with $f(g,h) = gh^{-1}$, so it gives the continuity of group opreation. However, it is not so easy to talk about the the topology of quoitent space G/H directly, it is better to see the quotient space as a **homogeneous space** under the action of G, hence we need to talk about acion of the topological group:

Definition 3.2. Let G be a topological group and X be a topological space, a continous action of G on X is a well-defined action in the sense of group together with a continous operation, that means there exists a continous map

$$\phi : G \times X \to X, \quad (g,x) \mapsto g \cdot x$$

such that for any $g,h \in G$ and $x \in X$, we have

(1) $e \cdot x = x$

(2) $g \cdot (h \cdot x) = (gh) \cdot x$

In the context of toplogical groups or Lie groups, we alwasys omit "continous" when it comes to actions, it is a convention sometimes in the liter

ature. It is similar to the algebraic action, we can define **stabilizer** of a point $x \in G$

$$I_x = \{g \in G \mid g \cdot x = x\}$$

It is sometimes called **isotropy group** at x, and it can be written as the preimage

$$I_x = \phi_x^{-1}(\{x\}), \quad \phi_x(g) = g \cdot x$$

with ϕ_x a continous map by fixing variable, hence the isotropy group is always a **closed subgroup** of G if X is Hausdorff (T1 is enough, but manifold makes sense). Similarly, we can define **orbit** of x by

$$Gx = O_x = \{g \cdot x \mid g \in G\}$$

which is the image of ϕ_x on the contrary, hence it gives a natural bijection

Lemma 3.6 (stabilizer-orbit)**.**
Let G be a compact group and X be a Hausdorff space, If G acts continously on X, then for any $x \in G$, there exists a homemomorphism $G/I_x \cong Gx$.

Proof. The proof is just based on the universal property of quotient space, and the stablizier-orbit theorem in group theory, we just need to check the natural bijection induced by ϕ_x is indeed a homemomorphism. $\square$

In particular, the orbit gives a equivalence reliation on X by

$$x \sim y \iff y = g \cdot x, \text{ for some } g \in G$$

hence it gives a quotient space denoted by X/G and we call it **orbit space**, it is natrually endowed with the quoitent topology with a (continous) quoitent map:

$$\pi : X \to X/G, x \mapsto Gx$$

the following lemma is useful and important, it shows that the topology of group action is well-behaved from the view of techinc:

Lemma 3.7. The natural quotient map $\pi : X \to G/X$ is open.

Proof. Suppose that U is an open set of X, then we can prove

$$\pi^{-1}(\pi(U)) = \bigsqcup_{g \in G} gU$$

and then by translation each gU is open, so we can finish the proof by the definition of quotient topology that $\pi(U)$ is open if its preimage is open. □

Then we can conclude

Proposition 3.8. Let G be a topological group and H be a subgroup, then

(1) H is closed if and only if G/H is Hausdorff.

(2) If G is connected, then G/H is connected.

(3) If G/H and H are both connected, then G is connected.

Proof. □

Neighborhood system

As we have mentioned before, topological group is a homogeneous space, so the local topology at one point can be translated to generate the global topology, for some convenience, we always study the local topology at identity of the group.

Formally, for any group G we define the **neighborhood system** (at x) to be the collection of subsets containing x, denoted by $\mathcal{V}_x$:

$$A \in \mathcal{V}_x \iff x \in A \wedge A \subset G$$

Hence we can induce a topology on G by neighborhood system (i.e. determine the open voisinage of identity).

Proposition 3.9. Let G be a group with $\mathcal{V}_e$ as the neighborhood system at identity e, if $\mathcal{V}_e$ satisfies the following conditions:

1. For any $U, V \in \mathcal{V}_e$, there exists $W \in \mathcal{V}_e$ such that
$$W \subset U \cap V.$$
2. If $a \in U \in \mathcal{V}_e$, then there exists $V \in \mathcal{V}_e$ such that
$$Va \subset U.$$
3. For each $U \in \mathcal{V}_e$, there exists $V \in \mathcal{V}_e$ such that
$$V^{-1}V \subset U.$$
4. For each $U \in \mathcal{V}_e$ and each $x \in G$, there exists $V \in \mathcal{V}_e$ such that
$$x^{-1}Vx \subset U.$$

then there exist a unique topology $\mathcal{T}$ such that $\mathcal{V}_e$ is the set of all open voisinage of e, and $(G, \mathcal{T})$ is a topological group.

Proof. The topology $\mathcal{T}$ is uniquely defined by translation:
$$\mathcal{B} = \{gU \mid g \in G, U \in \mathcal{V}_e\}$$
we just need to prove that $\mathcal{B}$ is the topology basis. Notice the condition (1) and (2) ensure that $\mathcal{B}$ generates a topology, the condition (3) is the requirement of continuity of multiplication and inverse; finally, the condition (4) is obligator for non-abelian group to ensure the continuity. □

In partiuclar, we can naturally choose subgroup as the element in the system, the condition can be simplified as following:

Corollary 3.10. Let G be a group with $\mathcal{U}$ as a collection of subgroups of G, if $\mathcal{U}$ satisfies the following conditions:

1. For any $U, V \in \mathcal{U}$, there exists $W \in \mathcal{U}$ such that
$$W \subset U \cap V.$$
2. For each $U \in \mathcal{U}$ and each $x \in G$, there exists $V \in \mathcal{U}$ such that
$$x^{-1}Vx \subset U.$$

then there exist a unique topology $\mathcal{T}$ such that $\mathcal{U}$ is a basis of open voisinage of e, and $(G, \mathcal{T})$ is a topological group.

Here are some classic examples of topological groups constructed by neighborhood system:

Example 3.1. We consider the usual topology on $\mathbb{Q}$, then the completion of $\mathbb{Q}$ is just $\mathbb{R}$, it is another method to construct real number (see in Bourkabi's book). We review that we always write the viosinage of 0 as the form of $(-\varepsilon, \varepsilon)$, which motivates us to consider the system of subgroup:

$$\mathcal{U} = \{\frac{1}{n}\mathbb{Z} \mid n \in \mathbb{N}^*\}$$

pay attention to the choices here! We can not take $n\mathbb{Z}$ to be an open set (consider it in $\mathbb{R}$, it is closed and not open). Then we can verify the topology induced by $\mathcal{U}$ is just the subspace topology induced by $\mathbb{R}$, i.e. it induces a usual absolute value on field $\mathbb{Q}$:

$$|x|_\infty = \begin{cases} x & x \geq 0 \\ -x & x \leq 0 \end{cases}$$

It is natural to ask that can we define other topological stucture on $\mathbb{Q}$ such that we can get other different completion? The answer is yes, and a classic example is the $p-adic$ **number field.**

Example 3.2. We fix a prime number p and then we can define a subgroup

$$\mathbb{Z}_{(p)} = \{\frac{m}{n} \mid m, n \in \mathbb{Z}, p \nmid n\}$$

and we let $\mathcal{U} = \{p^t\mathbb{Z}_{(p)} \mid t \in \mathbb{Z}\}$ to be the collection of subgroups, which generates an unique toplogy (verify!). Notice that we have a unique chain in the system:

$$\ldots p^2\mathbb{Z}_{(p)} \subset p\mathbb{Z}_{(p)} \subset \mathbb{Z}_{(p)} \subset p^{-1}\mathbb{Z}_{(p)} \subset p^{-2}\mathbb{Z}_{(p)} \subset \ldots$$

motivates from the above example, we consider each subgroup gives a method to measure the size of an elemnt is some sense, for example we consider the following number when $p = 3$:

$$1, \quad 4, \quad \frac{1}{2}, \quad \frac{2}{5}$$

they are all in $\mathbb{Z}_{(p)}$, and we consider the distance of them to 0 can be bounded by α (consider $(-\varepsilon, \varepsilon)$ gives the set of all elements with distance to 0 less than ε), and then we consider some other numbers:

$$\frac{1}{3}, \quad \frac{2}{3}, \quad \frac{7}{6}$$

they are all in $\frac{1}{3}\mathbb{Z}_{(p)}$ but not in $\mathbb{Z}_{(3)}$, so we can think that their distance to 0 is bouned by β, and it is natural to give a realtion $\alpha < \beta$, which motivates us to give a new method to consider the metric on $\mathbb{Q}$ by conisder the local factorization of p.

Hence we can define something to denote α and β just now:

$$v_p : \mathbb{Z} - \{0\} \to \mathbb{N}, \quad m \mapsto \max\{k : p^k | m\}$$

for example $v_3(6) = 2$, $v_3(-2) = 0$. Then the method of measuring can be extended to $\mathbb{Q}$ as following:

$$v_p : \mathbb{Z} - \{0\} \to \mathbb{Z}, \quad \frac{m}{n} \mapsto v_p(m) - v_p(n)$$

And we make a convention that $v_p(0) = +\infty$, it is called **p-adic valustion**. for example $v_3(\frac{1}{3}) = -1$, $v_3(\frac{9}{2}) = 2$. Then we can get a measure of number

$$p^k\mathbb{Z}_{(p)} \iff \{x \in \mathbb{Q} \mid v_p(x) \geq p^k\}$$

but a question occurs here is that $v_3(1) = 0$ and $v_3(\frac{1}{3}) = -1$, as what we discussed just now, actually we hope the measure of 1 should be less than the measure of $\frac{1}{3}$ (here $0 > -1$), hence the valuation v_p can not be treated as the absolute value directly, hence we modify it as following:

$$|-|_p : \mathbb{Q} \to \mathbb{R}_+, \quad x \mapsto \frac{1}{e^{v_p(x)}}$$

the convention just now forces $|0|_p = 0$, and again we can get a clear measure of numbers:

$$x \in p^k\mathbb{Z}_{(p)} \iff |x|_p \leq e^{-k}$$

hence $|1|_3 < |\frac{1}{3}|_3$, which admits our original idea.

Remark 3.1. it is just a short introduction of p-adic number from the view of topologyical group, and here are sommething to supplement if to go deeper:

(1) The metric of is induced by the absolute value:

$$d(x, y) = |x - y|_p$$

we can verify that is a **ultrametric (non-archimedean metric)**, i.e. a metric satifying the strong triangle inequality:

$$d(x.z) \leq \max d(x, y), d(y, z), \quad \forall x, y, z \in \mathbb{Q}$$

which induces different analysis structure on $\mathbb{Q}$.

(2) Same situation with $\mathbb{Q}$ equipped by the usual absolute value, $(\mathbb{Q}, |-|_p)$ is not complete as a metric space, we can complete it to get the **p-adic number field** $\mathbb{Q}_p$. (The completion process is same with what we do for the real number, and we can prove that the completion is unique up to isometry.)

(3) By **Ostrowski's theorem**, any non-trival absolute value on $\mathbb{Q}$ is equivalent to either the usual absolute value or some p-adic absolute value. In detail, if $|-|$ is a non-trival absolute value on $\mathbb{Q}$, then either there is some $c > 0$ such that

$$|x| = |x|_\infty^c, \quad \forall x \in \mathbb{Q}$$

or there is some prime p and some $c > 0$ such that

$$|x| = |x|_p^c, \quad \forall x \in \mathbb{Q}$$

3.2 Homogeneous Spaces

Review that a group action $G \circlearrowleft X$ is **transtive** if for any $x, y \in X$, there exists $g \in G$ such that $g \cdot x = y$, i.e. there is only one orbit.

Definition 3.3. A **homogeneous space** is a Hausdorff topological space X with a transtive continous action of a group G, in particular, we call it **G-homogeneous space** to emphasize the acting group G.

Before talking about the properties of homogeneous space, we consider a basic example but important example: rotatio action on sphere $SO(n) \circlearrowleft \mathbb{S}^{n-1}$

$$SO(n) \times \mathbb{S}^{n-1}, \quad (A, v) \mapsto Av$$

the operation is defined by matrix multiplication, so it defines a continous action, then we can verify the following results:

1. **The action is transitive.** When $n = 2$, it is clear since $SO(n)$ is just the set of rotations of plane, so we prove the case $n \geq 3$ by induction. We assume that the action is transitive for $n-1$,...
2. **The isotropy group at e_1 is isomorphic to $SO(n-1)$.** Suppose that $A \in SO(n)$ is matrix such that $Ae_1 = e_1$, then by caculation we can veify that A is of the form

$$M = \begin{pmatrix} 1 & 0_{1\times n} \\ 0_{n\times 1} & A' \end{pmatrix}, \qquad A' \in SO(n-1).$$

   Then it is easy to establish a isomorphism and a hoemomorphism.
3. **There exists a homeomorphism $SO(n)/SO(n-1) \cong \mathbb{S}^{n-1}$.** By the stablizer-orbit lemma, we can deduce a continous bijection $SO(n)/SO(n-1) \to \mathbb{S}^{n-1}$, so it is enough to verify that $SO(n)$ is compact.
4. **$SO(n)$ is connected.** When $n = 2$, a classic homeomorphism $SO(2) \cong \mathbb{S}^1$ shows that $SO(2)$ is connected, then we can prove the case $n \geq 3$ by induction.

From this example, we can see here sphere $\mathbb{S}^n$ is a homogeneous space under the action of $SO(n+1)$, we are very faimilar with the topology of sphere, but we are not fimilar the topology of $SO(n)$, so homogeneous space gives us a method to understand the topology of group by the homogeneous space. Conversely, if we are fimilar with the structure of the topological group, we can understand a strange space by this method, anyway homogeneous space gives a deep relation

$$\{\text{topological group}\} \sim \{\text{space with same local structure}\}$$

Theorem 3.1. Let G be a Lie group and X be a G-homogeneous space, if I_x the isotropy group at $x \in X$, then there exists a hoemomorphism

$$G/I_x \cong X$$

3.3 The Pontryagin Duality

We consider the category of locally compact abelian groups:

$$\mathbf{LCA}\ \Big|\ \begin{array}{l}\text{objects: locally compact abelian groups } (G,+)\\ \text{morphisms: continuous group homomorphisms}\end{array}$$

without talking about the topology of the group, we can simiarly consider the pure algebraic structure:

$$\mathbf{Ab}\ \Big|\ \begin{array}{l}\text{objects: abelian group } (G,+)\\ \text{morphisms: group homomorphisms}\end{array}$$

Clearly it is same as category of $\mathbb{Z}$-modules, hence it is an abelian category with tools of homological algebra, so we can choose a certain objects of abelian groups to construct a duality.

$$\hat{G} := \mathrm{Hom}_{\mathbb{Z}}(G, \mathbb{R}/\mathbb{Z})$$

Clearly, $\mathbb{R}/\mathbb{Z}$ is an injective $\mathbb{Z}$-module, hence the Hom-functor will keep exactness. In particular, it will have a good behavior when group is finite.

Proposition 3.11. If G is a finite abelian group, then $\hat{G}$ is isomorphic to G.

Proof. □

However, the duality fails when G is infinite. For example, if $G = \mathbb{Z}$, then $\hat{\mathbb{Z}} \cong \mathbb{R}/\mathbb{Z}$, but if we take dual again

$$\widehat{\mathbb{R}/\mathbb{Z}} = \mathrm{End}_{\mathbb{Z}}(\mathbb{R}/\mathbb{Z}) \cong \prod_p \mathbb{Z}_p$$

clearly it is larger than $\mathbb{Z}$, hence it actually fails to be a duality. What we have done is to give a exact restriction on the categroy of abelian groups to make the duality work: It depends on the topology of the group, so we need to consider the topology of the dual group firstly. Formally, we define dual group as following:

Definition 3.4. Let $G \in \mathbf{LCA}$, we define its **dual group** as G^* or $\hat{G}$ by

$$G^* := \mathrm{Hom}_{\mathbf{LCA}}(G, \mathbb{S}^1)$$

i.e. the group of continous group homomorphism from G to the circle group ($\mathbb{S}^1 \cong \mathbb{R}/\mathbb{Z}$). Among G^*, the elelment are called **characters** of G, and we usually denote it by

$$\chi : G \to \mathbb{S}^1$$

with operation defined by pointwise multiplication:

$$(\chi_1 + \chi_2)(g) := \chi_1(g) \cdot \chi_2(g)$$

Generally a Hom-functor is not necessary to keep category itself, it will maps to a category of sets, so we should verify that LCA is a good condition to make the functor closed.

Lemma 3.12. Let $G \in \mathbf{LCA}$, then its dual group G^* endowed with the compact-open topology is a locally compact abelian group, in particular

$$V(K, \varepsilon) = \{\chi \in G^* : \chi(K) \subset U_\varepsilon\}$$

for any compact subset $K \subset G$ and $U_\varepsilon = \{e^{it} \mid t \in (-\varepsilon, \varepsilon)\}$, it forms a neighborhood basis of $0 \in G^*$.

Proof. It is easy to verify that G^* is indeed a abelian group under defined opreation, to verify it is a topological group, we need to verify the family $\mathcal{B}$ of all $V(K, varepsilon)$ forms a neighborhood basis of identity of G^*: (1) zero character $g \mapsto 1$ is in any set clearly; (2) $V(K, r) \cap V(L, t) = V(K \cup L, \min(r, t))$, so it is stable under finite intersection; (3) $-V(K, \varepsilon) = V(K, \varepsilon)$ since $U_\varepsilon^{-1} = U_\varepsilon$; (4) $V(K, r) + V(L, t)$ is the subset of $V(K \cup L, r+t)$, so it is stable under opreation. Hence it forms a neighborhood system of identity by (1)-(4).

Finally, we need to verify that G^* is locally compact by Ascoli's theorem... $\square$

Remark 3.2. In particular, we denot evalution map by

$$\mathrm{ev} : G \times G^* \to \mathbb{S}^1, \quad (g, \chi) \mapsto \chi(g)$$

the compact-open topology is **the coarsest topology** such that the evaluation map is continuous (page 43 [1]).

A convention is that G^* always means the dual group with compact-open topology, so we finish the level of objects in the category, and then it is naturally to consider the functoriality, or the level of morphisms.

Lemma 3.13. Let $G, H \in \mathbf{LCA}$, and $f : G \to H$ be a continuous group homomorphism, then it induces a natrual map

$$f^* : H^* \to G^*, \quad \chi \mapsto \chi \circ f$$

which is a continuous (under the compact-open topology) group homomorphism.

Proof. For any two characters $\chi_1, \chi_2 \in H^*$, we have

$$f^*(\chi_1 + \chi_2)(g) = (\chi_1 + \chi_2)(f(g)) = \chi_1(f(g)) \cdot \chi_2(f(g)) = f^*(\chi_1)(g) \cdot f^*(\chi_2)(g)$$

for any $g \in G$, so $f^*(\chi_1 + \chi_2) = f^*(\chi_1) + f^*(\chi_2)$ it is a group homomorphism. To prove the continuity, we take a basic open set $V_G(K, \varepsilon)$, then

$$\begin{aligned} (f^*)^{-1}(V_G(K, \varepsilon)) &= \{\chi \in H^* \mid \chi \circ f \in V_G(K, \varepsilon)\} \\ &= \{\chi \in H^* \mid \chi(f(K)) \subset U_\varepsilon\} \\ &= V_H(f(K), \varepsilon) \end{aligned}$$

here $f(K)$ is compact since f is continous, so the preimage of any voisinage of identity is open, hence f^* is continous by translation. $\square$

It is easy to verify that $(f \circ g)^* = f^* \circ g^*$, hence we have constructed a contravariant functor insider the category **LCA**.

Lemma 3.14. If $f : G \to H$ is a surjective continuous group homomorphism, then $f^* : H^* \to G^*$ is injective; If $f : G \to H$ is both injective and open, then $f^* : H^* \to G^*$ is surjective.

Proof. If f is surjective, we take any $\chi_1, \chi_2 \in H^*$ such that $f^*(\chi_1) = f^*(\chi_2)$, we assume that $\chi_1 \neq \chi_2$, then there exists $h \in H$ such that $\chi_1(h) \neq \chi_2(h)$, since f is surjective, there exists $g \in G$ such that $f(g) = h$, hence $f^*(\chi_1)(g) \neq f^*(\chi_2)(g)$, so it is absurd.

If f is injective, then injective module $\mathbb{S}^1$ allows us to extend a charcter $\chi : G \to \mathbb{S}^1$ to $\tilde{\chi} : H \to \mathbb{S}^1$ such that $\tilde{\chi} \circ f = \chi$. We need to verify the continuity of $\tilde{\chi}$: notice that f is open, so $f(G)$ is the open subgroup of H, hence we can prove that $\tilde{\chi}$ is continous on $f(G)$, then we can conclue that $\tilde{\chi}$ is continous on H by translation (a group homomorphism is continous on some open subgroup, then the morphism is just continous). □

Theorem 3.2 (Pontryagin-Van Kampen,1934)**.**
Let $G \in$ **LCA** and G^* be its dual group, then evluation map gives a natural ismoprhism in **LCA** by fixing one variable:

$$\mathrm{ev} : G \to (G^*)^*, \quad g \mapsto \mathrm{ev}_g$$

where $\mathrm{ev}_g : G^* \to \mathbb{S}^1$ is defined by $\mathrm{ev}_g(\chi) = \chi(g)$.

Proof. □

Here is the slogan of the duality:

every LCA-group is the dual group of its dual group.

The conclusion is similar with the finite-dimension vector space, hence it will be a strong tool to study the structure of LCA-groups and the analysis on it. In the sense of category theory, the Pontryagin duality is a contravariant equivalence with inverse itself:

$$(-)^* : \mathbf{LCA} \to (\mathbf{LCA})^{op}$$

Consequences

It is wonderful to see what we can get from the duality, we make a convention that G in this section always means a LCA-group.

Proposition 3.15. In the sense of topological isomophism

$$G_1 \oplus G_2 \cong G_1 \times G_2$$

and

$$(G_1 \times G_2)^* \cong G_1^* \times G_2^*$$

Proof. In the sense of category of abelian groups, we have

$$(G_1 \times G_2)^* \cong G_1^* \oplus G_2^* \cong G_1^* \times G_2^*$$

So we just need to verify the isomophism is homemomorphism under the compact-open topology. □

Proposition 3.16. If G is compact, then G^* is discrete; If G is discrete, then G^* is compact.

Proof. □

Proposition 3.17. If G is compact group, then

- G is metrizable if and only if G^* is countable.

- G is connected if and only if G^* is torsion-free.

4 Classical Groups and Lie algebras

Matrix groups are the most important examples of Lie groups and Lie algebras, this section is some preparation for the general case.

4.1 Exponential map

Remember that $M_n(K) \cong K^{n^2}$ for any field (even any ring), in particualr we consider $K = \mathbb{C}$ or $\mathbb{R}$, then the space of endomorphisms or matirces is just a finite-dimensional vector space over the field, so we can define the typical norm on it:

$$\begin{aligned}
\|A\| &= \sqrt{\sum_{i,j} |a_{ij}|^2} \quad \text{(euclidean norm)} \\
\|A\|_\infty &= \max_{i,j} |a_{ij}| \quad \text{(Max norm)} \\
\|A\|_1 &= \sum_{i,j} |a_{ij}| \quad \text{(Sum norm)}
\end{aligned}$$

Moreover, we can consider the norm of $M_n(K)$ from another view, we view it as the linear continous operator inside K^n, hence in particular the norm of K^n can induce the norm of operator:

$$\|A\|_{op} = \sup_{x \in K^n - \{0\}} \frac{\|Ax\|}{\|x\|}$$

Anyway, the space of endomrphisms on finitie-dimensional vector space or the matrix space is still finite-dimensional, as a Banach space, all the norms on $M_n(K)$ is equivalent, with the basic knowledge of topology on it, we can talk about some whether some series in analysis makes sense or not, in particular, exponential map:

Definition 4.1. For a matrix $A \in M_n(K)$, the exponential of A is defined by

$$\exp(A) = \sum_{k \geq 0} \frac{A^k}{k!}$$

or sometimes we denote it by e^A.

The definition is analogous to the exponential map in $\mathbb{C}$, and we can verify that it is well-defined for any A by

$$\|\exp(A)\| \leq \sum_{k \geq 0} \frac{\|A\|^k}{k!} = \exp(\|A\|) < +\infty$$

so exponential map still makes sense for linear algebra! Some algebraic properties can be conclude to handle different calculation, and they are important.

Proposition 4.1.
Let $K = \mathbb{C}$ or $\mathbb{R}$, then

(1) $ge^X g^{-1} = e^{gXg^{-1}}$ for any $g \in \mathrm{GL}_n(K)$ and $X \in M_n(K)$.

(2) If $XY = YX$, then $e^{X+Y} = e^X e^Y$.

(3) e^X is invertible with inverse e^{-X}.

Proof. (1) It is easy to check that

$$(gXg^{-1})^k = gX^k g{-}1$$

thus by definition we have the equality. For (2), we can expand e^{X+Y} by definition and binomial formula, and the commutativity of X and Y allows us to rerrange the terms. For (3), we see the X as the complex matrix and then by Jordan form, we have

$$gXg^{-1} = D(I + N)$$

under the base change of g, where D is diagonal and N is nilpotent such that $DN = ND$, thus

$$ge^X g^{-1} = e^D e^{I+D}$$

with e^D is diagonal and e^{I+D} an upper-triangular matrix with non-zero diagonal entries, so we can conclude that e^X is invertible with inverse e^{-X} by the property (2). $\square$

There are many other proof of the propoistion, and each of them gives a new view about the matrix group and Lie group, for example, we can consider the exponential map as the solution of ODE:

> **Theorem 4.1.** For any $X \in M_n(K)$, the one-parameter
>
> $$\Phi_X : \mathbb{R} \to M_n(K), \qquad t \mapsto e^{tX}$$
>
> is the unique **global C^1 solution** of the initial value problem of ODE:
>
> $$\frac{d}{dt} Y(t) = X\Phi_Y(t), \qquad \Phi_X(0) = I$$
>
> with Y the matrix-valued function of t.

Proof. The uniqueness of the solution can be proved by Cauchy-Lipschitz theorem, for the existence, we should veirfy that Φ_X is exactly the solution: For any $t \in \mathbb{R}$

$$\frac{\Phi_X(t+h) - \Phi_X(t)}{h} = \frac{e^{(t+h)X} - e^{tX}}{h} = e^{tX}\frac{(e^{hX} - I)}{h}$$

The convergence of the expoential series is uniform, which allows us to exchange the limit and the sum, thus we can deduce the derivative of Φ_X is exactly $X\Phi_X$, hence Φ_X is the solution of the ODE (surely it is C^1, Φ_X is derivative so it is continous, that implies that $X\Phi_X$ is continuous too). $\square$

Remark 4.1. For property (2) in propoistion 4.1, here we can consider the ODE

$$Z'(t) = (X+Y)Z(t), \qquad Z(0) = I$$

we can verify that $t \mapsto e^{X+Y}$ and $t \mapsto e^X e^Y$ are both solutions when X and Y commute, thus they are equal by the uniqueness of the solution.

Remark 4.2. Actually, the theorem contain an simple information is that

$$\frac{d}{dt}(e^{tX}) = Xe^{tX}$$

so we notice the amazing similarity with the case in euclidean space, which allows us to consider the one-parameter subgroup as a dynamic system, and the exponential map is the flow of the system, this is the original ideal of the exponential map in Lie theory.

Another method to prove that e^X is invertible is to consider its determinant, which is natural to motivate the following result:

Theorem 4.2 (Jacobi's formula)**.** For any $A \in M_n(K)$, we have

$$\det(e^A) = e^{\mathrm{tr}(A)}$$

Proof. By Schur's lemma, any $X \in M_n(\mathbb{C})$ can be upper-triangularized by some $g \in \mathrm{GL}_n(\mathbb{C})$ such that $X = gTg^{-1}$ the upper-trianglue matrix T with diagonal entries $\lambda_1, ..., \lambda_n$, thus

$$\det(e^X) = e^{\lambda_1} \cdots e^{\lambda_n} = e^{\lambda_1 + \cdots + \lambda_n} = e^{\mathrm{tr}(X)}$$

$\square$

Another analytic aspects of the exponential map is asymptotic expansion, that means we can manuiplate the exponential map as what we do in analysis, but first of all we prove that the exponential map is similarly a smooth map by the requirement of the expansion:

Proposition 4.2. For any $X \in M_n(K)$, the one-parameter

$$\Phi_X : \mathbb{R} \to GL_n(K), \quad t \mapsto e^{tX}$$

defines a real-analytic homomorphism, and we have the following expansion

(1) $\exp(tX)\exp(tY) = \exp(t(X+Y) + \frac{1}{2}[X,Y] + O(t^3))$

(2) $\exp(tX)\exp(tY)\exp(-tX)\exp(-tY) = \exp(t^2[X,Y] + O(t^3))$

for any $X, Y \in M_n(K)$, with bracket $[X,Y] = XY - YX$ the commutator of X and Y.

Proof. For any $t_0 \in \mathbb{R}$, we can expand Φ_X at t_0 by definition:

$$e^{tX} = e^{t_0 X} e^{(t-t_0)X} = e^{t_0 X} \sum_{k=0}^{\infty} \frac{(t-t_0)^k}{k!} X^k$$

The series converges uniformly on any compact interval, so the map is real-analytic. To prove the formula, we just need to expand the left-hand side by definition and rearrange the terms. $\square$

Remark 4.3. If $XY = YX$, then the commutator $[X,Y] = 0$, thus the expansion in (1) is exact, which is just the property (2) in proposition 4.1, hence we get the same result by using the expansion.

Above properties of one-parameter allows us to consider the general case in matrix group, later we will see that the matrix group is indeed a smooth manifold, i.e. a Lie group.

Theorem 4.3. Let $\gamma : \mathbb{R} \to GL_n(\mathbb{R})$ be a one-parameter subgroup, then there exists a unique $X \in M_n(\mathbb{R})$ such that $\gamma(t) = e^{tX}$ for any $t \in \mathbb{R}$, so any one-parameter subgroup of $GL_n(\mathbb{R})$ is exactly real-analytic.

Proof. convolution... $\square$

4.2 Correspondence between lie group and lie algebra

Formally, we define the real-general linear group as following:

$$GL_n(\mathbb{R}) := \{A \in M_n(\mathbb{R}) \mid \det(A) \neq 0\}$$

it is the subset of $M_n(\mathbb{R})$, while the topology of $GL_n(\mathbb{R})$ is just the subspace topology induced by $M_n(\mathbb{R}) \cong \mathbb{R}^{n^2}$, hence we can conlude something about the topology of the group:

- $GL_n(\mathbb{R})$ is a open subset of $M_n(\mathbb{R})$, it is immediately by

  $$GL_n(\mathbb{R}) = \det^{-1}(\mathbb{R} - \{0\})$$

  since determinant is a continuous map, so it can be viewed as a open subset of euclidean space $\mathbb{R}^{n^2}$, so it is a smooth manifold.

- Following the above result, since it can be viewed as an open subset of euclidean space, so it is a "bon espace" in the sense of topology, i.e. semi-locally simply connected, locally path-connected, locally compact, and Hausdorff.

- Consider the opreation of the group
$$m(A,B) = AB^{-1} = \frac{1}{\det(B)} A \cdot adj(B)$$
where $adj(B)$ is the adjoint matrix of B, which is a polynomial of entries of B, hence this map is smooth as polynomial map, which shows that $GL_n(\mathbb{R})$ is a topological group, and in particular, with the good smooth structure, it is a Lie group.

- As the topological group, the local structure is same , so we can just consider the point $I \in GL_n(\mathbb{R})$. As a differential manifold, we can consider the tangent space at I, so we can identify the tangent space
$$T_I(\mathrm{GL}_n(\mathbb{R})) \cong \mathbb{R}^{n^2} \cong M_n(\mathbb{R})$$
Actually, we can view a strong result: For any $X \in M_n(\mathbb{R})$, we define a curve on manifold by
$$\gamma_X : \mathbb{R} \to GL_n(\mathbb{R}), \quad t \mapsto e^{tX}$$
by exponential map, this is smooth natrual curve with derivative $\gamma'_X(0) = X$, hence we prove that $M_n(\mathbb{R})$ is exactly the tangent space of $GL_n(\mathbb{R})$ at I, and we realize the connection between the local structure of the Lie group and its tangent space by one-parameter subgroup, that is the basic ideal of Lie theory.

We conclude above discussion by the following definition, or exactly the **vocabulary**, because they are just te new notation of manifold and vector space, but they are the basic object of Lie theory, so we need to give them a name:

Definition 4.2. A **lie group** is a smooth manifold G with a group structure such that the group operation $m : G \times G \to G$ defined by $m(g,h) = gh^{-1}$ is smooth.

A **real lie algebra** is a $\mathbb{R}$-vector space $\mathfrak{g}$ with a bilinear map $[-,-] : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$ such that for any $x, y, z \in \mathfrak{g}$ and $a, b \in \mathbb{R}$, we have

(1) $[x,y] = -[y,x]$ (antisymmetry)

(2) $[x,[y,z]] + [y,[z,x]] + [z,[x,y]] = 0$ (Jacobi identity)

with above discuss we can see that $GL_n(\mathbb{R})$ is exactly a Lie group, while its tangent space at identity $M_n(\mathbb{R})$ is a vector space, actually it is a Lie alegbra by defining the bracket as following:
$$[X,Y] = XY - YX$$
and we say $\mathfrak{h}$ is a **sub(-lie) algebra** of $\mathfrak{g}$ if $\mathfrak{h}$ is a subspace of $\mathfrak{g}$ and is closed under the bracket, i.e.
$$\forall x, y \in \mathfrak{h}, [x,y] \in \mathfrak{h}$$
To study other matrix groups, we can just consider the submanifold of $GL_n(\mathbb{R})$, i.e. the subgroup of $GL_n(\mathbb{R})$ with the subspace topology, hence we can get the following result:

Theorem 4.3. Let G be a closed subgroup of $GL_n(\mathbb{R})$, then G is a submanifold of $GL_n(\mathbb{R})$, and the tangent space of G at identity is a Lie algebra with

$$T_I(G) = \{M \in M_n(\mathbb{R}) \mid e^{tM} \in G, \forall t \in \mathbb{R}\}$$

it is a **subalgebra** of $M_n(\mathbb{R})$ with the bracket defined by commutator.

Proof. $\square$

which motivates the following definition:

Definition 4.1. A **linear lie group** is a closed subgroup of $GL_n(\mathbb{R})$ for some n, and its lie algebra is defined as its tangent space

$$\mathrm{Lie}(G) = \{M \in M_n(\mathbb{R}) \mid e^{tM} \in G, \forall t \in \mathbb{R}\}$$

and we usually denote it by frak character $\mathfrak{g}$.

We invite the Lie algebra of $GL_n(\mathbb{R})$ by consider its tangent space at identity, conversely, exponential map allows us to know the local structure of the group by its Lie algebra, which can be done beyond the general manifold

Theorem 4.4. The exponential map $\exp : M_n(\mathbb{R}) \to GL_n(\mathbb{R})$ is a local diffeomorphism at 0, in particular, it is a diffeomorphism from some neighborhood of 0 to the image.

Proof. $\square$

Let us consider some examples related to the general linear group:

Example 4.1. We view $\mathbb{C}$ as a real vector space, then there exists a natural embedding

$$\iota : M_n(\mathbb{C}) \to M_{2n}(\mathbb{R}), \quad A + iB \mapsto \begin{pmatrix} A & -B \\ B & A \end{pmatrix}$$

it is a injective continous linear map, and we can verify that

$$(A+iB)(x+iy) = 0 \iff \begin{pmatrix} A & -B \\ B & A \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

which shows that restriction of ι on $GL_n(\mathbb{C})$ takes image in $GL_{2n}(\mathbb{R})$, and image $\iota(GL_n(\mathbb{C}))$ is indeed a closed subgroup of $GL_{2n}(\mathbb{R})$, hence it is a linear lie group, and its lie algebra is exactly the image of ι on $M_n(\mathbb{C})$, which motivates the following diagram

$$\begin{tikzcd} M_n(\mathbb{C}) \arrow[r, "\iota"] \arrow[d, "\exp"'] & M_{2n}(\mathbb{R}) \arrow[d, "\exp"] \\ \mathrm{GL}_n(\mathbb{C}) \arrow[r, "\iota"] & \mathrm{GL}_{2n}(\mathbb{R}) \end{tikzcd}$$

similarly with general linear group, the closed subgroup of $\mathrm{GL}_n(\mathbb{C})$ is also a submanifold of $\mathrm{GL}_n(\mathbb{C})$, and the tangent space at identity is the subalgebra of $M_n(\mathbb{C})$ with the bracket defined by commutator, hence we can get a large class of linear lie groups and lie algebras by this method, it is a big case of linear lie groups.

Example 4.2. (Affine group) We consider a closed subgroup of $\mathrm{GL}_{n+1}(\mathbb{R})$ defined by

$$G = \left\{ \begin{pmatrix} A & b \\ 0 & 1 \end{pmatrix} \mid A \in GL_n(\mathbb{R}), b \in \mathbb{R}^n \right\}$$

and we consider the group action $G \circlearrowright \mathbb{R}^n$ defined by

$$\begin{pmatrix} A & b \\ 0 & 1 \end{pmatrix} \cdot x = \begin{pmatrix} A & b \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ 1 \end{pmatrix} = Ax + b$$

This action is transitive so $\mathbb{R}^n$ is a homogeneous space under G, and we can establish a homeomorphism

$$G/\operatorname{GL}_n(\mathbb{R}) \cong \mathbb{R}^n$$

It is a classic example about Erlangen progamme, here G is a important geometric group, we usually call it the **affine group** $\mathrm{Aff}_n(\mathbb{R})$.

4.3 Adjoint map

Review a bsic fact about the finite-dimensional vector space: any norm on a finite-dimensional Banach space is equivalent, hence it is natural to consider the abstract exponential map on linear opreators

Definition 4.3. Let V be a finite-dimensional vector space over $\mathbb{R}$ or $\mathbb{C}$, we define the exponential map $\mathrm{Exp} : \mathrm{End}(V) \to \mathrm{GL}(V)$ by

$$\mathrm{Exp}(T) = \sum_{k \geq 0} \frac{T^k}{k!}$$

where $T^k = T \circ \cdots \circ T$ is the k-th composition of T.

The definition is well-defined when we consider $\mathrm{End}(V)$ as a Banach space, and we can verify that the properties in proposition 4.1 still hold for the abstract exponential map. In particular, if we choose a basis of V and denote it by $\mathcal{B}$, then we can see their connection by the following commutative diagram:

$$\begin{tikzcd}
\mathrm{End}(V) \arrow[r, "\mathrm{Exp}"] \arrow[d, "{[\cdot]_{\mathcal{B}}}"'] & \mathrm{GL}(V) \arrow[d, "{[\cdot]_{\mathcal{B}}}"] \\
M_n(K) \arrow[r, "\exp"'] & \mathrm{GL}_n(K)
\end{tikzcd}$$

or simply the equality

$$[\mathrm{Exp}(T)]_{\mathcal{B}} = \exp([T]_{\mathcal{B}})$$

Our first aim is to construct the lie alegbra from exactly a lie group.

We consider the conjugation action of a linear lie group G on itself, and then for any $g \in G$, there exists a diffeomorphism by the smooth structure of G

$$c_g : G \to G, \quad h \mapsto ghg^{-1}$$

with $c_g(I) = I$, hence it allows us to consider the differential of c_g at identity,

$$(Dc_g)_I : T_I(G) \to T_I(G), \quad X \mapsto gXg^{-1}$$

the reason is as following: for linear lie group, the tangent space at identity is just Lie(G) by theorem 4.3, so for any $X \in \mathrm{Lie}(G)$, there exists a one-parameter subgroup $\gamma_X : \mathbb{R} \to G$ defined by $\gamma_X(t) = e^{tX}$ with $X = \frac{d}{dt}\big|_{t=0}\gamma_X(t)$, hence we consider the parameterziation of the conjugation

$$\frac{d}{dt}\Big|_{t=0}c_g(\gamma_X(t)) = \frac{d}{dt}\Big|_{t=0}g\gamma_X(t)g^{-1} = gXg^{-1}$$

which shows that $(Dc_g)_I(X) = gXg^{-1}$ exactly, hence by the construction we induce a representation

$$\mathrm{Ad} : G \to \mathrm{GL}(\mathfrak{g}), \quad g \mapsto Ad_g = (Dc_g)_I$$

and we call the representation Ad the **adjoint representation** of G.

Notice that $\mathfrak{g}$ is finite-dimensional, so GL($\mathfrak{g}$) is also a linear Lie group, with its Lie algebra End($\mathfrak{g}$), hence we can consider the differential of Ad at identity

Proposition 4.5. Let G be a linear lie group with lie algebra $\mathfrak{g}$, then the direction derivative of the adjoint representation at identity is

$$(D\,\mathrm{Ad})_I(X) = \frac{d}{dt}\Big|_{t=0}\mathrm{Ad}_{\exp(tX)} = \mathrm{ad}_X$$

with $\mathrm{ad}(X) : \mathfrak{g} \to \mathfrak{g}$ defined by $\mathrm{ad}_X(Y) = XY - YX$ the commutator of X and Y.

Proof. caculation... □

Hence we get the basic idea of the adjoint

Definition 4.4. Let G be a linear lie group with its lie algebra $\mathfrak{g}$, then

- The adjoint representation of G is defined by

  $$\mathrm{Ad} : G \to \mathrm{Aut}(\mathfrak{g}), \quad g \mapsto \mathrm{Ad}_g$$

  with $\mathrm{Ad}_g(X) = gXg^{-1}$ for any $X \in \mathfrak{g}$.

- The adjoint representation of Lie algebra $\mathfrak{g}$ is defined by

  $$\mathrm{ad} : \mathfrak{g} \to \mathrm{End}(\mathfrak{g}), \quad X \mapsto \mathrm{ad}_X$$

  with $\mathrm{ad}_X(Y) = [X, Y]$ for any $Y \in \mathfrak{g}$.

Remark 4.4. An application bettween two lie algebras $f : \mathfrak{g} \to \mathfrak{h}$ is a **morphism of Lie algebra** if it is a linear map and preserves the bracket

$$f([X,Y]_{\mathfrak{g}}) = [f(X), f(Y)]_{\mathfrak{h}}$$

In particular, the isomorphism in the categroy of lie algebras is called **automorphism**, and we use $\mathrm{Aut}(\mathfrak{g})$ to denote the group of automorphisms of $\mathfrak{g}$, which is subspace of $\mathrm{GL}(\mathfrak{g})$, the latest one is the set of all bijective linear maps on $\mathfrak{g}$.

with above discussion, we can describe a core properties of the adjoint representation, which is the connection between the local structure of the group and its lie algebra:

Theorem 4.4. Let G be a linear lie group with its lie algebra $\mathfrak{g}$, then for any $X \in \mathfrak{g}$, we have

$$\mathrm{Ad}_{\exp(X)} = \mathrm{Exp}(\mathrm{ad}_X)$$

i.e. the following diagram commutes:

$$\begin{tikzcd} \mathfrak{g} \arrow[r, "\mathrm{ad}"] \arrow[d, "\exp"'] & \mathrm{End}(\mathfrak{g}) \arrow[d, "\mathrm{Exp}"] \\ G \arrow[r, "\mathrm{Ad}"] & \mathrm{GL}(\mathfrak{g}) \end{tikzcd}$$

Proof. □

5 Lie Algebra

In this section all vector space will be finite-dimensional.

5.1 Ideal

Definition 5.1. Let $\mathfrak{g}$ be a lie algebra with bracket, then $\mathfrak{h} \subset \mathfrak{g}$ is an ideal if $\mathfrak{h}$ is a linear subalgebra and $[g, h] \in \mathfrak{h}$ for any $g \in \mathfrak{g}$ and $h \in \mathfrak{h}$.

Ideal is like the ideal in ring, it allows us to talk about the quotient of Lie algebra, i.e. we consider the quotient of linear space firstly

$$\pi : \mathfrak{g} \to \mathfrak{g}/\mathfrak{h}, \quad X \mapsto X + \mathfrak{h}$$

and then π is a lie algebra morphism if and only if $\mathfrak{g}/\mathfrak{h}$ is a well-defined lie algebra, that allows us to write

$$[x + \mathfrak{h}, y + \mathfrak{h}] = [\pi(x), \pi(y)] = \pi([x, y]) = [x, y] + \mathfrak{h}$$

which ensures us to define the bracket on the quotient linear space such that $\mathfrak{g}/\mathfrak{h}$ is a lie algebra, and similarly we have the isomorhism theorem.

Theorem 5.1. If $f : \mathfrak{g} \to \mathfrak{h}$ is a lie algebra morphism, then $\ker f$ is an ideal of $\mathfrak{g}$ and $\operatorname{Im} f$ is a subalgebra of $\mathfrak{h}$, and we have the isomorphism of lie algebra

$$\mathfrak{g}/\ker f \cong \operatorname{Im} f$$

Remark 5.1. Here are some quick notes about the ideal:

1. The intersection of ideals and sum of ideals are still ideals.
2. The intersection of ideal I and subalgebra $\mathfrak{g}$ will **be the ideal of subalgebra**, and it is necessary to be the ideal of the original algebra.
3. Two important examples of ideal, the first one is the **centre of a lie algebra**

   $$Z(\mathfrak{g}) = \{X \in \mathfrak{g} \mid [X, Y] = 0, \forall Y \in \mathfrak{g}\}$$

   and the second one is the **derived algebra** of a lie algebra

   $$[\mathfrak{g}, \mathfrak{g}] = \{\sum_{i=1}^{n} c_i[x_i, y_i] \in \mathfrak{g} \mid x_i, y_i \in \mathfrak{g}, c_i \in k, n \in \mathbb{N}\}$$

4. We can define the ideal generated by two subsets A and B by

   $$[A, B] = \operatorname{span}\{[a, b] \mid a \in A, b \in B\}$$

   A notation should be clarified is that sometimes we write $\mathfrak{a}\mathfrak{b}$ to instead of $[\mathfrak{a}, \mathfrak{b}]$, the reason is that the bracket is just a non-associative product defined on a module.

Definition 5.2. Let $\mathfrak{g}$ be a lie algebra, and we define

- The **central series** of $\mathfrak{g}$ is defined by

$$\mathcal{C}(\mathfrak{g}) = [\mathfrak{g}, \mathfrak{g}], \quad \mathcal{C}^{n+1}(\mathfrak{g}) = [\mathfrak{g}, \mathcal{C}^n(\mathfrak{g})]$$

  for any $n \geq 1$.

- The **derived series** of $\mathfrak{g}$ is defined by

$$\mathcal{D}(\mathfrak{g}) = [\mathfrak{g}, \mathfrak{g}], \quad \mathcal{D}^{n+1}(\mathfrak{g}) = [\mathcal{D}^n(\mathfrak{g}), \mathcal{D}^n(\mathfrak{g})]$$

  for any $n \geq 1$.

This is two types of decreasing chains of ideals, sometimes we swap the noatation as following:

$$\mathcal{C}^n(\mathfrak{g}) = \mathfrak{g}^n, \quad \mathcal{D}^n(\mathfrak{g}) = \mathfrak{g}^{(n)}$$

for any $n \geq 1$, the reason is same as the notation of derived algebra. Another useful notation is for quotient algebra, we can write

$$\mathcal{C}^n(\mathfrak{g}/\mathfrak{h}) = \mathcal{C}^n(\mathfrak{g}) + \mathfrak{h}/\mathfrak{h}, \quad \mathcal{D}^n(\mathfrak{g}/\mathfrak{h}) = \mathcal{D}^n(\mathfrak{g}) + \mathfrak{h}/\mathfrak{h}$$

Definition 5.1. Let $\mathfrak{g}$ be a lie algebra, then

- $\mathfrak{g}$ is **nilpotent** if there exists some n such that $\mathcal{C}^n(\mathfrak{g}) = 0$.

- $\mathfrak{g}$ is **solvable** if there exists some n such that $\mathcal{D}^n(\mathfrak{g}) = 0$.

Here are some results immediately by the definition:

1. nilpotent implies solvable since $\mathcal{D}^n(\mathfrak{g}) \subset \mathcal{C}^n(\mathfrak{g})$ for any n.
2. If $\mathfrak{g}$ is nilpotent and non-zero, then $Z(\mathfrak{g})$ is non-zero.
3. If $[\mathfrak{g}, \mathfrak{g}]$ is nilpotent, then $\mathfrak{g}$ is solvable.
   (Hint: by definition we can prove $\mathcal{D}^{n+1}(\mathfrak{g}) \subset \mathcal{C}^n([\mathfrak{g}, \mathfrak{g}])$)

The two properties will keep stable under the morphism of lie algebra, that is the following result:

Proposition 5.2. Let $\mathfrak{g}, \mathfrak{h}$ and $\mathfrak{f}$ be lie algebras satisying the short exact sequence

$$0 \to \mathfrak{f} \to \mathfrak{g} \to \mathfrak{h} \to 0$$

then $\mathfrak{g}$ is nilpotent (resp. solvable) if and only if $\mathfrak{f}$ and $\mathfrak{h}$ are nilpotent (resp. solvable), equivalently it is enough to prove the three properties:

(1) The subalgebra of a nilpotent (resp. solvable) lie algebra is still nilpotent (resp. solvable).

(2) The quotient of a nilpotent (resp. solvable) lie algebra is still nilpotent (resp. solvable).

(3) If $\mathfrak{g}/\mathfrak{h}$ and $\mathfrak{h}$ are nilpotent (resp. solvable), then $\mathfrak{g}$ is nilpotent (resp. solvable).

Proof. □

upper-triangular matrix

We have seen that many matirx ring is exactly a lie algebra, and each linear lie group matches a lie algebra, which motivates to consider the representation of an abstract lie algebra.

Definition 5.2.

Lemma 5.3. For any representation $\rho : \mathfrak{g} \to \mathrm{End}(V)$ with V non-zero, if $\rho(x)$ is nilpotent for any x, then there exists some non-zero vector $v \in V$ such that $\rho(x)(v) = 0$ for any x.

Theorem 5.1 (Engel's theorem)**.** Let $\mathfrak{g}$ be a lie algebra, then $\mathfrak{g}$ is nilpotent if and only if ad_x is nilpotent for any $x \in \mathfrak{g}$.

Proof. It is clearly all adjoint map is nilpotent in $\mathrm{End}(\mathfrak{g})$ if $\mathfrak{g}$ is nilpotent, so it is enough to prove the converse. If $\mathfrak{g} = 0$ or $\dim \mathfrak{g} = 1$, then $\mathfrak{g}$ is nilpotent, so we can assume for any $\dim \mathfrak{g} = n \geq 2$, if ad_x is nilpotent for any x, then $\mathfrak{g}$ is nilpotent.

By lemma 5.3 there exists $y \in \mathfrak{g}$ such that $\mathrm{ad}_x(y) = [x, y] = 0$ for any $x \in \mathfrak{g}$, so it means $Z(\mathfrak{g}) \neq 0$. Then we can consider the quotient algebra $\mathfrak{g}/Z(\mathfrak{g})$ with dimension less than n, and any adjoint map of $\mathfrak{g}/Z(\mathfrak{g})$ is nilpotent if any adjoint map of $\mathfrak{g}$ is nilpotent, hence by induction $\mathfrak{g}/Z(\mathfrak{g})$ is nilpotent, so $\mathfrak{g}$ is nilpotent since $Z(\mathfrak{g})$ is always nilpotent. □

Theorem 5.2 (Lie's theorem)**.** Let $\mathfrak{g}$ be a **solvable** lie algebra over an **algebraically closed field with characteristic zero**, then for any finite-dimensional representation $\rho : \mathfrak{g} \to \mathrm{End}(V)$, there exists $v \in V - 0$ such that

$$\rho(x)(v) = \lambda(x) \cdot v$$

for some scalar function $\lambda : \mathfrak{g} \to K$ and any $x \in \mathfrak{g}$.

Proof. □

5.2 Killing form

Definition 5.3. A **non-abelian** lie algebra $\mathfrak{g}$ is simple if $\mathfrak{g} \neq 0$ and the only ideals are 0 and $\mathfrak{g}$.

Remark 5.2. The condition of non-abelian is necessary, otherwise the definition of ideal will be reduced to the linear subspace, and the unique case of "simple" algebras is one-dimensional lie algebra.

Proposition 5.4. The lie algebra $\mathfrak{g}$ is simple if and only if the adjoint representation $\mathrm{ad} : \mathfrak{g} \to \mathrm{End}(\mathfrak{g})$ is irreducible.

Proof. Suppose that the adjoint representation is not irreducible, then there exists a proper non-zero subspace $V \subset \mathfrak{g}$ such that $\mathrm{ad}_X(V) \subset V$ for any $X \in \mathfrak{g}$, i.e.

$$\forall X \in \mathfrak{g}, \forall Y \in V, \quad [X, Y] \in V$$

it shows that V is an proper non-zero ideal of $\mathfrak{g}$, hence $\mathfrak{g}$ is not simple. Conversely, notice that if $\mathfrak{h}$ is an ideal of $\mathfrak{g}$, then

$$\forall X \in \mathfrak{g}, \forall Y \in \mathfrak{h}, \quad [X, Y] \in \mathfrak{h}$$

i.e. which shows that $\mathfrak{h}$ is an invariant subspace of the adjoint representation, hence if $\mathfrak{g}$ is simple, then the adjoint representation is irreducible. □

Definition 5.4. The radical of a lie algebra $\mathfrak{g}$ is defined as the maximal sovlable ideal of $\mathfrak{g}$, and we denote it by $\mathrm{rad}(\mathfrak{g})$.

Remark 5.3. The definition is well-defined since **the sum of two solvable ideals is still a solvable ideal**, alternatively, we can define the radical ideal as

$$\mathrm{rad}(\mathfrak{g}) = \sum \mathfrak{h}$$

where the sum runs over all solvable ideals $\mathfrak{h}$ of $\mathfrak{g}$, and the finite-dimensional condition ensures the choice of ideals is finite, hence the sum is exactly an ideal.

Definition 5.3. The killing form of a lie algebra $\mathfrak{g}$ is the trace form of its adjoint representation, i.e. it is the bilinear form defined by

$$\kappa(x, y) = \mathrm{tr}(\mathrm{ad}_x \circ \mathrm{ad}_y)$$

Remark 5.4. Here are some properties of the killing form:

1. The killing form is symmetric since the trace is symmetric.
2. The killing form is invariant under the adjoint representation

$$\kappa([x, y], z) + \kappa(y, [x, z]) = 0$$

Proposition 5.5. Let $\mathfrak{g}$ be a lie algebra, then the following statements are equivalent:

(a) The only abelian ideal is $\{0\}$.

(b) $\operatorname{rad}(\mathfrak{g}) = \{0\}$

If the base field is of characteristic 0

(c) The killing form of $\mathfrak{g}$ is non-degenerate.

(d) $\mathfrak{g}$ is a direct sum of simple lie algebras.

Proof. $\square$

6 Field and Galois Theory

To study the structure of the field, consider a natural map: Let K be a field

$$\phi : \mathbb{Z} \to K, \quad 1 \mapsto 1_K$$

It is a well-defined homomorphism, which invites a ideal of $\mathbb{Z}$

$$\ker \phi = \{n \in \mathbb{Z} | n \cdot 1_k = 0_k\}$$

By the structure of $\mathbb{Z}$, so there exists a integer $p \in \mathbb{Z}$ such that $\ker \phi = p\mathbb{Z}$,which gives the definition of the characteristic of the field.

Definition 6.1. For any field K, we define $\text{char}(K)$ be the characteristic of the field: either 0 or the smallest integer $n \in \mathbb{N}$ such that $n \cdot 1_k = 0$, by the natural map equivalently

$$\begin{cases} \text{char}(K) = 0 \iff \text{im}\phi \cong \mathbb{Z} \\ \text{char}(K) = p \iff \text{im}\phi \cong \mathbb{Z}/p\mathbb{Z} \end{cases}$$

The characteristic of a field is either zero or a prime number. Suppose that $\text{char}(K) = p \neq 0$, if $p = ab$ is not a prime, then we will get two zero divisor $a \cdot 1_K$ and $b \cdot 1_K$, but a field can not conatain any zero divisor, so p must be prime. An amazing thing in a field (or ring) with non-zero characteristic p is that we can write a equation

$$(x + y)^p = x^p + y^p$$

it has an interesting name: **freshman's dream**, this is an equality that would be written by someone who has studied very little mathematics or is just beginning to learn it.

Another natural map is **Frobenius endomorphism**, let K be a field with non-zero characteristic p, then we define

$$\sigma : K \to K, \quad x \mapsto x^p$$

It is a well-defined injective field homomorphism, so it is a endomorphism, but it is not necessary surjective.

Example 6.1. Consier $K = \mathbb{F}_p(x)$ a field of rational functions. $\text{char}(K) = p$ since $\text{im}\phi = \mathbb{F}_p$, and it is easy to observe that there exists no $f \in \mathbb{F}_p(x)$ such that $f^p(x) = x$, so the Frobenius map here is not surjective.

A good case is finite field, in that case frobenius endomorphism is furthermore a automorphism, so the frobenius map will be in galois group and it reflects the structure of the finite field.

6.1 Finite extension

For the beginning, notice a classic isomorphism:

$$\mathbb{R}[X]/(X-a) \cong \mathbb{R}$$

For any $a \in \mathbb{R}$, and we review the isomorphism given in complex number field:

$$\mathbb{R}[X]/(X^2+1) \cong \mathbb{C}$$

Clearly, $\mathbb{C}$ is a larger field containning $\mathbb{R}$, but we wonder how we can get it from the algebra structure. Here it motivates us to consider the question from the polynoimal structure defined on a field. Firstly we need some lemmas, it is very clear after commutative ring:

Lemma 6.1. Let L be a field containning K as a subfield, then L is a vector space over K.

This lemma allows us to give a rough classification of field extension:

Definition 6.2. If L is a field containing K as a subfield.

- L/K or $K \subset L$ is denoted a **field extension** of K.
- L/K is a **finite extension** if L is a finite-dimensional vector space over K. The **degree** of the finite extension is defined as $[L:K] := \dim_K L$.
- $a \in L$ is **algebraic** over K if there exists a non-zero polynomial $f \in K[X]$ such that $f(a) = 0$, L/K is an **algebraic extension** if every element in L is algebraic over K.
- a is **transcendental** over K if it is not algebraic over K.

Another lemma is the key to construction of field, the reason is that in a PID ring a ideal is maximal if and only if it is generated by an irreducible element.

Lemma 6.2. If K a field and $p \in K[X]$, then p is irreducible if and only if $K[X]/(p)$ is a field.

Hence the a contrsuction of field extension can be given by the quotient of polynoimal ring over a field by a good choice of ideal:

> **Proposition 6.3.** Let K be a field and I be a princiapl ideal generated by a monic irreducible polynoimal $p \in K[X]$ of degree d, Let $L = K[X]/I$, then
> (1) L is a field and K can be embedded in L , so K can be identified as a subfield of L.
> (2) p has a root β in L, exactly $\beta = X + I \in L$
> (3) If $g \in K[X]$ and β is a root of g in L, then $p|g$.

(4) p is the unique monic irreducible polynoimal in $K[X]$ having β as a root in L.
(5) L is a finite extension of K of degree d with the basis $\{1, \beta, ..., \beta^{d-1}\}$.

Proof. (1) L is a field by above lemma, and we consider an embedding $i(a) = a + I$ for any $a \in K$, clearly it is injective and actually it is $\pi|_K$, where π is the canoncial map from $K[x]$ to $K[x]/I$.

(2) Suppose $p(x) = a_0 + ... + a_d x^d$, then in L, we have

$$\begin{aligned} p(\beta) &= a_0 + a_1\beta + ... + a_d\beta^d \\ &= a_0 + a_1(X + I) + ... + a_d(X + I)^d \\ &= a_0 + a_1(X + I) + ... + a_d(X^d + I) \\ &= p(X) + I = I \end{aligned}$$

Here $p(X) \in I$ and I is the zero element in L.

(3) If p does not divide g, then $gcd(p, g) = 1$ since p is irreducible, so PID ring $K[X]$ imples the existence of r, t in $K[X]$ such that

$$1 = s(x)p(x) + t(x)g(x)$$

put $x = \beta$ in L, then $1 = 0$ leads to a contradiction.

(4) immediately from (3). For (5) we use euclidean divison for polynoimal, any $f \in K[X]$, there exists $q, r \in K[X]$ with $\deg r < d$ such that $f(x) = q(x)p(x)+r(x)$, then $f+I = r+I$ in L, and if $r(x) = b_0 + ... + b_k x^k$, $k < d$, by opreations of ideal

$$r + I = b_0 + b_1\beta + ... + b_k\beta^k$$

so $\{1, \beta, ..., \beta^{d-1}\}$ spans L. They are linearly independent because if we assume they are linearly dependent, that means there exists a polynoimal $h \in K[X]$ of degree $< d$ such that h has β as the root, but by (3) we know that $p|h$, which leads to a contradiction since $d = \deg p \leq \deg h$. $\square$

Review the example in the beginning, the complex number field can also be constructed by adjoining element i to $\mathbb{R}$ such that $i^2+1 = 0$, and we alaways write a complex number of the form $a+ib$ with a, b real numbers, which is another construction of field extension, and more generally we have the following proposition:

Proposition 6.4. Let K be a field and $f \in K[X]$ irreducible, adjoin element $c \notin K$ such that c is a root of f, then $K[c]$ is a field containning K as a subfield, and it can be written as $K(c)$.

$$K(c) = \{a_0 + a_1c + ... + a_kc^k | a_1, ..., a_k \in K, k = \deg f - 1\}$$

Proof. $K[c]$ naturally is a ring with same identity with K, so we just need to find the inverse. For any polynoimal $p \in K[X]$ with degree less than $\deg f$, it is co-prime with

f since f is irreducible, so by Bezout's theorem for polynoimal, there exists $r, t \in K[X]$ such that

$$p(x)r(x) + t(x)f(x) = 1$$

hence we take $x = c$ then immediately $p(c)r(c) = 1$, so in $K[c]$ we find the inverse of $p(c)$. and notice that $K[c]$ is a field garantee any rational polynoimal can have the form of polynoimal, so $K[c] = K(c)$ evidently. $\square$

Here we find two method to extension the field, the first method is consider the quotient of polynomial ring, and we can embed K into it; the second is more direct, we just add some element in the field.

Theorem 6.1 (Structure of field extension)**.**
Let L/K be field extension of K and $a \in L$ is algebraic, then
(1) There exists a unique **monic** irreducible polynoimal in $K[X]$ having a as a root, formally we call it **minimal polynoimal** of a over K, and denote it by Π_a.
(2) If $I = (\Pi_a)$, then there exists an isomorphism

$$\phi : K[X]/I \to K(a)$$

with $X + I \mapsto a$ and $c + I \mapsto c$ for all $c \in K$.
(3) Let L'/K is another field extension and $a' \in L'$ is also a root of Π_a, then there exists an isomrphism $\psi : K(a) \to K(a')$ such that $\psi|_K = \mathrm{id}_K$ and $\psi(a) = a'$.

Proof. It needs to consider the **valuation map**, we define $h : K[X] \to L$ by $p \mapsto p(a)$, then clearly $\ker h$ contains all polynoimals having a as a root, then notice that $K[X]$ is a principal ideal ring, so $\ker h$ must be the form (p) with monic $p \in K[X]$. By first isomorphism theorem, we know that $K[X]/I \cong K(a)$, then by Lemma 6.2 p must be irreducible, hence we prove (1) and (2), and we can draw commute diagram to finish (3):

$$\begin{tikzcd}
K[X] \arrow[r, "h"] \arrow[d, "h'"'] \arrow[dr, "\pi"] & K[a] \\
K[a'] & K[X]/I \arrow[l, "\simeq"'] \arrow[u, "\simeq"']
\end{tikzcd}$$

where $h'(p) = p(a')$, and π is the canoncial map. $\square$

Remark 6.1. The theorem shows a induction

$$\{\text{algebraic elements in } L/K\} \rightsquigarrow \{\text{monic irreducible polynoimal over } K\}$$

conversely it is not right since the field extension L may be not enough large, which refers the **"closure"** in the sense of algebraic element. For statement (3), a simple example is $\mathbb{Q}(\sqrt{2}) = \mathbb{Q}(-\sqrt{2})$, and monic irreducible polynoimal $X^2 - 2 \in \mathbb{Q}[X]$.

6.2 Algebraic Closed field

6.3 Normal extension and Automorphism

Example 6.1. We consider the field extension $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$, here the minimal polynoimal of $\sqrt[3]{2}$ over $\mathbb{Q}$ is $X^3 - 2$, we can sovle it in $\mathbb{C}$ with roots:

$$\sqrt[3]{2}, \quad \sqrt[3]{2}\omega, \quad \sqrt[3]{2}\omega^2 \quad \text{with } \omega = e^{2\pi i/3}$$

by above theorem we can connclude $\mathbb{Q}(\sqrt[3]{2}) \cong \mathbb{Q}(\sqrt[3]{2}\omega) \cong \mathbb{Q}(\sqrt[3]{2}\omega^2)$, and it also implies a **question**: quotient ring such as $\mathbb{Q}[X]/(X^3 - 2)$ can not construct a field containing all roots of irreducible polynoimal such as $X^3 - 2$.

However, we can construct a larger field since $\mathbb{Q}(\sqrt[3]{2})$ has been a field: equivalently, we can adjoin ω to $\mathbb{Q}(\sqrt[3]{2})$, or consider the irreducible polynoimal $X^2 + X + 1$ over $\mathbb{Q}(\sqrt[3]{2})$, then we can get a larger field $\mathbb{Q}(\sqrt[3]{2}, \omega)$ containing all roots of $X^3 - 2$.

Above is a typical example to motivate the definition of splitting field, and we can give a more general definition:

Suppose L/K is a extension and a polynoimal $f \in K[X]$, we say f **splits** over K if there exists $c, a_1, \ldots, a_n \in K$ such that f can be factorized as linear terms:

$$f(x) = c(x - a_1)(x - a_2) \cdots (x - a_n)$$

above examples shows that a extension L/K can not always contain all roots of a polynomial, and we can construct a larger field.

Theorem 6.5 (splitting field)**.**
Let K be a field and $f \in K[X]$ with degree $n \geq 1$, then there exists a smallest field extension L/K such that f splits over L, formally such a field extension is called **splitting field** of f over K, and it satisfies:

- $L \cong K[r_1, \ldots, r_n]$ with $r_1, ..., r_n$ are the roots of f in L, it shows that the splitting field is unique up to isomorphism.

- As a finite extension, $[L : K] \leq n!$

Proof. □

Remark 6.2. L.Kronecker proved that for any field K and $f \in K[X]$, there exists a extension L/K such that f splits over L. The smallest of such extension is the splitting field of f, and an extremely larger field is the algebraic closed field of K, although the result is a little old in nowdays.

For the example at the begining, the experience of $\mathbb{C}$ gives us a hint that $\mathbb{Q}(\sqrt[3]{2})$ can not contain the complex number ω, which reminds us to consider the roots of the polynomial in algebraic closure.

Definition 6.1. Let L/K be an algebraic extension, we define

$$\mathrm{Hom}_{K-alg}(L, \bar{K}) = \{f : L \hookrightarrow \bar{K} \mid f|_K = id_K\}$$

to be K**-embedding of** L.

Remark 6.3. It is natural to consider the category of field extensions of K, we denote it by $\mathbf{Ext}_K$. Its objects are the extension $K \subset L$, or we can see it as a K-algebra by an embedding $\tau : K \hookrightarrow L$. Hence its morphism is just the morhism of K-algebra, for example if we consider $f : L \to L'$, then f must be injective since it is a morphism of field, and $f(r) = rf(1) = r$ for any $r \in K$, in particular $L' = \bar{K}$ we get the definition of K-embedding of L. Hence here we get a (non-full) subcategory of $\mathbf{CAlg}_K$.

For any $L/\mathbb{Q}$ algebraic extension, we consider f is a embedding, then we have

$$f(X^3 - 2) = f(X)^3 - 2 \qquad f(X^2 + X + 1) = f(X)^2 + f(X) + 1$$

for any $X \in L$. which allows us to calculate that

$$\mathrm{Hom}_{\mathbb{Q}-alg}(\mathbb{Q}(\sqrt[3]{2}), \bar{\mathbb{Q}}) = \{f_1, f_2, f_3\}$$

with $f_i(\sqrt[3]{2}) = \sqrt[3]{2}\omega^{i-1}$ for $i = 1, 2, 3$.

$$\mathrm{Hom}_{\mathbb{Q}-alg}(\mathbb{Q}(\sqrt[3]{2}, \omega), \bar{\mathbb{Q}}) = \{f_{1,1}, f_{1,2}, f_{1,3}, f_{2,1}, f_{2,2}, f_{2,3}\}$$

with $f_{i,j}(\sqrt[3]{2}) = f_i(\sqrt[3]{2})$ and $f_{i,j}(\omega) = \omega^j$ for $i = 1, 2, 3$ and $j = 1, 2$. We can obeserve that the second case forms exactly a group since the image of $f_{i,j}$ is $\mathbb{Q}(\sqrt[3]{2}, \omega)$ itself, and in particular it can be identified as S_3 (notice $f_{2,1}$ is of order 3 and $f_{1,2}$ is of order 2, they are not commute).

Definition 6.2. Let L/K be a field extension, we define

$$\mathrm{Gal}(L/K) = \mathrm{Hom}_{K-alg}(L, L) = \{\sigma \in \mathrm{Aut}(L) \mid \sigma|_K = \mathrm{id}_K\}$$

to be the **Galois group** (or **automorphism group**) of L/K, it is a subgroup of $\mathrm{Aut}(L)$ which fixes K.

Notice a clear inclusion

$$\mathrm{Gal}(L/K) \subset \mathrm{Hom}_{K-alg}(L, \bar{K})$$

which is suffisant to prove that when $L = \mathbb{Q}(\sqrt[3]{2}, \omega)$, the inclusion turns to be an equality. while $L = \mathbb{Q}(\sqrt[3]{2})$, we can only get $\mathrm{Gal}(L/\mathbb{Q}) = \{f_1\}$, hence the difference between Galois group and the embeddings is the key to distinguish the extensions.

Theorem 6.2. Let L/K be a algebraic extension, then the following conditions are equivalent:

1. L is a splitting field of of a set of polynoimals in $K[X]$.
2. $\mathrm{Gal}(L/K) = \mathrm{Hom}_K(L, \bar{K})$
3. For any $a \in L$, the minimal polynomial $\Pi_a \in K[X]$ splits over L.

If one of the above conditions is satisfied, we say L/K is a **normal extension**.

Remark 6.4. Let $a \in L$ and Π_a is the minimal polynomial of a, then K-embedding induces a bijection to the roots of Π_a in $\bar{K}$, denote the set by R_a

$$\mathrm{Hom}_K(L, \bar{K}) \to R_a, \quad f \mapsto f(a)$$

exactly

6.4 Separable extension

Example 6.2. Let us consider the function field $K = \mathbb{F}_p(t)$

- $X^p - t$ is irreducible in $K[X]$ since K is the fraction field of $\mathbb{F}_p[t]$, and

  $$\mathbb{F}_p[t][X] \cong \mathbb{F}_p[t, X] \cong \mathbb{F}_p[X, t]$$

  allow us to prove the irreducibility in $\mathbb{F}_p[t][X]$, so it holds in fraction field as well.

- consider $L = K[X]/(X^p - t)$, then we can get a extension $L \cong \mathbb{F}_p(t^{1/p})$, which allows to factorize the minimal polynomial

  $$X^p - t = (X - t^{1/p})^p$$

  That shows L is a splitting field of $X^p - t$ over K, so L/K is a normal extension of degree p.

- We exactly construct a extension L/K such that the minimal polynomial has repeated root, in this case the $\bar{K}$-embedding fails to distinguish the roots and $\mathrm{Gal}(L/K) = \mathrm{Hom}_K(L, \bar{K}) = \{id\}$.

Review the case in $\mathbb{R}[X]$, although we do not prove the FTA exactly, but the our experience in $\mathbb{C}$ enough to conclude that any irreducible polynomial in $\mathbb{R}[X]$ can not have multiple roots in $\mathbb{C}$, since for the given distinct roots $R = \{r_1, ..., r_k\}$ with multiplicity $\{m_1, ..., m_k\}$, we can uniqely determine the monic polynomial of the form

$$\prod_{r_i \in \mathbb{R} \cap R} (X - r_i)^{m_i} \cdot \prod_{\text{otherwise}} (X^2 + b_j x + c_j)^{m_j}$$

since the complex roots always appear in pairs, so we can roughly conclude the result. Even more, we can not find a counterexample in $\mathbb{Q}[X]$ as well, this section is to distinguish such a field.

Definition 6.3. Let A be a k-algebra, an endomorphism $\partial : A \to A$ is called a **derivation** if it satisfies the Leibniz rule:

$$\forall f, g \in A, \quad \partial(fg) = \partial(f)g + f\partial(g)$$

It is easy to verify that

$$\begin{aligned}
&\partial(\lambda) = 0, \quad \forall \lambda \in k \\
&\partial(f^n) = nf^{n-1}\partial(f), \quad \forall f \in A, n \in \mathbb{N}
\end{aligned}$$

We can use the derivation to define the formal derivative of a polynoimal, which is a good view that does not depend on the the exactly form of the polynoimal.

$$\Omega_{L/K} = 0 \iff L/K \text{ is separable}$$

This is the general explanation of the separable extension by Kähler differentials. Exactly in the definition we take $A = k[X]$, and then endomorphism will be determined by the image of X by the universal property of polynomial ring, then we can get the formal derivative of any polynoimal

Definition 6.4. Let $\partial(X) = 1$, then for any $f = \sum_n a_n X^n \in k[X]$, we can define the formal derivative of f by

$$f'(X) := \partial(f) = \sum_n n a_n X^{n-1}$$

and a polynoimal f is called **separable** if f is coprime with its formal derivative f' in $k[X]$.

Notice that $k[X]$ is a PID, so the coprime condition is equivalent to the existence of $r, s \in k[X]$ such that

$$1 = r(X)f(X) + s(X)f'(X)$$

we describle the separable polynomial without talking about roots of polynoimal, and similarly we can give a definition of roots of a polynoimal via arithemtic of polynoimal:

Definition 6.5. Let $f \in k[X]$ and $a \in k$, then a is a **root** of f if there exists $k \in \mathbb{N}^*$such that

$$(X - a)^n \mid f(X)$$

and the **order of the root** is defined as the largest integer m such that

$$(X - a)^m \mid f(X) \wedge (X - a)^{m+1} \nmid f(X)$$

in particular, a is a **simple root** if the order of the root is 1, and it is a multiple root if the order of the root is greater than 1.

Remark 6.5. Following this definition, $f(a) = 0$ if a is a root of f, the coverse is not right in general, the ring should at least be an integral domain **(sure for field)** or the polynoimal should be monic, the proof depends on the division of polynoimal.

Proposition 6.6. For polynoimal $f \in k[X]$, the following conditions are equivalent:

1. f is separable.
2. f has no multiple root in $\bar{k}$.
3. f and f' has no common root in a algebraic closure $\bar{k}$ of k.
4. f has exactly $\deg f$ distinct roots in $\bar{k}$.
5. $\bar{k}[X]/(f)$ is reduced and isomorphic to $\bar{k}^{\deg f}$.

Proof. (1) $\Rightarrow$ (2): If f has a multiple root a, then $(X-a)^2 \mid f(X)$, so $f(X) = (X-a)^2 g(X)$ for some $g \in k[X]$, then we can calculate the formal derivative of f:

$$f'(X) = 2(X-a)g(X) + (X-a)^2 g'(X)$$

hence we get $(X-a)|f(X)$ and $(X-a)|f'(X)$, which contradicts the coprime condition.

(2) $\Rightarrow$ (3): in $\bar{k}$, we can factorized f as linear terms, and the formal derivative of f can be caculated of the form

$$f'(X) = \sum_a \frac{f(X)}{X-a}$$

where a runs over all roots of f, we suppose that b is a common root of f', then $f'(b) = f(X)/(X-b)|_{b=0} = 0$ implies that $X-b$ divides $f(X)/-b$, which gives a contradiction.

(3) $\Rightarrow$ (4): similar argument as above.

(4) $\Rightarrow$ (5): The isomorphism can be proved by the **chinese remainder theorem**, and the product of fields is reduced surely.

(5) $\Rightarrow$ (4): Notice that $\bar{k}[X]/(X-a)^2$ is not reduced with $\overline{X-a}^2 = 0$ in the quotient, and for any f we can factorized it to get a isomorphism

$$\bar{k}[X]/(f) \cong \prod_a \bar{k}[X]/(X-a)^{m_a}$$

with a distinct roots of f and m_a the multiplicity, so surely m_a should be all 1.

(4) $\Rightarrow$ (1): caculation. $\square$

actually we can characterize the irreducible polynomial if it is separable, it depends on the characteristic of the field.

Proposition 6.7. Let f be an irreducible polynomial in $k[X]$, then f is inseparable if and only if $f' = 0$, which shows that
- If $\mathrm{Char}(k) = 0$, any irreducible polynomial is separable.
- If $\mathrm{Char}(k) = p$, any irreducible polynomial is of the form $g(X^{p^r})$ with g a separable irreducible polynomial and $r \in \mathbb{N}$, and the representation is unique.

Proof. In $k[X]$ (a UFD), f and f' coprime means their gcd associates to 1, or equivalently the gcd is a unit. When f is irreducible, then f is a prime element in $k[X]$, so the gcd is either f or a unit. If f is inseparable, then the gcd is f, so $f|f'$ implies $f' = 0$ since $\deg f' < \deg f$.

□

Definition 6.3. Let L/K be a algebraic extension, then L/K is called a **separable extension** if for any $a \in L$, the minimal polynomial Π_a of a over K is separable.

Remark 6.6. Any subextension of a separable extension is separable: Let $K \subset E \subset L$ be a tower of algebraic extension, and L/K is separable. If $a \in E$, then $\Pi_{a,E}$ divides $\Pi_{a,L}$, and factor of separable polynoimal is still separable (verify).

By proposition above, we can conclude that **any algebraic extension of a field with characteristic zero is separable**, in particular the extension of $\mathbb{Q}$, it is the core object in number theory. Hence, it is hopeless to find a example of inseparable extension of $\mathbb{Q}$ or $\mathbb{R}$, when we change our attention to field of non-zero characteristic, we can find a such example like beginning of this section.

Theorem 6.3. Let L/K be a finite extension, then the following conditions are equivalent:

1. L/K is separable.
2. $|\,\mathrm{Hom}_K(L, \bar{K})| = [L : K]$
3. $\bar{K} \otimes_K L$ is reduced and is isomorphic to $\bar{K}^{[L:K]}$.

Proof. □

Remark 6.7. When the extension is infinite, the condition 2 makes no sense, but it is still make sense to consider the condition 3. In the example of $L = \mathbb{F}_p(t^{1/p})$, we can find that the degree is p while the number of embedding is 1, so it is another way to show that L is an inseparble extension.

Corollary 6.8 (Primitive element theorem)**.** Any finite separable extension is a simple extension, that is, for any finite separable extension L/K, there exists $a \in L$ such that $L = K(a)$, such a element a is called a **primitive element** of the extension.

Proof. ☐

Remark 6.8. If $L = K(a)$ a simple algebraic extension, we denote the roots of the minimal polynomial Π_a in $\bar{K}$ by R, then we can get a natural bijection:

$$\mathrm{Hom}_K(L, \bar{K}) \to R, \quad f \mapsto f(a)$$

It is well-defined since embedding sends a to another root, and the map is always injective: For any $f, g \in \mathrm{Hom}_K(L, \bar{K})$ with $f(a) = g(a)$, take any $x \in L = K(a)$, then there exists a polynoimal $p \in K[X]$ such that $x = p(a)$, so we can calculate

$$f(x) = f(p(a)) = p(f(a)) = p(g(a)) = g(p(a)) = g(x)$$

which shows that $f = g$.

Hence the simple algebraic extension allows to identify the conjuations of the primitive element with the K-embeddings of the extension, and surely simple algebraic extension is not necessary separable, for example $L = \mathbb{F}_p(t^{1/p})$.

6.5 Galois extension

Let K/k be a finite extension, with $\mathrm{Aut}(K/k)$ the automorphism group then we can consider the group action $\mathrm{Aut}(K/k) \circlearrowleft K$ defined by

$$\sigma \cdot a := \sigma(a)$$

the fixed point of the action will be remarkable:

Lemma 6.9. Let $S \subset \mathrm{Aut}(K/k)$ be a non-empty subset, then

$$K^S := \{x \in K \mid \sigma(x) = x \text{ for all } \sigma \in S\}$$

is a intermediate field of K/k.

Proof. $\sigma \in \mathrm{Aut}(K/k)$ fixes k, so $k \subset K^S$. For any $x, y \in K^S$ and $\sigma \in S$, we can calculate

$$\sigma(x + y) = \sigma(x) + \sigma(y) = x + y$$

$$\sigma(xy) = \sigma(x)\sigma(y) = xy$$

it is enough to show the reuslt. ☐

Theorem 6.4. Let K/k be a finite extension, then the following conditions are equivalent:

1. K/k is normal and separable.
2. For any $a \in K$, the minimal polynomial of a splits over K with
$$\Pi_a(X) = \prod_{\sigma \in \mathrm{Aut}(K/k)} (X - \sigma(a))$$
3. K is a splitting field of a separable polynomial in $k[X]$.
4. $|\mathrm{Aut}(K/k)| = [K : k]$
5. Fixed field $K^{\mathrm{Aut}(K/k)} = k$.

Proof. (1) implies (2): Since K/k is normal, then $\mathrm{Aut}(K/k)$ is just the k-embedding onto itself, that means $\sigma(a)$ runs over all roots of Π_a in $\bar{k}$ exactly, and since K/k is separable, the roots are of one-multiplicity, so Pi_a splits like the form in (2).

(2) implies (3): Π_a is a separable polynomial in $K[X]$ and splits over K, so we can consider the splitting field of all minimal polynomials.

(3) implies (1), it is immediate.

(3) implies (4): Since (1) and (3) are equivalent, so it is enough to show that (1) implies (4), and it can be proved by cardinal
$$|\mathrm{Aut}(K/k)| = |\mathrm{Hom}(K, \bar{k})| = [K : k]$$
The first equality holds since K/k is normal, and the second equality holds since K/k is separable.

(4) implies (5): Set $k' = K^{\mathrm{Aut}(K/k)}$, then we can prove that $\mathrm{Aut}(K/k') = \mathrm{Aut}(K/k)$, and then we can prove that $k' = k$ by proving that
$$[K : k'] \leq [K : k] = |\mathrm{Aut}(K/k)| = |\mathrm{Aut}(K/k')| \leq [K : k']$$
the first inequality holds since $k \subset k'$, and the last inequality holds since for any finite extension K/k we have
$$\mathrm{Aut}(K/k) \subset \mathrm{Hom}(K, \bar{k}), \quad |\mathrm{Hom}(K, \bar{k})| \leq [K : k]$$

(5) implies (1): For any $a \in K$, we can set consider the orbit of a, it contains the distinct roots of Π_a in K, and then we set $f(X) = \prod_{r \in Orb(a)} (X - r)$. For any $\sigma \in \mathrm{Aut}(K/k)$, we can view that $\sigma(f) = f$ since σ just permutes the roots
$$\sigma(f) = \prod_{r \in Orb(a)} (X - \sigma(r)) = \prod_{r \in Orb(\sigma(a))} (X - r) = f$$
that means any coefficient of f will lie in $K^{\mathrm{Aut}(K/k)} = k$, i.e. $f \in k[X]$. Hence Π_a divides f since $f(a) = 0$, and Π_a is separable since f is separable. So we can actually show (5) implies (3), and (3) implies (1) as before. □

The equivalent statement motivates us to give the definition:

Definition 6.6. An algebraic extension K/k is called a **Galois extension** if it is normal and separable, and in this case we call $\mathrm{Aut}(K/k)$ the **Galois group** of the extension, and denote it by $\mathrm{Gal}(K/k)$.

Another **alternative definition** of galois extension is an algebraic extension K/k such that $K^{\mathrm{Aut}(K/k)} = k$, which is equivalent to the definition above by the theorem without limitation on the finiteness of the extension.

To establish the correspondence between the intermediate fields of a galois extension and the subgroups of the galois group, we investigate an example carefully:

Example 6.2. Let $K = \mathbb{Q}(\sqrt[3]{2}, \omega)$ be the extension of $\mathbb{Q}$, it is exactly a galois extension with $\mathrm{Gal}(K/\mathbb{Q}) = S_3$. We set $a = \sqrt[3]{2}$, then we can denote the basis by

$$1, a, a^2, \omega, a\omega, a^2\omega$$

Exactly the galois group can be generated by two elements $g = f_{1,2}$ and $h = f_{2,1}$, with

$$g(a) = a\omega, \quad g(\omega) = \omega$$

$$h(a) = a, \quad h(\omega) = \omega^2$$

so we can find that g is of order 3 and fixes $\mathbb{Q}(\omega)$, while h is of order 2 and fixes $\mathbb{Q}(\sqrt[3]{2})$, i,e. we construct two intemediate fields in according to the subgroups

$$K^G = \mathbb{Q}(\omega), \quad G = \langle g \rangle \cong \langle (1,2,3) \rangle \cong A_3$$

$$K^H = \mathbb{Q}(\sqrt[3]{2}), \quad H = \langle h \rangle \cong \langle (1,2) \rangle$$

Similarly, we can recover all possible intermediate fields by considering the subgroups of S_3, and it can be drawn as following

$$\begin{tikzcd}
& \mathbb{Q}(a,\omega) \arrow[dl, dash] \arrow[d, dash] \arrow[dr, dash] \arrow[drr, dash] & & \\
\mathbb{Q}(\omega) \arrow[dr, dash] & \mathbb{Q}(a) \arrow[r, dotted, dash, "\sim"] \arrow[d, dash] & \mathbb{Q}(a\omega) \arrow[r, dotted, dash, "\sim"] \arrow[dl, dash] & \mathbb{Q}(a\omega^2) \arrow[dll, dash] \\
& \mathbb{Q} & &
\end{tikzcd}$$

where the intermediate fields correspond to all non-trival proper subgroups of S_3:

$$A_3, \langle (1,2) \rangle, \langle (2,3) \rangle, \langle (1,3) \rangle$$

Notice that A_3 is the unique normal subgroup of S_3, while $\mathbb{Q}(\omega)$ is the unique sub-galois extension of $K/\mathbb{Q}$, so it motivates us to consider the general case.

Proposition 6.10. Let K/k be a galois extension with L a intermediate field, then

1. K/L is a galois extension.
2. $\mathrm{Gal}(K/L)$ a is subgroup of $\mathrm{Gal}(K/k)$, if extension K/k is finite, then the index of galois group is $[L : k]$.
3. For any $\sigma \in \mathrm{Gal}(K/k)$, we have conjugation
$$\mathrm{Gal}(K/\sigma(L)) = \sigma \,\mathrm{Gal}(K/L)\sigma^{-1}$$
4. L/k is galois if and only if $\mathrm{Gal}(K/L)$ is a normal subgroup of $\mathrm{Gal}(K/k)$, and in this case we have an isomorphism
$$\mathrm{Gal}(L/k) \cong \mathrm{Gal}(K/k)/\,\mathrm{Gal}(K/L)$$

Proof. (1) For any $a \in K$, Let $\Pi_a \in k[X]$ be the minimal polynomial of a over k, and $M_a \in L[X]$ be the minimal polynomial of a over L, then M_a divides Π_a, it implies that M_a is separable since Π_a is separable. What's more, the roots of M_a are also the roots of Π_a, and Π_a takes all roots in K, so does M_a, hence K/L is normal amd separable.

(2) Since $k \subset L$, hence $\mathrm{Gal}(K/L) \subset \mathrm{Gal}(K/k)$, and it is not hard to verify it is exactly a subgroup, for index we can conclude that

$$[\mathrm{Gal}(K/k) : \mathrm{Gal}(K/L)] = [K : k]/[K : L] = [L : k]$$

The first equality holds since the two extensions are galois, and the second holds by tower law.

(3) For any σ we have

$$\begin{aligned} f \in \mathrm{Gal}(K/\sigma(L)) &\iff f(\sigma(x)) = \sigma(x), \quad \forall x \in L \\ &\iff \sigma^{-1} \circ f \circ \sigma|_L = id_L \\ &\iff \sigma^{-1} \circ f \circ \sigma \in \mathrm{Gal}(K/L) \\ &\iff f \in \sigma \,\mathrm{Gal}(K/L)\sigma^{-1} \end{aligned}$$

(4) If L/k is galois, then $\mathrm{Gal}(L/k) = \mathrm{Hom}_k(L, \bar{k})$ since it is normal, hence for any $\sigma \in \mathrm{Gal}(K/k)$, when consider $\sigma|_L$ as the restriction map, it will be contained in the k-embedding of L, so $\sigma(L) = L$, which shows that the following map is well-defined

$$\mathrm{Gal}(K/k) \to \mathrm{Gal}(L/k), \quad \sigma \mapsto \sigma|_L$$

the kenel of the map is exactly $\mathrm{Gal}(K/L)$, so it is enough to prove it is surjective.

Conversely, if $\mathrm{Gal}(K/L)$ is normal, then for any $\sigma \in \mathrm{Gal}(K/k)$ we have

$$\sigma(L) = K^{\mathrm{Gal}(K/\sigma(L))} = K^{\sigma \,\mathrm{Gal}(K/L)\sigma^{-1}} = K^{\mathrm{Gal}(K/L)} = L$$

so again above map is well-defined and surjective, so any $\tau \in Gal(L/k)$, there exists $\sigma \in \text{Gal}(K/k)$ such that $\tau(L) = \sigma|_L(L) = L$, which shows that L/k is normal, so it is galois as well since the sub-extension of a separable extension is separable.

□

Definition 6.7. Let $(\mathcal{F}, \leq_F)$ and $(\mathcal{G}, \leq_G)$ be two partially ordered sets, a application $r : \mathcal{F} \to \mathcal{G}$ is called **order-preserving (isotone)** if for any $x, y \in \mathcal{F}$ with $x \leq_F y$, we have $r(x) \leq_G r(y)$. Conversely, is is called **order-reversing (antitone)** if $r(x) \geq_G r(y)$.

The galois correspondence can be stated by the language of category, firstly any partial ordered set $(P, \leq)$ can be viewed as a categroy as following:

$$\mathcal{C}_P \left| \begin{array}{l} \text{objects: } x \in P \\ \text{morphisms: } \text{Hom}(x, y) = \{*\} \text{ if } x \leq y, \text{ otherwise } \emptyset \end{array} \right.$$

hence an isotone (antitone) bijection between two partially ordered sets can be viewed as an (anti)euquiavlence between the corresponding categories, so we can restate the galois correspondence as following:

Theorem 6.5 (Galois correspondence)**.**
Let K/k be a galois extension, let $\mathcal{I}$ be the set of intermediate fields of K/k and $\mathcal{S}$ be the set of subgroups of $\text{Gal}(K/k)$, then they are all partial ordered sets by inclusion $\subset$, and there exists an antitone bijection between them by

$$L \mapsto \text{Gal}(K/L) \quad \text{with inverse} \quad H \mapsto K^H$$

In another word, the antitone as a functor gives an equivalence

$$\mathcal{C}_{\mathcal{I}} \simeq \mathcal{C}_{\mathcal{S}}^{op}$$

Proof. □

As we have seen in the previous example ($a = \sqrt[3]{2}$), we can consider the galois extension tower

$$\mathbb{Q} \subset \mathbb{Q}(\omega) \subset \mathbb{Q}(a, \omega)$$

with Galois groups $\text{Gal}(\mathbb{Q}(\omega)/\mathbb{Q}) \cong \mathbb{Z}/(2)$ and $\text{Gal}(\mathbb{Q}(a, \omega)/\mathbb{Q}) \cong \mathbb{Z}/(3)$, while in the theory of group the symmetric group can be viewed as a semi-direct product

$$S_3 = \mathbb{Z}/(3) \rtimes \mathbb{Z}/(2)$$

so it is natural to ask whether the sturcture of galois group can be reflected in a extension tower.

Definition 6.8. Let K/k and L/k be two extensions contained in an extensionΩ/k, then we define the composite of K and L over k by

$$KL = \langle K \cup L \rangle = \bigcap_{K \cup L \subset E \subset \Omega} E$$

as the samllest subextension of Ω/k containing both K and L.

Remark 6.9. For any two algebraic extensions, we can always take $\Omega = \bar{k}$ to define, Furthermore if K and L are finite extensions, then we always have

$$[KL : K] \leq [K : k][L : k]$$

since if we write $K = k(a_1, \ldots, a_n)$ and $L = k(b_1, \ldots, b_m)$, clearly $KL \subset k(a_i b_j)$ since $K \cup L \subset k(a_i b_j)$. In the category of $\mathbf{Ext_k}$

Remark 6.10. Composite of extensions is **stable** under the separable and normal condition, hence the composite of galois extensions is still a galois extension, if the composite is exactly a pushout, we can talk something about the structure of galois group.

Proposition 6.11. Let K_1 and K_2 be the separable extensions of k, denote $K_{12} = K_1K_2$, then the following conditions are equivalent:

1. The canconical map $K_1 \otimes_k K_2 \to K_{12}, x_1 \otimes x_2 \mapsto x_1x_2$ is an isomorphism.

2. $K_1 \otimes_k K_2$ is a integral domain.

3. For any $a \in K_2$, the minimal polynomial of a over k is still irreducible in $K_1[X]$.

   If K_1/k and K_2/k are finite

4. $[K_{12} : k] = [K_1 : k][K_2 : k]$

   If K_1/k or K_2/k are galois (i.e. normal here)

5. $K_1 \cap K_2 = k$

If one of above conditions holds, we say that K_1 and K_2 are **linearly disjoint** over k, and K_{12} is the **extension étrangère**.

Remark 6.11. The normal condition here is necessary, consider the previous example $a = \sqrt[3]{2}$, we take $K_1 = \mathbb{Q}(a)$ and $K_2 = \mathbb{Q}(a\omega)$, then we can find that $K_{12} = \mathbb{Q}(a, \omega)$ exactly, while

$$9 = [K_1 : k][K_2 : k] \neq [K_{12} : k] = 6$$

although $K_1 \cap K_2 = \mathbb{Q}$ exactly.

Proof. □

Corollary 6.12. Along above notation, if K_{12}/k and K_1/k are finite galois extensions, and K_{12} is a extension étrangère, then galois group of the composite extension is **the inner semi-direct product**

$$\mathrm{Gal}(K_{12}/K_1) \rtimes \mathrm{Gal}(K_{12}/K_2) \cong \mathrm{Gal}(K_{12}/k)$$

In particular the product is reduced to diredct product **if and only if** K_2/k is a galois extension.

Proof. $\square$

7 Theory of Algebraic Number

7.1 Norm, Trace and Discriminant

For any finite extension L/K of degree n, we can view element $a \in L$ as a K-linear map

$$m_a : L \to L, \quad x \mapsto ax$$

in pariticular, it is an automorphism of L if $a \neq 0$, and it satifies several properties:

- $m_{a+b} = m_a + m_b$
- $m_{ab} = m_a \circ m_b$
- $m_{c \cdot a} = c \cdot m_a$

with $a, b \in L$ and $c \in K$. Since L is a finite-dimensional, it is natural to consider m_a as a matrix and define two invariants in linear algebra

- Trace of a over K: $\mathrm{Tr}_{L/K}(a) = \mathrm{Tr}(m_a)$

- Norm of a over K: $\mathrm{N}_{L/K}(a) = \det(m_a)$

exactly what we do can be concluded as the following

$$L \hookrightarrow \mathrm{End}_K(L) \to K$$

the first map refers to a **regular representation** of L over K, and the second map is what we are familar with, and we can verify some properties of trace and norm:

- $\mathrm{Tr}_{L/K}(a+b) = \mathrm{Tr}_{L/K}(a) + \mathrm{Tr}_{L/K}(b)$
- $\mathrm{Tr}_{L/K}(c \cdot a) = c \cdot \mathrm{Tr}_{L/K}(a)$
- $\mathrm{N}_{L/K}(ab) = \mathrm{N}_{L/K}(a) \cdot \mathrm{N}_{L/K}(b)$
- $\mathrm{N}_{L/K}(c \cdot a) = c^n \cdot \mathrm{N}_{L/K}(a)$
- $\mathrm{Tr}_{L/K}(c) = nc$ and $\mathrm{N}_{L/K}(c) = c^n$

with $a, b \in L$ and $c \in K$. We can also give a more explicit description of trace and norm by the minimal polynoimal of a over K, moreover, for any endomorphism m_a we can consider its **characteristic polynomial** $\chi_a(t) = \det(t \cdot \mathrm{id}_L - m_a)$, and conclude the following result:

Proposition 7.1. For any L/K finite extension with degree n and $a \in L$,

(a) $\chi_a = (\Pi_a)^d$ with $d = [L : K(a)]$.

(b) Suppose $\Pi_a(t) = t^m + c_{m-1}t^{m-1} + ... + c_0$, then

$$\mathrm{Tr}_{L/K}(a) = -dc_{m-1} \qquad \mathrm{N}_{L/K}(a) = (-1)^n c_0^d$$

Proof. (a) It is clear that $\deg \Pi_a = [K(a) : K]$, we denote it by m, then

$$\Pi_a(t) \ = t^m + c_{m-1}t^{m-1} + ... + c_0$$

for some $c_i \in K$, and $\{1, a, ..., a^{m-1}\}$ is a basis for $K(a)/K$, we suppose that $\{e_1, .., e_d\}$ is a basis for $L/K(a)$, then we can conclude a basis of L/K is

$$e_1, \dots, e_1 a^{m-1} \mid e_2, \dots, e_2 a^{m-1} \mid \cdots \mid e_d, \dots, e_d a^{m-1}$$

so we can write the matrix of m_a under this basis

$$m_a \sim \mathrm{diag}(A_1, \dots, A_d), \quad A_i = \begin{pmatrix} 0 & 0 & \dots & -c_0 \\ 1 & 0 & \dots & -c_1 \\ 0 & 1 & \dots & -c_2 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & -c_{m-1} \end{pmatrix}$$

By caclulate the characteristic polynomial of A_i we can get exactly

$$\chi_a(t) = \prod_{i=1}^{d} \det(t \cdot I - A_i) = (\Pi_a(t))^d$$

(b) By vieta's formula we can conclude that

$$\chi_a(t) = t^n - \mathrm{Tr}_{L/K}(a)t^{n-1} + \cdots + (-1)^n \mathrm{N}_{L/K}(a)$$

then by formula of (a) we can caclculate the result. □

The invariant behaves well in separable extension (especially, Galois extension), for example the number fields, the reason is that we can talk about the conjugations of elements with invariant.

Proposition 7.2. Let L/K be a separable finite extension of degree n, then

- $\mathrm{N}_{L/K}(a) = \prod_\sigma \sigma(a)$

- $\mathrm{Tr}_{L/K}(a) = \sum_\sigma \sigma(a)$

- $\chi_a(X) = \prod_\sigma (X - \sigma(a))$
where $a \in L$ and $\sigma \in \mathrm{Hom}_K(L, \bar{K})$ runs over all K-embeddings

Proof. We consider a relation of equivalence for $\Sigma = \mathrm{Hom}_K(L, \bar{K})$ by

$$\sigma_1 \sim_a \sigma_2 \iff \sigma_1(a) = \sigma_2(a)$$

it holds for any finite extension, then we calculate the class of equivalence:

$$\begin{aligned}[\sigma] &= \{\tau \in \Sigma \mid \tau(a) = \sigma(a)\} \\ &= \{\tau \in \Sigma \mid \tau(p(a)) = \sigma(p(a)), \forall p \in K[X]\} \\ &= \{\tau \in \Sigma \mid \tau = \sigma \text{ on } K(a)\}\end{aligned}$$

which shows that $|[\sigma]| = |\mathrm{Hom}_{K(a)}(L, \bar{K})|$, in particular, if L/K is separable, then $L/K(a)$ is also separble, so exactly $|[\sigma]| = [L : K(a)]$, it implies

$$\prod_{\sigma}(X - \sigma(a)) = \prod_{[\sigma] \in \Sigma/\sim_a} (X - \sigma(a))^{[L:K(a)]} = \Pi_a(X)^{[L:K(a)]} = \chi_a(X)$$

Here $\sigma(a)$ runs over all roots of Π_a since the restriction map

$$res : \mathrm{Hom}_K(L, \bar{K}) \to \mathrm{Hom}_K(K(a), \bar{K}), \quad \sigma \mapsto \sigma|_{K(a)}$$

is srujective.

□

Definition 7.1. Let L/K be a finite extension of degree n
- The **trace pairing** of the extension is a symmetric K-bilinear form

$$Q_{L/K}(x, y) = \mathrm{Tr}_{L/K}(xy)$$

- we define the **discriminant** of a family $\{e_1, \ldots, e_n\}$ in L by

$$\mathrm{Disc}(e_1, ..., e_n) = \det(\mathrm{Tr}_{L/K}(e_i e_j)) = \det(Q_{L/K}(e_i, e_j))$$

It is better to consider the discriminant in a separable extension, although we always use the **"free condition"** in the extension of $\mathbb{Q}$, the reason is as following:

Proposition 7.3. Let L/K be a finite extension, then the following conditions are equivalent:

1. L/K is separable.
2. $\mathrm{Tr}_{L/K} \neq 0$.
3. The trace pairing $Q_{L/K}$ is non-degenerate.

Proof. □

Remark 7.1. Consider the inseparable example of $L = \mathbb{F}_p(t^{1/p})$, we can calculate $\mathrm{Tr}_{L/K}(u^k) = 0$ with $u = t^{1/p}$ and $k = 0, 1, ..., p-1$, which shows that the trace is identically zero, so the discriminant of any basis is zero, which provides no information about the extension.

Proposition 7.4. Let L/K be a finite separable extension of degree n, and $a_1, ..., a_n$ be a basis of L/K, then the discriminant of the basis

$$\mathrm{Disc}(a_1, ..., a_n) = \det(\mathrm{Tr}_{L/K}(\sigma_i(a_j)))^2$$

with $\sigma_i \in \mathrm{Hom}_K(L, \bar{K})$ n distinct K-embeddings of L.

Proof. Review that for any two n order square matrices A, B, we have

$$(AB)_{i,j} = \sum_{k=1}^{n} A_{i,k}B_{k,j}$$

In particular

$$(A^T A)_{i,j} = \sum_{k=1}^{n} A_{k,i}A_{k,j}$$

so For any (i, j)-entry of trace matrix, applying proposition 7.2 we can calculate

$$\mathrm{Tr}_{L/K}(a_i a_j) = \sum_{k=1}^{n} \sigma_k(a_i)\sigma_k(a_j) = \sum_{k=1}^{n} \sigma_k(a_i)\sigma_k(a_j) = \sum_{k=1}^{n} = (A^T A)_{i,j}$$

where $A = \mathrm{Tr}_{L/K}(\sigma_i(a_j))_{1\leq i,j\leq n}$. □

Corollary 7.5. Let L/K be a finite separable extension of degree n, then the discriminant of any basis of L/K is no-zero, in particular, if a is the primitive element such that $L = K(a)$ and $\{1, ..., a^{n-1}\}$, then

$$\mathrm{Disc}(1, a, ..., a^{n-1}) = \prod_{i<j}(r_i - r_j)^2$$

where r_i are the distinct n roots of Π_a in $\bar{K}$.

Proof. Review that

$$(PAP^T)_{i,j} = \sum_{1\leq k,l\leq n} P_{i,k}P_{j,l}A_{k,l}$$

For any two n order square matrices A, P. For any basis $\{e_1, ..., e_n\}$, there exists a invertible matrix P such that $e_i = \sum_{j=1}^{n} P_{ij}a^{j-1}$, since $\{1, \ldots, a^{n-1}\}$ is a basis, then

$$\begin{aligned}
\mathrm{Tr}_{L/K}(e_i e_j) &= \mathrm{Tr}_{L/K}\left(\left(\sum_{k=1}^{n} P_{ik}a^{k-1}\right)\left(\sum_{l=1}^{n} P_{jl}a^{l-1}\right)\right) \\
&= \sum_{k,l=1}^{n} P_{ik}P_{jl}\mathrm{Tr}_{L/K}(a^{k+l-2}) \\
&= \sum_{k,l=1}^{n} P_{ik}P_{jl}\mathrm{Tr}_{L/K}(a^{k+l-2})
\end{aligned}$$

which shows that the trace matrix of the basis is $P^T AP$, where $A = (\mathrm{Tr}_{L/K}(a^{i-1}a^{j-1}))_{i,j}$, which suffice to calculate that

$$\mathrm{Disc}(e_1, ..., e_n) = |\det(P)|^2 \,\mathrm{Disc}(1, a, \ldots, a^{n-1})$$

Hence just need to calculate the discriminant of the basis $\{1, ..., a^{n-1}\}$, and prove that it is non-zero, which finishes the proof. By the above proposition, we can calculate

$$\mathrm{Disc}(1, a, \ldots, a^{n-1}) = \begin{vmatrix} 1 & \sigma_1(a) & \ldots & \sigma_1(a)^{n-1} \\ 1 & \sigma_2(a) & \ldots & \sigma_2(a)^{n-1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \sigma_n(a) & \ldots & \sigma_n(a)^{n-1} \end{vmatrix}^2$$

where $\sigma_i(a)$ runs over all distinct roots since L/K is separable, hence we can conclude the result by the **Vandermonde determinant formula**.

Remark 7.2. A remakrable fact in the proof is the scale of the discrimniant, suppose that $\{e_1, \ldots, e_n\}$ and $\{a_1, \ldots, a_n\}$ are two family with $[e_1, \ldots, e_n] = P[a_1, \ldots, a_n]$ for some endomorphiosm P, then

$$\mathrm{Disc}(e_1, \ldots, e_n) = |\det(P)|^2 \mathrm{Disc}(a_1, \ldots, a_n)$$

□

7.2 Number fields

To motivate the algebraic approach, we firstly consider a typical example in number theory, it will be the basis of the discussion.

- The finite extension of $K/\mathbb{Q}$ is called a **number field**, typically it an separable extension since $\mathrm{Char}(\mathbb{Q}) = 0$, so we can apply **primitive element** theorem to conclude that $K = \mathbb{Q}(a)$ for some $a \in K$.
- The element $a \in K$ is called an **algebraic integer** if there exists a monic polynomial $f(x) \in \mathbb{Z}[x]$ such that $f(a) = 0$. Notice that the following conditions are equivalent:
  (a) a is an algebraic integer.
  (b) minimal polynomial $\Pi_a(X) \in \mathbb{Z}[X]$.
  (c) $\mathbb{Z}[a]$ is a finitely generated $\mathbb{Z}$-module.

  Proof. □
- Following the above definition, we can verify a similar result with the algebraic numbers: (a) sum of two algebraic integers is an algebraic integer, (b) product of two algebraic integers is an algebraic integer.

Proof. ☐

- It is trival that algebraic integers are algebraic numbers, so similarly we can define
$$\tilde{\mathbb{Q}} = \{x \in \bar{\mathbb{Q}} : x \text{ is an algebraic integer}\}$$
It is a subring of algebraic closed field $\bar{\mathbb{Q}}$ or $\mathbb{C}$, and then we can define the ring of integers of number field K as
$$\mathcal{O}_K := \tilde{\mathbb{Q}} \cap K$$
equivalently, it is the set of all algebraic integers contained in K.

So we skecth the basic notation and definitions, the ring of integers of number field is special and important, since it is analogous to $\mathbb{Z}$ in $\mathbb{Q}$, and in particualr number theory is the study of "integers", although here the integraility is not clear, so we will focus on it.

Theorem 7.1. Let $K/\mathbb{Q}$ be a number field with $\mathcal{O}_K$ its ring of intgers, then

1. $(\mathcal{O}_K, +)$ is a free $\mathbb{Z}$-module of rank $[K : \mathbb{Q}]$, hence it is a noetherian ring.
2. $\mathcal{O}_K$ is integrally closed in K (following the definition).
3. Any non-zero prime ideal in $\mathcal{O}_K$ is maximal, i.e. the krull dimension of $\mathcal{O}_K$ is 1.

Proof. ☐

To study the arithmetic of "integers" we should understand the factorization of elements in $\mathcal{O}_K$, unfortunately, the ring of integers are not always UFD, typcial example is $\mathcal{O}_K = \mathbb{Z}[\sqrt{-5}]$ with $K = \mathbb{Q}(\sqrt{-5})$, in it we have
$$6 = 2 \times 3 = (1 + \sqrt{-5}) \times (1 - \sqrt{-5})$$
even more terrible, for the simplest example $K = \mathbb{Q}(\sqrt{d})$ (d is square-free), we have the following known results:

- When $d < 0$, $\mathcal{O}_K$ is a UFD if and only if d is one of the following integer:
$$-1, -2, -3, -7, -11, -19, -43, -67, -163$$
It is owe to the **Heegner-Baker-Stark theorem**, and the proof is non-trivial.

- When $d > 0$, it is conjectured by Gauss that there are infinitely many d such that $\mathcal{O}_K$ is a UFD, but it is still open (till 2026...).

Hence it is almost hopeless to consider the factorization of elements in $\mathcal{O}_K$. After dedekind and Kummer introduced the concept of ideal, a more general factorization is possible, in particular it holds for any number fields.

Theorem 7.2. For any number field K with ring of integers $\mathcal{O}_K$, any non-zero proper ideal $I \subsetneq \mathcal{O}_K$ can be factored into prime ideals, i.e. there exist prime ideals $\mathfrak{p}_1, \cdots, \mathfrak{p}_n$ such that

$$I = \mathfrak{p}_1 \cdots \mathfrak{p}_n$$

and the factorization is unique up to permutation of the prime ideals.

For example we consider $\mathcal{O}_K = \mathbb{Z}[\sqrt{-5}]$, we have the following factorization of ideals:

$$(6) = \mathfrak{p}_2^2 \mathfrak{p}_3 \mathfrak{p}_3'$$

with

$$\mathfrak{p}_2 = (2, 1 + \sqrt{-5}), \quad \mathfrak{p}_3 = (3, 1 + \sqrt{-5}), \quad \mathfrak{p}_3' = (3, 1 - \sqrt{-5})$$

then we can find that the factorization of 6 is unique in the sense of ideals:

$$(2) = \mathfrak{p}_2^2, \quad (3) = \mathfrak{p}_3 \mathfrak{p}_3'$$

while

$$(1 + \sqrt{-5}) = \mathfrak{p}_2 \mathfrak{p}_3, \quad (1 - \sqrt{-5}) = \mathfrak{p}_2 \mathfrak{p}_3'$$

To prove the theorem, some preparation should be done since the result is eseentailly not trival, firstly we should consider the division of ideals:

Definition 7.2. Let $\mathfrak{a}$ and $\mathfrak{b}$ be two non-zero ideals in $\mathcal{O}_K$, we define

$$\mathfrak{a} | \mathfrak{b} \iff \mathfrak{b} \subset \mathfrak{a}$$

This definition can be done for any commutative ring, but the properties of rings of integers ensures the following results, which is the key to the proof.

Lemma 7.6. $\mathfrak{p}$ is a prime ideal of $\mathcal{O}_K$ if and only if for any ideals $\mathfrak{a}$ and $\mathfrak{b}$ in $\mathcal{O}_K$, we have $\mathfrak{p} | \mathfrak{a}\mathfrak{b} \implies \mathfrak{p} | \mathfrak{a}$ or $\mathfrak{p} | \mathfrak{b}$.

Proof. $\square$

Lemma 7.7. For any non-zero ideals $\mathfrak{a} \in \mathcal{O}_K$, there exists non-zero prime ideals $\mathfrak{p}_1, \cdots, \mathfrak{p}_n$ such that $\mathfrak{p}_1 \cdots \mathfrak{p}_n \subset \mathfrak{a}$.

Proof. $\square$

Lemma 7.8. For any prime ideal $\mathfrak{p}$ in $\mathcal{O}_K$, we define its inverse ideal as

$$\mathfrak{p}^{-1} = \{x \in K : x\mathfrak{p} \subset \mathcal{O}_K\}$$

then for any non-zero ideal $\mathfrak{a}$ in $\mathcal{O}_K$, we have $\mathfrak{a} \subsetneq \mathfrak{a}\mathfrak{p}^{-1}$. In particular, $\mathfrak{p}\mathfrak{p}^{-1} = \mathcal{O}_K$.

Proof. $\square$

With the two lemma we can finish the proof:

Proof of theorem 7.2. For the existence, we set a set S of ideals which can not be factorized into prime ideals, if S is empty then we are done, **hence we assume S is non-empty**. We consider the partial ordered set $(S, \subset)$, then the noetherian condition ensures the existence of maximal element, denoted by $\mathfrak{a}$.

$\mathfrak{a}$ can not be prime (i.e. maximal), otherwise it is already factorized, hence there exists a maxiaml ideal (i.e prime) such that $\mathfrak{a} \subsetneq \mathfrak{p}$, which implies

$$\mathfrak{a} \subsetneq \mathfrak{a}\mathfrak{p}^{-1} \subsetneq \mathfrak{p}\mathfrak{p}^{-1} = \mathcal{O}_K$$

the chain holds trus by lemma 7.8, hence we construct an element $\mathfrak{a}\mathfrak{p}^{-1} \notin S$, which allows us to factorize

$$\mathfrak{a}\mathfrak{p}^{-1} = \mathfrak{p}_1 \cdots \mathfrak{p}_r$$

then $\mathfrak{a} = \mathfrak{a}\mathcal{O}_K = \mathfrak{a}\mathfrak{p}^{-1}\mathfrak{p}$ shows that the maximal element can be factorized, which is a contradiction.

For the uniqueness, we assume that there are two factorization for an ideal $\mathfrak{a}$

$$\mathfrak{a} = \mathfrak{p}_1 \cdots \mathfrak{p}_n = \mathfrak{q}_1 \cdots \mathfrak{q}_m$$

then $\mathfrak{p}_1 | \mathfrak{a}$ shows that there exists i such that $\mathfrak{p}_1 | \mathfrak{q}_i$, prime ideals are maximal so $\mathfrak{p}_1 = \mathfrak{q}_i$ exactly, and we can cancel then by multiplying $\mathfrak{p}^{-1}$, then we can repeat the process to conclude that $n = m$ and $\mathfrak{p}_i = \mathfrak{q}_{\sigma(i)}$ for any permutation $\sigma \in S_n$. $\square$

with unique factorization, we can extend the definition of norm from elements to ideals.

Definition 7.1. For any non-zero ideal $\mathfrak{a}$ in $\mathcal{O}_K$, we define its **(aboslute) norm** by

$$N(\mathfrak{a}) = |\mathcal{O}_K/\mathfrak{a}|$$

where $\mathcal{O}_K/\mathfrak{a}$ is called the **residue ring** of $\mathfrak{a}$.

Remark 7.3. The definition should be clarified, any non-zero ideal $\mathfrak{a}$ is exactly a $\mathbb{Z}$-submodule of $\mathcal{O}_K$, while $\mathcal{O}_K$ is free and $\mathbb{Z}$ is PID, hence $\mathfrak{a}$ is a free $\mathbb{Z}$-module with rank $\leq n = [K : \mathbb{Q}]$. We can take any non-zero element x of $\mathfrak{a}$ such that we have a chain

$$(x) \subset \mathfrak{a} \subset \mathcal{O}_K$$

principal ideal must be a free $\mathbb{Z}$-module of rank n by an $\mathbb{Z}$-module isomorphism with $\mathcal{O}_K$, hence the chain forces $\mathfrak{a}$ is also a free $\mathbb{Z}$-module of rank n, hence the residue ring is exactly finite.

Proposition 7.9. Let K be a number field with ring of integers $\mathcal{O}_K$

(a) For non-zero ideals $\mathfrak{a}$ with fractorization $\mathfrak{a} = \mathfrak{p}_1^{r_1} \cdots \mathfrak{p}_n^{r_n}$

$$N(\mathfrak{a}) = N(\mathfrak{p}_1)^{r_1} \cdots N(\mathfrak{p}_n)^{r_n}$$

(b) Norm is **multiplicative**, i.e. $N(\mathfrak{a}\mathfrak{b}) = N(\mathfrak{a})N(\mathfrak{b})$ for any two non-zero ideals.

(c) Let $\mathfrak{a}$ be a non-zero ideal with $\mathbb{Z}$-basis $\{a_1, \cdots, a_n\}$, then

$$N(\mathfrak{a}) = \sqrt{\frac{d_a}{d_K}}$$

(d) For principal ideal, $N((a)) = |N_{K/\mathbb{Q}}(a)|$ with a non-zero in $\mathcal{O}_K$.

Proof. If the factorization of a non-zero ideal $\mathfrak{a}$ is $\mathfrak{a} = \mathfrak{p}_1^{r_1} \cdots \mathfrak{p}_n^{r_n}$, then by Chinese remainder theorem, the residue ring can be decomposed as

$$\mathcal{O}_K/\mathfrak{a} \cong \mathcal{O}_K/\mathfrak{p_1}^{r_1} \times \cdots \times \mathcal{O}_K/\mathfrak{p_n}^{r_n}$$

hence it is enough to prove that for any prime ideal $\mathfrak{p}$ we have $N(\mathfrak{p}^k) = N(\mathfrak{p})^k$. For any $i \geq 1$, $\mathfrak{p}^i/\mathfrak{p}^{i+1}$ is a $\mathcal{O}_K/\mathfrak{p}$-vector space since $\mathfrak{p}$ is a $\mathcal{O}_K$-module,

$$\mathfrak{p}^i/\mathfrak{p}^{i+1} \cong \mathcal{O}_K/\mathfrak{p} \otimes_{\mathcal{O}_K} \mathfrak{p}^i$$

consider $k = \mathcal{O}_K/\mathfrak{p}$, then k is the residue field of local ring $\mathcal{O}_{K,\mathfrak{p}}$, hence

$$\begin{aligned}
\mathfrak{p}^i/\mathfrak{p}^{i+1} &\cong (k \otimes_{\mathcal{O}_K} \mathcal{O}_{K,\mathfrak{p}}) \otimes_{\mathcal{O}_K} \mathfrak{p}^i \\
&\cong k \otimes_{\mathcal{O}_{K,\mathfrak{p}}} \mathfrak{p}^i\mathcal{O}_{K,\mathfrak{p}} \\
&\cong k
\end{aligned}$$

which shows that $\mathfrak{p}^i/\mathfrak{p}^{i+1}$ is one-diemnsional over k **(an alternative proof can be found in [Tall] Lemma 6.20 , with direct discussion isomorphism)**, hence we can conclude that

$$N(\mathfrak{p}^k) = |\mathcal{O}_K/\mathfrak{p}| \cdot |\mathfrak{p}/\mathfrak{p}^2| \cdots |\mathfrak{p}^{k-1}/\mathfrak{p}^k| = N(\mathfrak{p})^k$$

which finishes the proof of (a), and (b) is a direct consequence of (a).

For (c), □

Similarly with Gauss integer, the norm of an ideal can be used to determine whether it is prime, and it will open a complicated story about the arithemtic of number fields.

Lemma 7.10. Let $\mathfrak{p}$ be a non-zero prime ideal of $\mathcal{O}_K$, then

(a) there exists a unique prime number p such that $p \in \mathfrak{p}$.

(b) $N(\mathfrak{p}) = p^e$ for some integer $1 \leq e \leq n$.

(c) $\mathfrak{p} \cap \mathbb{Z} = (p)$.

Proof. Notice $\mathcal{O}_k/\mathfrak{p}$ is a finite field if $\mathfrak{p}$ is prime (in $\mathcal{O}_K$ it is maxiaml), which proves (b). For (c), notice that $\mathfrak{p} \cap \mathbb{Z}$ is pre-image of $\mathfrak{p}$ under the inclusion $\mathbb{Z} \hookrightarrow \mathcal{O}_K$, and the contraction of a prime ideal is either zero or a prime ideal. $\mathfrak{p} \cap \mathbb{Z}$ can't be zero, in fact we take any non-zero $x \in \mathfrak{p} \subset \mathcal{O}_K$, then using the minimal polynomial $\Pi_x \in \mathbb{Z}[X]$, we can deduce that $f = \Pi_x - \Pi_x(0) \in \mathbb{Z}[X]$ is a polynomial such that $f(x) = -\Pi_x(0) \in \mathbb{Z}$, which shows that$f(x) \in \mathfrak{p} \cap \mathbb{Z}$, and it proves (b). For (a), (c) shows the existsence of prime number p exactly, if there are another prime number $q \in \mathfrak{p}$ then

$$N(\mathfrak{p}) \text{ divise } N(p) = p^n, \quad N(\mathfrak{p}) \text{ divise } N(q) = q^n$$

it show taht $N(\mathfrak{p}) = 1$, it is impossible. $\square$

The properties of prime ideals invite some vocabulary, for any prime number p, there exists a factorization in sense of principla ideal

$$(p) = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_k^{e_k}$$

a prime ideal $\mathfrak{p}$ is called **lying over** p if $\mathfrak{p}|(p)$, and the **ramification index** of $\mathfrak{p}$ is defined as

$$e = \max\{l \in \mathbb{N} : \mathfrak{p}^l|(p)\}$$

and the **residue index** of $\mathfrak{p}$ is defined as the extension degree

$$f = [\mathcal{O}_K/\mathfrak{p} : \mathbb{F}_p]$$

exactly such $\mathfrak{p}$ must be one of $\mathfrak{p}_i$ with e_i as its ramification index, and residue index is just the expoent f_i such that $N(\mathfrak{p}_i) = p^{f_i}$, hence we immediately we have the following relation

Corollary 7.11.

$$[K : \mathbb{Q}] = \sum_{\mathfrak{p}_i} e_i f_i$$

where $\mathfrak{p}_i$ runs over all prime ideals lying over p, with e_i and f_i as its ramification index and residue index respectively.

We can talk more about the factorization of prime numbers, which is the basis of the study of arithemtic of number fields.

1. For any number field K, as a separable extension there eixsts $a \in K$ such that $K = \mathbb{Q}(a)$ and $\mathcal{O}_K = \mathbb{Z}[a] \cong \mathbb{Z}[X]/(\Pi_a(X))$. We want to study $(p) = p\mathcal{O}_K$ for some **integral prime** p, so it is natural to consider the quotient $\mathcal{O}_K/p\mathcal{O}_K$: if the quoitent is a field or just integral domain, we can determine that (p) is prime.

2. Otherwise, we consider $\mathcal{O}_K$ as a $\mathbb{Z}$-module

$$\mathcal{O}_K/p\mathcal{O}_K \cong (\mathbb{Z}/(p)) \otimes_{\mathbb{Z}} \mathbb{Z}[X]/(\Pi_a) \cong \mathbb{F}_p[X]/(\overline{\Pi}_a)$$

we consider the factorization of $\bar{\Pi}_a$ in $\mathbb{F}_p[X]$, it will be unique since $\mathbb{F}_p[X]$ is a PID, so by chinese remainder theorem we can decompose the quotient as a product

$$\mathcal{O}_K/p\mathcal{O}_K \cong \prod_i \mathbb{F}_p[X]/(\bar{Q}_i^{e_i})$$

notice that the structure is analogoue to the factorization of ideals $(p) = \prod_i \mathfrak{p}^{e_i}$, so it is natural to find $\mathfrak{p}_i$ by consider the lifting of each irreducible factor $\bar{Q}_i$, so it motivates us to define

$$I_p(Q) = p\mathcal{O}_K + Q(a)\mathcal{O}_K$$

where $Q \in \mathbb{Z}[X]$ is a monic polynomial such that $\bar{Q} \in \mathbb{F}_p[X]$ is a factor of $\bar{\Pi}_a$.

3. Notice that $\bar{a} \mapsto \bar{X}$ gives the first isomorphism in 2, so if $\bar{Q}$ is a irreducible factor of $\bar{\Pi}_a$, then we can conclude

$$I_p(Q)/p\mathcal{O}_K \cong (Q)/(Q^e)$$

this is the maximal ideal in local ring $\mathbb{F}_p[X]/(Q^e)$, so by this isomorphism we can conclude that $I_p(Q)$ **is the maximal (so prime) ideal.**

4. The lifting Q is not unique, but $I_p(Q)$ **does not depend on the choice of** Q: In fact if we set $Q' \in \mathbb{Z}[X]$ as another lifting of $\bar{Q}$, then $Q(a) - Q'(a) = 0 \mod p$, which shows that $Q(a) \in I_p(Q')$ and $Q' \in I_p(Q)$ respectively, hence $I_p(Q) = I_p(Q')$.

5. In fact as we have known, the factorization of (p) is unique, so in priori we can write the isomorphism $\mathcal{O}_K/p\mathcal{O}_K \cong \prod_i \mathcal{O}_K/\mathfrak{p}_i^{e_i}$, and $I_p(Q)$ as we have shown is a prime ideal lying over p, so we can conclude directly the following result (It is another view to prove the uniqueness of factorization of ideals)

Theorem 7.3 (Dedkind-Kummer)**.**
Let K be a number field with primitive element a, and $p \in \mathbb{Z}$ a prime, then

(a) The bijection $Q \mapsto I_p(Q)$ gives a correspondence

$$\{\text{monic factors of } \overline{\Pi}_a \in \mathbb{F}_p[X]\} \leftrightarrow \{\text{ideal dividing } (p)\}$$

$$\{\text{ irreducible factors}\} \leftrightarrow \{\text{prime ideals}\}$$

(b) The norm of $I_p(Q)$ is $p^{\deg(Q)}$. When $\bar{Q}$ is a irreducible factor, $\deg(Q)$ is exactly the residue index of $I_p(Q)$.

(c) there exists integer $e_i \in \mathbb{N}$ such that

$$(p) = \prod_i I_p(Q_i)^{e_i}$$

where $\bar{Q}_i$ runs over all distinct irreducible factors of $\bar{\Pi}_a$ in $\mathbb{F}_p[X]$, and e_i is the multiplicity of $\bar{Q}_i$ in $\bar{\Pi}_a$, it is exactly the ramification index of $I_p(Q_i)$.

7.3 Minkowski's space

This theory corresponds to the geometry of numbers, the idea is to consider embedding of number fields into $\mathbb{R}$ or $\mathbb{C}$, we now consider a number field K of degree n, and review some basic notations:

1. an embedding $\sigma : K \to \mathbb{C}$ is called a **real embedding** if $\sigma(K) \subset \mathbb{R}$, otherwise it is called a **complex embedding**. Since $K/\mathbb{Q}$ is separable, hence the number of embeddings is exactly n.

2. It is easy to verify that if σ is an embedding, then $\bar{\sigma}$ is also an embedding, we say that two complex embeddings σ and σ' are conjugate if $\sigma' = \bar{\sigma}$. Hence the set of embeddings can be divided as following:

$$\Sigma = \Sigma_r \sqcup \Sigma_c \sqcup \bar{\Sigma}_c$$

where Σ_r is the set of real embeddings, and Σ_c is the set of distinct complex embeddings up to conjugation. In particular, we set

$$\Sigma_0 = \Sigma_r \sqcup \Sigma_c \cong \Sigma / \sim$$

to denote the set of distinct embeddings up to conjugation, and usually we write $n = r + 2s$ with

$$r = |\Sigma_r|, \quad s = |\Sigma_c|$$

3. For any $\mathbb{R}$-vector space V of dimension n with an inner product $\langle -, - \rangle$, an additive subgroup $L \subset V$ is **a complete lattice** if V/L is compact (or equivalently, L is discrete of rank n), and we can define the volume and covlume

$$\mathrm{covol}(L) = \mathrm{Vol}(V/L) = \mu(F)$$

where μ is the induced measure, and F is the fundamental domain. If $L = \oplus_{i=1}^n \mathbb{Z} v_i$ for some basis $\{v_1, \dots, v_n\}$, then we have

$$\mathrm{covol}(L) = \mathrm{Vol}(V/L) = \sqrt{\det G}$$

where G is the Gram matrix with entries $G_{ij} = \langle v_i, v_j \rangle$.

Definition 7.3. For any number field K of degree $n = r+2s$, we define the **Minkowski's space** of K as a $\mathbb{R}$-vector space $K_\infty = \mathbb{R}^n$ together with an inner product defined by

$$\langle x, y \rangle_m = \sum_{i=1}^{n} a_i \cdot x_i y_i$$

with

$$a_i = \begin{cases} 1 & \text{if } i \leq r \\ 2 & \text{if } i > r \end{cases}$$

Remark 7.4. The Minkowski's space always appears together with the embedding of K by

$$\iota : K \hookrightarrow \mathbb{R}^r \times \mathbb{C}^s \cong K_\infty, \quad x \mapsto (x_\sigma)_{\sigma \in \Sigma_0}$$

where $x_\sigma = \sigma(x)$, the inner product can be pulled back to K by

$$\langle x, y \rangle_M = \langle \iota(x), \iota(y) \rangle_m = \sum_{\sigma \in \Sigma_0} a_\sigma \cdot \mathrm{Re}(x_\sigma \overline{y_\sigma})$$

with

$$a_\sigma = \begin{cases} 1 & \text{if } \sigma \text{ real} \\ 2 & \text{if } \sigma \text{ complex} \end{cases}$$

It is not hard to verify that the inner product concides with the trace pairing if K is totally real ($s = 0$).

Remark 7.5. We can view that this inner product is the restriction of the Hermitian inner product on $\mathbb{C}^n$, i.e. it is same with the pulled back from the embedding

$$K \to \mathbb{C}^n, \quad x \mapsto (x_\sigma)_{\sigma\in\Sigma}$$

the reason to consider the restriction is that $\mathbb{R}$-vector space is exactly the place where the convex geometry works, and we always view ideal as a lattice in the Minkowski's space, such that we can talk about the volume and the density.

Theorem 7.4 (The structure of Minkowski's space)**.**
Along the notation above

(a) $K_\infty \cong K \otimes_{\mathbb{Q}} \mathbb{R}$.

(b) A subset of K_∞ is measurable if and only if it is lebesgue measuable, in particular we they are distict up to a constant

$$\mathrm{Vol}(S) = 2^s \cdot \mathrm{Vol}_{Lebesgue}(S)$$

(c) If $\mathfrak{a}$ is a non-zero ideal in $\mathcal{O}_K$, then $\iota(\mathfrak{a})$ is a lattice in K_∞ with

$$\mathrm{covol}(\iota(\mathfrak{a})) = \sqrt{|d_{\mathfrak{a}}|}$$

Moreover, the aboslute norm of $\mathfrak{a}$ can be defined **equivalently** by

$$N(\mathfrak{a}) = \frac{\mathrm{covol}(\iota(\mathfrak{a}))}{\mathrm{covol}(\iota(\mathcal{O}_K))}$$

Proof. (a)

(b) The inner product is the euclidean inner product with weight, hence the norm are equivalent to the standard norm. Notice that the lebesgue measure and induced measure on Minkowski's space are both Haar measure, so they are distinct up to a constant, so it is enough to compute the hypercube $S = [0,1]^n$ in both measures, and we have

$$\mathrm{Vol}(S) = \sqrt{\det(\langle e_i, e_j\rangle_m)_{i,j}} = 2^s$$

(c) Review that for square matrix A we have

$$(A^T\bar{A})_{i,j} = \sum_k A_{k,i}\overline{A_{k,j}}$$

by definition of the discriminant, if $a_1, \ldots, a_n$ is a basis of $\mathfrak{a}$, then we have

$$d_{\mathfrak{a}} = \det[\sigma_i(a_j)]^2$$

hence we set here A is the matrix with entries $A_{i,j} = \sigma_i(a_j)$, then we have

$$(\bar{A}^T A)_{i,j} = \langle a_i, a_j\rangle_M$$

hence $\text{covol}(\iota(\mathfrak{a})) = \sqrt{\det(\bar{A}^T A)} = |\det A| = |d_{\mathfrak{a}}|$, notice here $A \in M(n, \mathbb{C})$ exactly with $\det A \in \mathbb{C}$.

□

Remark 7.6. In sense of Minkowski's space, the discriminant of a number field K can be described as a geometric invariant, it is eaxctly the square of the volume of the fundamental domain with respect to the translation action $\mathcal{O}_K$ on K_∞.

Since ι is an inclusion, so we it has no problem to identify the subset of K (numbers) as the subset of K_∞ (vectors), from now on we will not distinguish them with a littble abusion of notation, i.e.

$$\text{Vol}(S) = \text{Vol}(\iota(S))$$

for any subset $S \subset K$ (or K_∞).

Theorem 7.12 (Minkowski's convex theorem)**.**
Let K be a number field of degree n with ring of integers $\mathcal{O}_K$, and let $\iota(S) \subset K_\infty$ be a convex symmetric measurable subset with

$$\mu(\iota(S)) > 2^n \cdot \mu(\iota(\mathcal{O}_K))$$

for any Haar measure on K_∞, then there exists a non-zero element $x \in \mathcal{O}_K$ such that $\iota(x) \in S$.

The proof is owe to the theory of lattice, should pay attention to the habite that we usually consider the lebesgue measure λ, and choose a proper convex body we can get some interesting results about number fields.

> **Theorem 7.5.** For any non-zero ideal $\mathfrak{a}$ in $\mathcal{O}_K$, there exists a non-zero element $a \in \mathfrak{a}$ such that
>
> $$|N_{K/\mathbb{Q}}(a)| \leq M_K N(\mathfrak{a})$$
>
> where $M_k = \frac{n!}{n^n}(\frac{4}{\pi})^s\sqrt{|d_K|}$ is defined as the **Minkowski's bound** of K.

Proof. We define a L^1 norm on K_∞ by

$$\|(x_1, \ldots, x_r, z_1, \ldots, z_s)\|_1 = \sum_{i=1}^{r} |x_i| + 2\sum_{i=1}^{s} |z_i|$$

and then under this norm we define a convex symmetric body by

$$C_R = \{x \in K_\infty : \|x\|_1 \leq R\}, \quad R > 0$$

then we can calculate the volume $\lambda(C_R) = 2^r(\frac{\pi}{2})^s\frac{R^n}{n!}$, and then we apply Minkowski's theorem here to conclude the lower bound of R such that $\mathfrak{a} \cap \iota^{-1}(C_R) \neq \emptyset$ will be

$$R^n \geq n! \cdot \frac{2^{n-r}}{\pi^s} \cdot N(\mathfrak{a})\sqrt{|d_K|}$$

under this condition, we set $a \in \mathfrak{a}$ to be a such element, to estimate the norm of a, we apply **the inequality of arithmetic and geometric means** to get

$$\begin{aligned}|N_{K/\mathbb{Q}}(a)| &= \prod_{\sigma\in\Sigma} |\sigma(a)| \\ &\leq \frac{1}{n^n}\|\iota(a)\|_1^n \\ &\leq \frac{R^n}{n^n}\end{aligned}$$

so it is enough to take the lower bound of R to finishe the proof. □

7.4 Logarithmic space and Unit theorem

This section is the complement of the the last section, and the notation will be the same as before for the sake of consistency.

Definition 7.4. The **logarithmic space** of K is defined as the $\mathbb{R}$-vector space $\mathbb{L} \cong \mathbb{R}^{r+s}$ together with the map

$$l : K^\times \to \mathbb{L}, \quad x \mapsto (a_\sigma \cdot \log|x_\sigma|)_{\sigma\in\Sigma_0}$$

the map is **a homomorphism onto a additive group**, i.e.

$$l(xy) = l(x) + l(y)$$

For any $x, y \in K^\times$.

Remark 7.7. The alternative definition is via the minkowski's space

$$K^\times \xrightarrow{\iota} K_\infty^\times \xrightarrow{\text{Log}} \mathbb{L}$$

with the logarithm map naturally defined by

$$\log : K_\infty^\times \to \mathbb{L}, \quad (x_1, \ldots, x_r, z_1, \ldots, z_s) \mapsto (\log|x_1|, \ldots, \log|x_r|, 2\log|z_1|, \ldots, 2\log|z_s|)$$

hence exactly here $l = \text{Log}\circ\iota$, based on the definition we can quickly investigate some properties:

- The kernel of l is the group of roots of unity $\mu(K)$, hence it motivates us to restrict the map l on $\mathcal{O}_K^\times$ for the structure.
- Define the norm map

  $$N : K_\infty \to \mathbb{R}, \quad x \mapsto \prod_{x \text{ real}} x \prod_{z \text{ complex}} |z|^2$$

  and trace map

  $$T : \mathbb{L} \to \mathbb{R}, \quad x \mapsto \sum_{i=1}^{r+s} x_i$$

such that the following diagram commutes

$$\begin{tikzcd}
K^\times \arrow[r, hook, "\iota"] \arrow[d, "\mathrm{N}_{K/\mathbb{Q}}"'] & K_\infty^\times \arrow[r, "\mathrm{Log}"] \arrow[d, "N"'] & \mathbb{L} \arrow[d, "T"'] \\
\mathbb{Q}^\times \arrow[r, hook] & \mathbb{R}^\times \arrow[r, "\log|-|"'] & \mathbb{R}
\end{tikzcd}$$

- To restrict the map, notice the following equivalent conditions

$$x \in \mathcal{O}_K^\times \iff \mathrm{N}_{K/\mathbb{Q}}(x) = \pm 1 \iff N(\iota(x)) = \pm 1 \iff T(l(x)) = 0$$

Hence it motivates us to define the **hyperplane**

$$H := \{x \in \mathbb{L} \mid x_1 + \cdots + x_{r+s} = 0\}$$

as the $r + s - 1$ dimensional subspace of $\mathbb{L}$

7.5 Class group and Class number

It is natural to consider the group structre of the ideal under multplication since the factorization of prime ideals is unqiue, similarly with $\mathbb{Z}$, it can not exactly forms a group under mutiplication, and it needs a "localization" to complete the elements, it is back to the dedekind's original definition of ideal.

Definition 7.5. Let K be a number field with ring of integers $\mathcal{O}_K$, a fractional ideal of K is a **non-zero** subset $\mathfrak{a} \subset K$ such that
(1) $\mathfrak{a}$ is an $\mathcal{O}_K$-submodule of K.
(2) There exits a non-zero element $x \in O_k$ such that $x\mathfrak{a} \subset \mathcal{O}_K$. The set of all fractional ideals of K is denoted by J_K.

Remark 7.8. The definition is a little axiomatic, here is some descriptions:

- condition (2) is equivalent to $\mathfrak{a}$ is finitely generated with rank equal to $[K : \mathbb{Q}]$, and in particular it is also a free $\mathbb{Z}$-module of rank $[K : \mathbb{Q}]$.
- The non-zero ideal $\mathfrak{a}$ is exactly a $\mathcal{O}_K$-submodule of $\mathcal{O}_K$ itself, hence naturally it is a fractional ideal, and we call it **integral ideal** to distinguish with the fractional ideal.
- Notice that $x\mathfrak{a} \subset \mathcal{O}_K$ is equivalent to $x\mathfrak{a}$ is an integral ideal, hence the definition can be expressed precisely by saying that a fractional ideal is a set of the form

$$xI := \{xa \mid a \in I\} \subset K$$

  where $x \in K - \{0\}$ and I is an intergal ideal, since as a fractional field $K = \mathrm{Frac}(\mathcal{O}_K)$, we can always write $x = a/b$ with $a, b \in \mathcal{O}_K$.

Lemma 7.13. For any fractional ideal $\mathfrak{a}$, there exists a unique factorization up to order

$$\mathfrak{a} = \mathfrak{p}_1^{r_1} \cdots \mathfrak{p}_n^{r_n}$$

where $\mathfrak{p}_\mathfrak{i}$ are prime ideals and r_i are integers.

Proof. By the definition of fractional ideal, there exists $x \in \mathcal{O}_K - \{0\}$ such that $x\mathfrak{a} \subset \mathcal{O}_K$ is an integral ideal, hence there exists a unique fractorziation of $x\mathfrak{a}$ up to order

$$x\mathfrak{a} = \mathfrak{p}_k^{r_1} \cdots \mathfrak{p}_k^{r}$$

notice $x\mathfrak{a} = (x)\mathfrak{a}$, hence there exists a unique fractorization for the principal ideal

$$(x) = \mathfrak{q}_1^{s_1} \cdots \mathfrak{q}_m^{s_m}$$

then we take the inverse of the principal ideal to rewrite $\mathfrak{a} = (x)^{-1}x\mathfrak{a}$, then we can conclude the result. □

Remark 7.9. There are serval imediate consequences of the properties:

1. J_K forms a abelian group under multiplication, it is clear that $\mathcal{O}_K = (1)$ is the identity, so for any fractional ideal $\mathfrak{a} = \mathfrak{p}_1^{r_n} \cdots \mathfrak{p}_n^{r_n}$, we can exactly find an inverse by $\mathfrak{a}^{-1} = \mathfrak{p}_1^{-r_1} \cdots \mathfrak{p}_n^{-r_n}$ since lemma 7.8 shows that $\mathfrak{a}\mathfrak{a}^{-1} = \mathcal{O}_K$. In the sense of monoid, the existence of inverse ensures uniqueness of inverse, hence J_K is exactly a group.
2. Indeed J_K is a free abelian group generated by the prime ideals, it is immediate by a natural bijection
$$v : J_K \to \mathbb{Z}^{(\mathcal{P})}, \quad \prod_{\mathfrak{p}\in\mathcal{P}} \mathfrak{p}^{a_\mathfrak{p}} \mapsto (a_\mathfrak{p})_{\mathfrak{p}\in\mathcal{P}}$$
where $\mathcal{P}$ is the set of all prime ideals of $\mathcal{O}_K$.
3. Hence we can extend the definition of **absolute norm** of ideals to J_K by
$$N : J_K \to \mathbb{R}_{>0}, \quad \mathfrak{a} \mapsto \prod_{\mathfrak{p}\in\mathcal{P}} N(\mathfrak{p})^{v_\mathfrak{p}(\mathfrak{a})}$$
where $v_\mathfrak{p} = \pi_\mathfrak{p} \circ v$ is exactly the projection of v at $\mathfrak{p}$-compoent, here naturally we extend the absolute norm as a group homomorphism, and invite the valuation of ideals.

To consider the ideal as a number, we define **the principal fractional ideal** are the fractional ideal of the form $(x) = x\mathcal{O}_K$ for some $x \in K - \{0\}$, and P_K the set of all principal fractional ideals, then we define a relation on J_K by
$$\mathfrak{a} \sim \mathfrak{b} \iff \exists x \in K - \{0\}, \mathfrak{a} = x\mathfrak{b} = (x)\mathfrak{b}$$
then the relation is exactly equivalent, and we can get the classes
$$[\mathfrak{a}] = \{x\mathfrak{a} \mid (x) \in P_K\}$$
and it gives a partition of J_K

Lemma 7.14. In each class of, there exists an integral ideal I such that
$$N(I) \leq M_K$$
where M_k is the Minkowski's bound of K, thus J_K/P_K is exactly a finite abelian group, and we call it the **class group** of K, and its order is called the **class number** of K.

Proof. J_K is an abelian group with P_K as its subgroup, and the relation defined above is exactly the relation defined by cosets of P_K, so it is enough to show the first statement, since there exists only finitely many ideal of $\mathcal{O}_K$ with norm less than a fixed number, and each class can be represented by an integral ideal.

For any class $[\mathfrak{a}]$, let $y \in \mathcal{O}_K$ such that $y\mathfrak{a}^{-1} \subset \mathcal{O}_K$ be a non-zero ideal of $\mathcal{O}_K$, the there exists a non-zero element $x \in y\mathfrak{a}^{-1}$ such that
$$|N_{K/\mathbb{Q}}(x)| \leq M_K N(y\mathfrak{a}^{-1})$$
we set $xy^{-1}\mathfrak{a} = \mathfrak{b}$, then $\mathfrak{b}$ is an integral ideal since $xy^{-1} \in \mathfrak{a}^{-1}$, hence we multiply the inequality both sides by $N(y^{-1}\mathfrak{a})$ to conclude that $\mathfrak{b}$ is exactly the ideal we hope for the class. $\square$

8 Analytic approach to number theory

8.1 Arithmetic function

8.1.1 Dirichlet convolution and multiplicative functions

An arithemetic function is a complex-valued function defined on the positive intgers $\mathbb{N}^*$, a remarkable operation on the set is defined as following:

Definition 8.1. Let f, g be two arithmetic functions, we define the **Dirichlet convolution** of f and g by

$$(f * g)(n) = \sum_{d|n} f(d)g(n/d)$$

such a operation is commutative and associative, and the identity element is the dirac function

$$\delta : \mathbb{N}^* \to \mathbb{C}, \quad n \mapsto \begin{cases} 1 & \text{if } n = 1 \\ 0 & \text{otherwise} \end{cases} = [\frac{1}{n}]$$

that shows the set of arithemetic functions forms a monoid under the convolution, furthermore some restrictions can make it a group.

Proposition 8.1. If f is an arithmetic function with $f(1) \neq 0$, then there exists a unique arithmetical function g such that

$$f * g = g * f = \delta$$

and g is given by the recursion formulas

$$g(1) = f(1)^{-1}, \quad g(n) = -g(1) \sum_{d|n, d<n} f(n/d)g(d)$$

Proof. □

Remark 8.1. We shall denote the set of all arithmetical functions by $\mathcal{A}$, and the set of all arithmetical functions with $f(1) \neq 0$ by $\mathcal{A}^*$, then $(\mathcal{A}, *)$ is a monoid, and $(\mathcal{A}^*, *)$ is a group. Notice that such a construction holds generally for any function $f : \mathbb{N}^* \to R$ with R a commutative ring, in this case $f(1) \in R^\times$ is the condition for the existence of inverse.

Example 8.1. 1-function is defined by $\mathbf{1}(n) = 1$ for any $n \in \mathbb{N}^*$, it is a function with $\mathbf{1}(1) = 1$, and we can calculate that

$$(\mathbf{1} * \mathbf{1})(n) = \sum_{d|n} \mathbf{1}(d)\mathbf{1}(n/d) = \sum_{d|n} 1 =: \tau(n)$$

τ is defines as the **divisor function** which counts the number of divisors. More generally, use some combinatorial trick we can prove that

$$\underbrace{\mathbf{1} * \cdots * \mathbf{1}}_{k \text{ times}}(n) = \tau_k(n) = \#\{(a_1, \ldots, a_k) \in \mathbb{N}^k \mid a_1 \cdots a_k = n\}$$

where τ_k is the **k-th power divisor function**, with a convention that $\tau_2 = \tau$ and $\tau_1 = \mathbf{1}$.

Example 8.1. Möbius function μ is defined as following:

$$\mu(n) = \begin{cases} 1 & n = 1 \\ (-1)^k & n = p_1 \cdots p_k \\ 0 & \text{otherwise} \end{cases}$$

a remarkable equality is that

$$\sum_{d|n} \mu(d) = \delta(n)$$

that shows $\mu \in \mathcal{A}^*$ with inverse $\mathbf{1}$, so $\tau_k^{-1} = \mu^{*k}$.

Example 8.2. Identity map is denoted by I with $I(n) = n$ for any $n \in \mathbb{N}^*$, it is a function with $I(1) = 1$, and we can calculate that

$$(I * I)(n) = \sum_{d|n} d \cdot \frac{n}{d} = n\tau(n)$$

which shows that $I * I = I\tau$, and more generally we have

$$\underbrace{I * \cdots * I}_{k \text{ times}}(n) = n\tau_k(n)$$

and its inverse is $\mu \cdot I$, in fact

$$(I * \mu I)(n) = \sum_{d|n} n\mu(n/d) = n\delta(n) = \delta(n)$$

Example 8.2. Euler totient function φ is defined by

$$\varphi(n) = \sum_{\substack{1 \leq k \leq n \\ (k,n)=1}} 1$$

and a famous formula is that

$$\sum_{d|n} \varphi(d) = n$$

that shows $\varphi * \mathbf{1} = I$, with I the identity function mapping n to n, so

$$\varphi(n) = (\varphi * \mathbf{1} * \mu)(n) = (I * \mu)(n) = n \sum_{d|n} \frac{\mu(d)}{d}$$

and the inverse of φ is $\mu I * \mathbf{1}$.

The example of Euler totient function can be seen as a motivation to define the convlution, since it plays an important role in number theory, and another property related to the Euler function can be concluded as following:

Definition 8.2. A arithmetic function f is called **multiplicative** if

$$f(mn) = f(m)f(n), \quad (m,n) = 1$$

and f is called **completely multiplicative** if

$$f(mn) = f(m)f(n), \quad \forall m,n \in \mathbb{N}^*$$

Remark 8.2. The multiplicative function f must be in $\mathcal{A}^*$, the reason is that $f(n) = f(1)f(n)$ for any $n \in \mathbb{N}^*$.

Example 8.3. The Euler totient function is multiplicative, which allows to deduce the formula for $n = p_1^{a_1} \cdots p_k^{a_k}$,

$$\varphi(n) = \prod_{i=1}^{k} (p_i - 1) p_i^{a_i - 1}$$

while φ is not completely multiplicative, since $\varphi(4) = 2 \neq \varphi(2)^2 = 1$.

Example 8.4. The Möbius function is multiplicative, but not completely multiplicative. Hence in example 8.2 we notice that $d \mapsto \mu(d)/d$ is multiplicative, which allows us to deduce the another formula for Euler totient function

$$\varphi(n) = n \prod_{p|n} (1 - p^{-1})$$

Theorem 8.1. The set of multiplicative functions, denote by $\mathcal{M}$, it forms a subgroup of $\mathcal{A}^*$ under the convolution.

Proof. It □

Remark 8.3. The completely multiplicative functions is not stable, for example 1-function $\mathbf{1}$ is completely multiplicative, while divisor function $\tau = \mathbf{1} * \mathbf{1}$ is mutiplicative but not completely $(\tau(p^2) = 3 \neq 4 = \tau(p)\tau(p))$.

Proposition 8.2. Let f be a multiplicative function, then

(a) f is completely multiplicative if and only if $f^{-1} = \mu \cdot f$.

(b) we have

$$\sum_{d|n} \mu(d) f(d) = \prod_{p|n} (1 - f(p))$$

and consequently $\varphi^{-1}(n) = \sum_{p|n}(1-p)$.

Proof. □

Proposition 8.3 (Euler's product)**.** Supppose f is an arithmetic function such that the series $\sum_{n\geq 1} f(n)$ converges absolutely, then $\prod_p \sum_{k\geq 0} f(p^k)$ converges abolutely, and

-If f is multpilicative, $\sum_{n\geq 1} f(n) = \prod_p \sum_{k\geq 0} f(p^k)$.

-If f is completely multiplicative, then $|f(p)| < 1$ for any prime p and

$$\sum_{n\geq 1} f(n) = \prod_p \frac{1}{1 - f(p)}$$

Proof. □

1

8.1.2 sommation formula and average

Let f be an arithmetic function, its average in many cases means a limit

$$\lim_{N\to\infty} \frac{\sum_{n\leq N} f(n)}{N} = \lim_{x\to\infty} \frac{\sum_{n\leq x} f(n)}{x}$$

and we are interested in the existence of such a limit, or the asymptotic behavior at infty.

Definition 8.3. Let f be an arithmetic function and g be a smooth function, we say that **the averge of** f **is of order** g if

$$\frac{1}{x} \sum_{n\leq x} f(n) \sim g(x) \qquad \text{as } x \to \infty$$

the order of average measures the growth of the average of arithmetic function, or behavior of the average at infty, by a comparison with a well-kown function.

Example 8.5 (distribution of prime numbers)**.** Let $\pi(n) = \sum_{p\leq n} 1$ be the prime counting function, a remarkable result is that

$$\lim_{x\to\infty} \frac{\pi(x)\log x}{x} = 1$$

in the language of average, we define the prime characteristic function by

$$\mathbf{1}_{\mathbb{P}}(n) = \begin{cases} 1 & \text{if } n \text{ is a prime} \\ 0 & \text{otherwise} \end{cases}$$

it means that the average of $\mathbf{1}_{\mathbb{P}}$ is of order $1/\log n$, which shows that the density of prime numbers is zero.

Theorem 8.4 (Euler-Maclaurin)**.** If f est class C^1 on $[a,b]$ with $a,b \in \mathbb{Z}$. then

$$\sum_{a\le n\le b} f(n) = \int_a^b f(t) + \psi(t)f'(t)\,dt + \frac{f(a)+f(b)}{2}$$

where $\psi(t) = t - [t] - 1/2$.

Example 8.3. If $x \geq 1$, then we have asymptotic equality:

(1) $\sum_{n\le x} \frac{1}{n} = \log x + C + O(1/x)$ with C the euler constant.

(2) $\sum_{n\le x} \frac{1}{n^s} = \zeta(s) + \frac{x^{1-s}}{1-s} + O(x^{-s})$ if $s > 1$ and $s \neq 0$.

Proof. In (1), we take $a = 1$ and $b = [x]$, apply the formula to $f(n) = 1/n$ we have

$$\begin{aligned}\sum_{n\le x} \frac{1}{n} &= \int_1^b \frac{dt}{t} + \int_1^b \psi(t)\frac{-1}{t^2}\,dt + \frac{1}{2} + \frac{1}{2b} \\ &\leq \log b + \frac{1}{2} - \underbrace{\int_1^b \frac{1}{2t^2}\,dt}_{\text{converges if } b\to\infty} + O(1/b)\end{aligned}$$

and notice that $b = x + O(1)$, so there exists a constant C such that (a) holds, and take the limit $x \to \infty$ we can conclude the value of C as the euler constant. □

Theorem 8.5 (Abel)**.** Let $a(n)$ be an arithmetical function and f be a function of class C^1 on $[y,x]$, then

$$\sum_{y<n\le x} a(n)f(n) = A(x)f(x) - A(y)f(y) - \int_y^x A(t)f'(t)\,dt$$

where $A(x) = \sum_{n\le x} a(n)$ with convention $A(x) = 0$ if $x < 1$.

Proof. One proof is based on Abel's sum, such a proof can be found in theorem 4.2 [**Apostol**]. Another proof is more direct, we notice that $x \mapsto A(x)$ is a step function and increasing, so the sum can be expressed as a Riemann-Stieltjes integral

$$\sum_{y<n\le x} a(n)f(n) = \int_y^x f(t)dA(t)$$

then we can apply the integration by parts to get the result. □

Remark 8.4. An other version of Euler-Maclaurin formula can be derived here by setting $a(n) = 1$, then we can get

$$\sum_{y<n\le x} f(n) = \int_y^x f(t) + (t-[t])f'(t)\,dt + f(y)(y-[y]) - f(x)(x-[x])$$

a comparison with theorem 8.4 shows that the two versions are equivalent.

Another method of sumnation is **Hyperbola method**, loosely speaking, we can express the sum as following

$$\sum_{n\le x}\sum_{d|n} f(d,n) = \sum_{qd\le x} f(d,qd)$$

the reason can be expressed geometrically

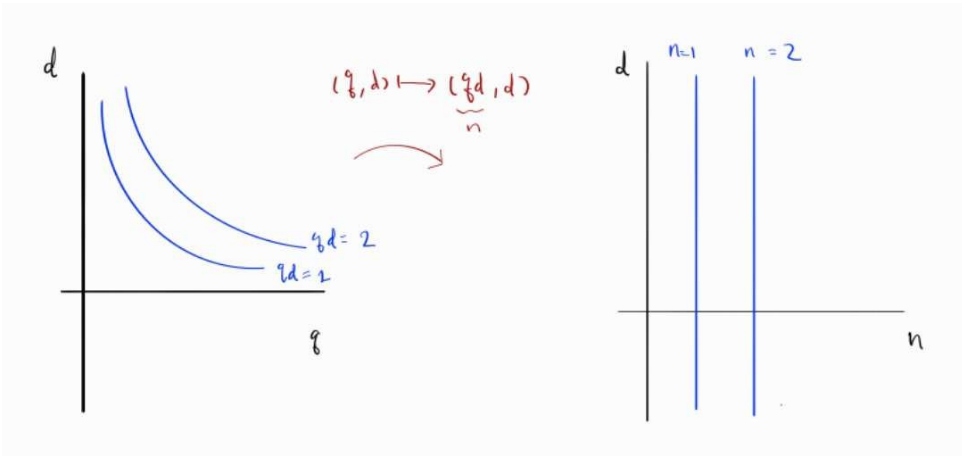

Figure 1: example $x = 2$

the sum on the left is indeed the number of integer points in a family of hyperbolas, it allows us to rewrite the sum of dirichlet convolution as following

$$\sum_{n\le x}(a*b)(n) = \sum_{n\le x}\sum_{d|n} a(d)b(n/d) = \sum_{qd\le x} a(d)b(q) = \sum_{d\le x} a(d)\sum_{q\le x/d} b(q)$$

Example 8.4. (Dirichlet's formula) For any $x \ge 1$, we have

$$\sum_{n\le x}\tau(n) = x\log x + (2C-1)x + O(x^{1/2})$$

where C is the euler constant, and a rougher estimate is

$$\sum_{n\le x}\tau(n) = x\log x + O(x)$$

Proof. By hyberbola method $\sum_{n\le x}\tau(n) = \sum_{d\le x}\sum_{q\le x/d} 1 = \sum_{d\le x}[\frac{x}{d}]$, so if we estimate $[\frac{x}{d}] \le \frac{x}{d}$ and apply the asymptotic formula for harmonic series we can get a rough estimate. To get a refined estimate, we reconsider $\sum_{n\le x}\tau(x) = \sum_{qd\le x} 1$ to count the lattice, notice that the number of lattice on the line $q = d$ are $[\sqrt{x}]$, and the distribution of lattice are symmetric with respect to the line, so we have

$$\begin{aligned}\sum_{n\le x}\tau(n) &= (2\sum_{qd\le x, d\le\sqrt{x}} 1) - [\sqrt{x}]^2\\ &= (2\sum_{d\le\sqrt{x}}[\frac{x}{d}]) - [\sqrt{x}]^2\\ &= \{2\sum_{d\le\sqrt{x}}\frac{x}{d} + O(1)\} - \{\sqrt{x}+O(1)\}^2\\ &= \{2x\sum_{d\le\sqrt{x}}\frac{1}{d}\} + O([\sqrt{x}]) - \{x + O(\sqrt{x})\}\\ &= 2x\{\log\sqrt{x} + C + O(1/\sqrt{x})\} - x + O(\sqrt{x})\\ &= x\log x + (2C-1)x + O(\sqrt{x})\end{aligned}$$

□

8.1.3 periodic arithmetic function

Definition 8.4. An arithmetic function $f : \mathbb{N} \to \mathbb{C}$ is called periodic if there exists a positive integer k such that

$$f(n+k) = f(n)$$

for all $n \in \mathbb{N}$, and we call the smallest such k the (fundamental) period of f.

Remark 8.5. The fundamental period is unique, and it divides any other period of f. In fact, if we set $\mathcal{P}$ as the set of all periods of f, as a non-empty subset of $\mathbb{N}$, it has a minimum element k. If N is another period, by the division we have $N = qk + r$ for some $q \in \mathbb{N}$ and $0 \le r < k$, hence we have

$$f(n) = f(n+N) = f(n+r+qk) = f(n+r)$$

for any $n \in \mathbb{N}$, hence r is also a period of f if $r \neq 0$, hence $\mathcal{P} = k\mathbb{N}^*$ exactly.

Example 8.5. A natural example is exponential function, for the sake of notation,

$$e(n) = e^{2\pi i n}$$

it is a 1-peridoic function.

Theorem 8.6 (finite Fourier series)**.**
Let f be an arithmetical function with period k, then there exists a unique arithmetical function $\hat{f}$ such that

$$f(m) = \sum_{n=1}^{k}\hat{f}(n)e(\frac{mn}{k})$$

In fact $\hat{f}$ is the fourier transform of f on $\mathbb{Z}/k\mathbb{Z}$ by

$$\hat{f}(n) = \frac{1}{k}\sum_{m=1}^{k} f(m)e(-\frac{mn}{k})$$

8.2 Character

Definition 8.5. Let G be a finite abelian group, a **character** of G is a group homomorphism $\chi : G \to \mathbb{C}^\times$, and we denote the set of all characters of G as $\hat{G}$, it forms a group under function multiplication

$$\forall g \in G : (\chi_1 \cdot \chi_2)(g) := \chi_1(g)\chi_2(g)$$

And we call $\hat{G}$ the **Character group** of G.

Here are some quick obeservations:

- **image of character should be in $\mathbb{S}^1$**: any elements $a \in G$ is of finite order, and homomorphism of groups makes $\chi(e) = 1$, hence $\chi(a)^n = 1$ with $n = ord(a)$.
- **Inverse of character is the complex conjugate**: $|\chi(n)| = 1$, which allows us to define the inverse
$$\chi^{-1}(n) = 1/\chi(n) = \bar{\chi}(n)$$
- In particular, $\chi(-1) = \pm 1$ since $\chi(-1)^2 = \chi(1) = 1$, which allows us to classify the character into two types: **even** character with $\chi(-1) = 1$ and **odd** character with $\chi(-1) = -1$.

Remark 8.6. We consider the categroy of all abelian groups, denoted by **Ab**, the character group can be generalized in the category by a contravariant functor $\hat{(-)} : \mathbf{Ab} \to \mathbf{Ab}^{op}$: For the object $G \in \mathbf{Ab}$, we have $\hat{G} = \mathrm{Hom}(G, \mathbb{C}^\times)$; For the morphism $f : G \to H$, it induces a morphism between dual groups $\hat{f} : \hat{H} \to \hat{G}$ by $\chi \mapsto \chi \circ f$, i.e.

$$\begin{tikzcd}
G \arrow[r, "f"] \arrow[dr, dashed, "\hat{f}(\chi)"'] & H \arrow[d, "\chi"] \\
& \mathbb{C}^*
\end{tikzcd}$$

However, the functor is not an equivalent since it is not full, that is the reason why we only consider the character group of finite abelian groups.

Remark 8.7. Another view of character is that the character is naturally **a one-dimensional complex representation** of G since $\mathbb{C}^* \cong \mathrm{GL}_1(\mathbb{C})$, and any character as such is irreducible.

Proposition 8.7. For any finite abelian group G, we have $\hat{G} \cong G$

Proof. For any finite abelian group G, there exists a decomposition of cyclic groups $G \cong \mathbb{Z}/n_1 \oplus \cdots \oplus \mathbb{Z}/n_r$, the universal property of direct sum allows $\widehat{G \oplus H} \cong \widehat{G} \times \widehat{H}$ hence we have

$$\hat{G} \cong \widehat{\mathbb{Z}/n_1} \oplus \cdots \oplus \widehat{\mathbb{Z}/n_r}$$

hence it is enough to prove the isomorphism for any cyclic group $\mathbb{Z}/n$. $\square$

Here is a useful lemma, which appears in many proofs, from the view of category, it shows that the induced morphism of a injection is a surjection.

Lemma 8.8. Let G be a finite abelian group and H be its subgroup, then for any character $\chi \in \hat{H}$, there exists exactly $[G : H]$ characters $\hat{G}$ that extend χ to G.

Proof. Consider restriction map $r : \hat{G} \to \hat{H}$ defined by $\chi \mapsto \chi|_H$, it is a group homomorphism, and we will show that r is surjective, then we consider the equivalence relation on $\hat{G}$ by

$$\chi_1 \sim \chi_2 \iff \chi_1|_H = \chi_2|_H$$

the relation gives that the class of χ is exactly the set of characters that extend χ to G, and the relation is actually same as the relation defined by cosets of $\bar{H}$, then we can deduce the number by isomorphism of dual groups:

$$\#[\chi] = [\hat{G} : \hat{H}] = [G : H]$$

Surjective: We take $g_1 \notin H$ and construct a subgroup of G by

$$G_1 = \langle H, g_1 \rangle = \{g_1^k h \mid k \in \mathbb{N}, h \in H\}$$

then we can extend the character χ by defining $\chi_1(g_1^k h) = \chi(h)$, i.e. we set $\chi_1(g_1) = 1$ and allow $\chi_1(ab) = \chi_1(a)\chi_1(b)$, then we can verify that $\chi_1 \in \widehat{H_1}$ and $\chi_1|_H = \chi$. Similarly we can construct a chain of subgroups by $G_k = \langle G_{k-1}, g_k \rangle$ with some $g_k \notin G_{k-1}$, so we have

$$G_0 = H \subsetneq G_1 \subsetneq \cdots \subsetneq G_n = G$$

the chain will stop since G is finite (or it is finitely generated $\mathbb{Z}$-module, so we can use noetherian condition). □

Remark 8.8. Notice that $\mathbb{C}^*$ is a divisible abelian group, then for any short squence of abelian groups

$$0 \to H \to G \to G/H \to 0$$

we have a conversion of exact sequence

$$0 \to \widehat{G/H} \to \hat{G} \to \hat{H} \to 0$$

we can talk more if we restrict our view on finite abelian groups.

Theorem 8.9. Let G be a finite abelian group, then any quotient group of G can be seen as a subgroup of G, furthermore we have

$$G \cong H \oplus G/H$$

Proof. We set $H^\perp = \{\chi \in \hat{G} \mid \chi(h) = 1, \forall h \in H\}$ as the complement of $\hat{H}$ in $\hat{G}$, it is natural in the sense of $\mathbb{Z}$-module, then we can prove that $H^\perp \cong \widehat{G/H}$. Notice that $H^\perp$ is exactly a subgroup of $\hat{G}$, and $H^\perp \cap \hat{H} = \{1_H\}$, so we have a direct sum

$$\hat{G} = \hat{H} \oplus H^\perp \cong \hat{H} \oplus \widehat{G/H}$$

by the isomprhism of dual groups, we can deduce the result. □

Proposition 8.10 (orthogonality)**.** Let G be a finite abelian group,

1. Fix $\chi \in \bar{G}$, then
$$\sum_{g\in G} \chi(g) = \begin{cases} |G| & \chi = 1 \\ 0 & \chi \neq 1 \end{cases}$$

2. Fix $g \in G$, then
$$\sum_{\chi\in \hat{G}} \chi(g) = \begin{cases} |G| & g = e \\ 0 & g \neq e \end{cases}$$

Proof. For the first case, if $\chi = 1$ then it is trivial, otherwise we can take $h \in G$ such that $\chi(h) \neq 1$, and the transaltion $g \mapsto h^{-1}g$ is a bijection of G, so we have

$$\sum_{g\in G} \chi(g) = \sum_{g\in G} \chi(h)\chi(h^{-1}g) = \chi(h) \sum_{g\in G} \chi(g)$$

then we can deduce the result. For the second case, we suppose that $g \neq e$, then there exists a character ψ such that $\psi(g) \neq 1$, and the translation $\chi \mapsto \psi^{-1} \cdot \chi$ is a bijection of $\hat{G}$, so we have
$$\sum_{\chi\in\hat{G}} \chi(g) = \psi(g) \sum_{\chi\in\hat{G}} (\psi^{-1} \cdot \chi)(g) = \psi(g) \sum_{\chi\in\hat{G}} \chi(g)$$
then we can deduce the result. ☐

Corollary 8.11. Let $a, b \in G$, then we have

$$\sum_{\chi\in\hat{G}} \chi(a)\overline{\chi(b)} = \begin{cases} |G| & a = b \\ 0 & a \neq b \end{cases}$$

Proof. ☐

A special result of orthogonality is about the structure of the character, we set **FinAb** to be the category of finite abelian groups, as we have mentioned before, there exists a contravariant functor inside the categroy, here we set D : **FinAb** $\to$ **FinAb**^{op}, the following result shows that the functor is actually an equivalence with $D^2 = Id$.

Theorem 8.2. Let G be a finite abelian group, there exists a natural isomorphism $\phi : G \to \hat{\hat{G}}$ defined by $g \mapsto \phi_g$ with $\phi_g(\chi) = \chi(g)$.

Proof. It is easy to verify that the map is a morphism, so it is enough to show injection since it is finite, and we consider the kernel.

$$\begin{aligned} g \in \ker \phi &\iff \phi_g(\chi) = 1, \forall \chi \in \hat{G} \\ &\iff \chi(g) = 1, \forall \chi \in \hat{G} \\ &\iff \sum_{\chi \in \hat{G}} \chi(g) = |\hat{G}| = |G| \\ &\iff g = e \end{aligned}$$

□

Remark 8.9. Notice that the map is «natrual» since it does not depend on the choice of groups, and we can verify the following diagram commutes for any morphism $f : G \to H$

$$\begin{tikzcd} G \arrow[r, "\phi_G"] \arrow[d, "f"'] & \widehat{\widehat{G}} \arrow[d, "D^2(f)"] \\ H \arrow[r, "\phi_H"] & \widehat{\widehat{H}} \end{tikzcd}$$

which ensures that the functor D is an equivalence, it is a special case of the **Pontryagin duality** for locally compact abelian groups, since every finite group can be endowed with the discrete topology.

8.2.1 Dirichlet Character and Gauss sum

Definition 8.6. Let $N \in \mathbb{N}^*$, a **Dirichlet character** modulo N is a character of the multiplicative group $(\mathbb{Z}/N\mathbb{Z})^\times$, i.e.

$$\chi : (\mathbb{Z}/N\mathbb{Z})^\times \to \mathbb{C}^\times$$

Here are some quick descriptions of this type of character:

- Let χ be a Dirichlet character modulo N, then we can extend χ to a function on $\mathbb{Z}$ by
$$\tilde{\chi}(a) = \begin{cases} \chi(\bar{a}) & \gcd(a, N) = 1 \\ 0 & \gcd(a, N) \neq 1 \end{cases}$$
It defines an arithmetic function that is **completely multiplicative** and N-**periodic**. Each character can be extended like that, so we sometimes do not distinguish the character and the arithemetic function.

- The order of $(\mathbb{Z}/N\mathbb{Z})^\times$ is $\varphi(N)$ with φ the Euler totient function, so the orthogonality of characters can be written as
$$\sum_{\chi \in \widehat{(\mathbb{Z}/N\mathbb{Z})^*}} \chi(a)\overline{\chi(b)} = \begin{cases} \varphi(N) & a \equiv b \mod N \\ 0 & a \not\equiv b \mod N \end{cases}$$
for any $a, b \in \mathbb{Z}$ with $(b, N) = 1$.

- We call a dirchlet character to be **principal or trival** if $\chi(g) = 1$ for any $g \in (\mathbb{Z}/N\mathbb{Z})^\times$, and we denote the principal character by χ_0.

Theorem 8.12. Let $f : \mathbb{N} \to \mathbb{C}$ be an arithemetic function is a Dirichlet character modulo N if it satisfies the following conditions:

(a) completely multiplicative

(b) N-periodic for some $N \in \mathbb{N}^*$

(c) $|f(n)| = 1$ if and only if $\gcd(n, N) = 1$

Proof. □

Example 8.6. Notice that $(\mathbb{Z}/3\mathbb{Z})^\times$ is of order 2, so there are exactly two characters, it can be given by the following table:

n	0	1	2
χ_1	0	1	1
χ_2	0	1	-1

while $(\mathbb{Z}/6\mathbb{Z})^\times \cong (\mathbb{Z}/2\mathbb{Z})^\times \times (\mathbb{Z}/3\mathbb{Z})^\times$ is of order 2, so the dirichlet character mod 6 can be given by the product of the chracter mod 2 (trival character) and mod 3 (χ_1 and χ_2 as above), and we write down the table

n	0	1	2	3	4	5
ψ_1	0	1	0	0	0	-1
ψ_2	0	1	0	0	0	1

Example 8.7. Another example is the character mod 9, there are exactly 6 characters, while in it there are two characters can be written down by character mod 3

n	0	1	2	3	4	5	6	7	8
$\tilde{\chi_1}$	0	1	-1	0	1	-1	0	1	-1
$\tilde{\chi_2}$	0	1	-1	0	1	-1	0	1	-1

for instane,

$$\tilde{\chi_1}(n) = (\chi_1 \cdot f_0)(n) = \begin{cases} \chi_1(n) & \gcd(n, 9) = 1 \\ 0 & \gcd(n, 9) \neq 1 \end{cases}$$

where f_0 is the trival character mod 9, it is clearly that $\tilde{\chi_1}$ is a 9-periodic function and completely multiplicative, and for any n coprime with 9 ($9 = 3^2$ so also coprime with 3), we have $\tilde{\chi_1}(n) = \chi_1(n) = \pm 1$, hence it is a dirichlet character mod 9, and the same for $\tilde{\chi_2}$.

The second example shows that for a dirichlet character mod N, if N is large then it may be determined by a dirichlet charcter mod the factors of N, which induces the following definition.

Lemma 8.13. Let χ be a dirichlet character mod N, and d a positive divisor of N, then the following statements are equivalant:

(a) χ factors through $(\mathbb{Z}/d\mathbb{Z})^{\times}$: there exists a character ψ_d mod d such that the following diagram commutes:

$$\begin{tikzcd} (\mathbb{Z}/N\mathbb{Z})^{\times} \arrow[r, "\chi"] \arrow[d, "\pi"'] & \mathbb{C} \\ (\mathbb{Z}/d\mathbb{Z})^{\times} \arrow[ur, "\psi_d"'] & \end{tikzcd}$$

where π is the natural projection sending $[x]_N$ to $[x]_d$.

(b) $\chi(a) = 1$ for any $a \in \mathbb{N}$ such that $\gcd(a, N) = 1$ and $a \equiv 1 \mod d$.

(c) there is a character ψ_d mod d such that in the sense of arithemetic function

$$\chi(n) = \psi_d(n)\chi_0(n)$$

If one of the above statements holds, we say that d is an **induced modulus** of χ.

Proof. □

Remark 8.10. Notice that N is alaways an induced modulus of χ, while 1 is an induced modulus of χ if and only if χ is trival. In above example, 3 is an induced modulus of $\tilde{\chi}_1$ and $\tilde{\chi}_2$.

Definition 8.1. Let χ be a dirichlet character mod N, the conductor of χ is the smallest induced modulus of χ.

The character χ is called **primitive** if its conductor is exactly N.

Remark 8.11. **Any non-trival charcter mod a prime number p is primitive**, since the divisor of p is only 1 and p, so the conductor can only be 1 or p, it is p since the character is non-trival.

Remark 8.12. By lemma, for any dirichlet chracter χ mod N, there exists a unique primitive character ψ mod conductor of χ, such that

$$\chi(n) = \psi(n)\chi_0(n)$$

the proof depends on the fact that conductor is the smallest induced modulus (cf. [**Apostol**], Theorem 8.18).

Notice that dirichlet character can be viewed as a periodic arithemetic function, which allows us to consider its fourier transform, here is the notation introduced by Gauss in the study of quadratic reciprocity.

Definition 8.2. The **Gauss sum** of a dirichlet character χ mod N is defined by

$$G(n,\chi) = \sum_{m=1}^{N} \chi(m)e(\frac{mn}{N})$$

and we write $\tau(\chi) = G(1,\chi)$ for convenience.

Lemma 8.14. If χ is a dirichlet character mod N, then for any $\gcd(n,N) = 1$ we have

$$G(n,\chi) = \bar{\chi}(n)\tau(\chi)$$

Proof. □

χ can be viewed as a function on $\mathbb{Z}/N\mathbb{Z}$, so by theorem 8.6 we can rewrite

$$G(n,\chi) = N \cdot \hat{\chi}(-n)$$

so we can desrible primitive character by gauss sum again, since we believe that fourier transform can keep the information of the original function.

Lemma 8.15. Along above notation, the following statements are equivalent:

(a) χ is primitive.

(b) $G(n,\chi) = \bar{\chi}(n)\tau(\chi)$ for any $n \in \mathbb{N}$.

(c) $G(n,\chi) = 0$ for any $\gcd(n,N) \neq 1$.

If one of the above statements holds, we call the Gauss sum is **separable**.

Proof. □

Proposition 8.16. Let χ be a primitive character mod N, then

$$|G(n,\chi)| = \begin{cases} \sqrt{N} & \gcd(n,N) = 1 \\ 0 & \gcd(n,N) \neq 1 \end{cases}$$

Proof. □

Remark 8.13. If χ is not primitive, then by Ramanujan sum we can show that

$$\tau(\chi) = 0$$

Finally, when the character is primitive, then the fourier expansion of character will be simple since the guass sum is separable, we can conclude that as following, which is important in many applications.

Theorem 8.3. Let χ be a primitive character mod N, then its fourier trnasform is

$$\hat{\chi} = \frac{\tau(\chi)\bar{\chi}(-1)}{N} \cdot \bar{\chi}$$

with the fourier expansion

$$\chi(n) = \frac{\tau(\chi)}{N} \sum_{m=1}^{N} \bar{\chi}(n) e(-\frac{mn}{N})$$

Proof. $\square$

8.2.2 Legendre Symbol and real character

A character of a finite abelian group is called **real character** if its image is contained in $\{\pm 1\}$, this parts is to introduce a typcial example of real character.

Historically, a classical diophantine problem is to solve

$$X^2 = b \mod p$$

with p a prime, and b is a given non-zero intger, equivalently, solve the quadratic equation in the finite field $\mathbb{F}_p$. For example, $X^2 - 2 = 0$ has no solution in $\mathbb{F}_5$ while it has solution in $\mathbb{F}_3$.

Lemma 8.17. Let p be an odd prime, then $\varphi : \mathbb{F}_p^* \to \mathbb{F}_p^*$ with $x \mapsto x^{\frac{p-1}{2}}$ defines a homomorphism of groups, with kernel S_p the set of all squares in $\mathbb{F}_p^*$, and the image is $\{1, -1\}$.

Proof. p is prime so $\frac{p-1}{2}$ is an integer, which is enough to show the map is a homomorphism. Notice that $S_p \subset \ker\varphi$, it is enough to show that $|S_P| = |\ker\varphi| = \frac{p-1}{2}$. After that, the cardianlity of the image is 2, and $\varphi(x)$ is exactly the roots of $X^2 - 1 = 0$, which implies the image is $\{1, -1\}$.

S_p is the image of morphism $f(x) = x^2$ inside $\mathbb{F}_p^*$, the kernel of f is exactly the solutions of $X^2 - 1 = 0$ in $\mathbb{F}_p$, they are exactly $-1, 1$ since there are at most two distinct solutions, hence $|S_p| = |\mathbb{F}_p^* / \ker f| = \frac{p-1}{2}$.

$\varphi(x) = 1$ means that x is a roots of $X^{\frac{p-1}{2}} - 1 = 0$ in $\mathbb{F}_p$, which has at most $\frac{p-1}{2}$ distinct solutions, hence $|\ker\varphi| \leq \frac{p-1}{2} = |S_p|$, so immediately $\ker\varphi = S_p$. $\square$

Definition 8.3. A integer a is called a **quadratic residue** modulo p if a is a square in $\mathbb{F}_p$, i.e. the diophantine equation

$$x^2 \equiv a \mod p$$

has a solution in $\mathbb{F}_p$. The **legendre symbol** is a function defined by

$$\left(\frac{a}{p}\right) = \begin{cases} 0 & a = 0 \mod p \\ 1 & a \text{ is a non-zero quadratic residue modulo } p \\ -1 & \text{otherwise} \end{cases}$$

where p is an odd prime and a is an integer.

Here are some quick descriptions:

- If $p = 2$, then any elements in $\mathbb{F}_2$ is square, so legndre symbol makes no sense to distinguish the quadratic residue, hence we only consider odd prime p.
- If we fixed the prime p, then the legendre symbol is a extension of the character on $(\mathbb{Z}/p\mathbb{Z})^\times$ defined by $a \mapsto a^{\frac{p-1}{2}}$ in the above lemma, exactly it is **a real primitive character mod** p.
- It is the unique non-trival real character mod p: In fact, $(\mathbb{Z}/p\mathbb{Z})^\times$ is a cyclic group of order $p-1$, so we can choose a generator g, then the character will depend only on $\chi(g) = \{\pm 1\}$, if $\chi(g) = 1$, then χ is trival, hence along the last statement, we can conclude.

Proposition 8.18. Let a be a non-zero square free integer and $m = 4|a|$, then there exists a unique dirichlet character χ_a mod m such that

$$\forall \text{ prime } p \nmid m, \quad \chi_a(p) = \left(\frac{a}{p}\right)$$

it will be non-trival if $a \neq 1$.

Proposition 8.19. Let p be an odd prime, then we have the following formula:

(a) Euler's criterion: $\left(\frac{n}{p}\right) = n^{\frac{p-1}{2}} \mod p$ for any intger n.

(b) Gauss's lemma: Let n be an integer coprime with p, and consider the following numbers:

$$n, 2n, 3n, \ldots, \frac{p-1}{2}n$$

For any above numbers, there exists a unique number $1 \leq a_k \leq p-1$ such that $a_k = kn \mod p$ with $1 \leq k \leq \frac{p-1}{2}$, then we have

$$\left(\frac{n}{p}\right) = (-1)^s$$

where s is the number of a_k that is greater than $\frac{p}{2}$.

Via the lemma, Gauss proved a remarkable result about the legendre symbol, so-called quadratic reciprocity, for sake of a proof, [**Apostol**] Theorem 9.8 sketches as such style. Here we will see legendre symbol as a character, and consider the gauss sum of other type, such that we can get a series of results about characters.

Lemma 8.20. The **quadratic gauss sum** is defined as

$$G_n = \sum_{k=1}^{n} e(\frac{k^2}{n})$$

then we have

$$G_n = \frac{1+i^{-m}}{1+i^{-1}}\sqrt{m} = \begin{cases} (1+i)\sqrt{m} & m \equiv 0 \mod 4 \\ \sqrt{m} & m \equiv 1 \mod 4 \\ 0 & m \equiv 2 \mod 4 \\ i\sqrt{m} & m \equiv 3 \mod 4 \end{cases}$$

Proof. $\square$

Lemma 8.21. The **generalized gauss sum** is defined as

$$G_n(a) = \sum_{k=1}^{n} e(\frac{ak^2}{n})$$

with $G_n(1) = G_n$ exactly, then we conclude following properties:

(a) If $\gcd(a,n) = d$, then $G_n(a) = d \cdot G_{n/d}(a/d)$.

hence we usually consider the case a coprime with n

(b) If a is coprime with $n = n_1 n_2$ and n_1 is corpime with n_2, then

$$G_n(a) = G_{n_1}(an_2)G_{n_2}(an_1)$$

(c) If n is fixed, then $a \mapsto G_n(a)$ is n-periodic and in particular if $n = p$ is an odd prime then

$$G_p(a) = \left(\frac{a}{p}\right) G_p = \left(\frac{a}{p}\right) \varepsilon_p \sqrt{p}$$

with a factor depening on p

$$\varepsilon_p = \begin{cases} 1 & p \equiv 1 \mod 4 \\ i & p \equiv 3 \mod 4 \end{cases}$$

(d) Let p be an odd prime, then the gauss sum of legendre symbol $\chi_p = \left(\frac{\cdot}{p}\right)$ is exactly G_p with

$$\varepsilon(\chi_p) = \varepsilon_p \sqrt{p}$$

Theorem 8.4 (Quadratic reciprocity)**.** Let p and q be two distinct odd primes, then we have

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2}\cdot\frac{q-1}{2}}$$

Proof. $\square$

Theorem 8.5 (Gauss sum)**.** Let n be a square-free odd, and χ be a real primitive character mod n, then

$$\tau(\chi) = \varepsilon_m \sqrt{m}$$

with

$$\varepsilon_m = \begin{cases} 1 & m \equiv 1 \mod 4 \\ i & m \equiv 3 \mod 4 \end{cases}$$

Proof. □

8.3 Riemann Zeta-function

8.3.1 Analytical Continuation: different methods

As a core object in number theory, Riemann defines the zeta-function in his short paper [**Riemann**] as following

$$\zeta(s) = \sum_{n\geq 1} \frac{1}{n^s}$$

The basic description can be given as following:

1. ζ defines a homomorphic function on the half-plane $\Re(s) > 1$, the reason is that the series is uniformly convergent on any compact subset of the half-plane, and $s \mapsto n^s$ is a entire function.

2. On the half-plane $\Re(s) > 1$, we have the Euler product expansion

$$\zeta(s) = \prod_{p \text{ prime}} \frac{1}{1 - p^{-s}}$$

   which encodes the information of prime numbers into this function, that is the reason why it plays an important role in number theory.

3. On the half-plane $\Re(s) > 0$, make use of the Gamma function we have the formula

$$\frac{\Gamma(1+s)}{n^s} = \int_0^\infty e^{-nt}\, t^s \,\frac{dt}{t}$$

   then some calculation can be done to arrange the formula in the form of Mellin transform:

$$\pi^{-s/2}\Gamma(s/2)\zeta(s) = \int_0^\infty \frac{\Theta(t) - 1}{2}\, t^{s/2-1}\,\frac{dt}{t} \tag{1}$$

   with theta function defined by

$$\Theta(t) := \sum_{n\in\mathbb{Z}} e^{-\pi n^2 t} \quad t > 0$$

   and by the poisson's formula it satisfies the functional equation

$$\Theta(t) = t^{-1/2}\Theta(1/t) \tag{2}$$

It can be proved in half plane by complex analysis with no diffcult, and the analytic continuation can be done by mellin transform, the result will be important in number theory, and we state as following:

Theorem 8.6. Define $Z(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)$ as the c**ompleted zeta-function**, then Z is holomorphic on half plane $\Re(s) > 1$ and it satisfies

1. Z has an meomorphic continuation the whole complex plane.
2. Z has the only simple poles at $s = 0$ and $s = 1$ with residue -1 and 1 respectively.
3. For any $s \neq 0, 1$, functional equation is $Z(1-s) = Z(s)$.

Proof. Set the auxiliary function $\psi(t) = \frac{\Theta(t)-1}{2}$ and by equation (2) we have

$$t^{1/2}(2\psi(t)+1) = 2\psi(1/t)+1, \quad t > 0$$

then we can rewrite the integral representation (1)

$$\begin{aligned} Z(s) &= \int_0^1 \psi(t)t^{\frac{s}{2}-1}dt + \int_1^\infty \psi(t)t^{\frac{s}{2}-1}dt \\ &= \int_1^\infty \psi(1/t)t^{-s/2}\frac{dt}{t} + \int_1^\infty \psi(t)t^{\frac{s}{2}}\frac{dt}{t} \end{aligned}$$

plug the reflection fomrula of ψ into the first integral, we can calculate as following

$$\begin{aligned} Z(s) &= \frac{1}{2}\int_1^\infty t^{-s/2-1/2}dt - \frac{1}{2}\int_1^\infty t^{s/2-1}dt + \int_1^\infty \psi(t)(t^{s/2-1} + t^{-s/2-1/2})dt \\ &= \frac{1}{s-1} - \frac{1}{s} + \int_1^\infty \psi(t)(t^{\frac{s}{2}} + t^{\frac{1-s}{2}})\frac{dt}{t} \\ &= \frac{1}{s-1} - \frac{1}{s} + F(s) + F(1-s) \end{aligned}$$

with $F(s) = \int_0^\infty \psi(t)t^{s/2}\,dt$, the integrand can be proved to be a entire function since ψ is rapidly decreasing, so it finishes the proof by a quick verification. $\square$

The consequence is immediately about the zeta function, and such a method can be seen as a standard model to get the analytic continuation and functional equation of L-fuctions.

Corollary 8.22. Riemann zeta function ζ has an analytic continuation to the whole complex plane with only a simple pole at $s = 1$ with residue 1, and it satisfies the functional equation

$$\zeta(1-s) = 2(2\pi)^{-s}\cos(\frac{\pi s}{2})\,\Gamma(s)\,\zeta(s) \qquad s \neq 1, 0, -1, -2, \ldots$$

Proof. By completed zeta function, the function

$$s \mapsto \frac{Z(s)}{\pi^{-\frac{s}{2}}\Gamma\left(\frac{s}{2}\right)}$$

defines an meomorphic function on the hole complex plane, which concides with ζ on the half-plane $\Re(s) > 1$, so it is the analytic continuation of ζ. Notice that Γ has a simple pole at $s = 0$, which cancels the simple pole of Z at $s = 0$, hence ζ is actually holomorphic at $s = 1$ with residue

$$\mathrm{Res}_{z=1}\zeta = \lim_{s\to 1} \frac{(s-1)Z(s)}{\pi^{-s/2}\Gamma(s/2)} = \frac{1}{\pi^{-1/2}\pi^{1/2}} = 1$$

and the funnctional equation can be derived via the formula of gamma function. □

8.3.2 The values on $\mathbb{Z}$ and critical strip

We rewrite the functional equation of zeta function here

$$\zeta(1-s) = 2(2\pi)^{-s}\cos(\frac{\pi s}{2})\,\Gamma(s)\,\zeta(s) \qquad s \neq 1, 0, -1, -2, \dots \tag{3}$$

it gives a symmetry of values, and determine when the values is zero.

Example 8.8. We set $s = 2$ in the functional equation, then we have

$$\zeta(-1) = -\frac{1}{2\pi^2}\Gamma(2)\zeta(2) = -1/12$$

since $\zeta(2) = -\pi^2/6$ and $\Gamma(2) = 1$. In another case, we hope the cosine term vanish, for example we set $s = 3$ then $\zeta(-2) = 0$, it gives a zero of zeta function.

Following the example, the functional equation (3) establishes a non-vanish relation between the negative odd integers and the positive even integers, to know them all we should firstly know one sides of them. To do that, a remarkable constant is introduced.

Definition 8.4. The function $z \mapsto \frac{z}{e^z-1}$ is holomorphic near the zeros, with the expansion of the series:

$$\frac{z}{e^z-1} = \sum_{n\geq 0} \frac{B_n}{n!} z^n$$

the n-th **Bernoulli's number** is defined as B_n, with respect to the coefficient of the series.

The original definition of Bernoulli's number is given by Jacob Bernoulli in his book "Ars Conjectandi" (1713), he defined the number by the summation of powers:

$$\sum_{k=1}^{m} k^n = \frac{1}{n+1}\sum_{j=0}^{n}\binom{n+1}{j} B_j \cdot m^{n+1-j}$$

for any $n, m \in \mathbb{N}$, and the two definitions can be shown to be equivalent by the method of **generating function**. The definition of power series is easy for us to calculate the values of Bernoulli's number, for Example

$$B_0 = 1, \quad B_1 = \frac{-1}{2}, \quad B_2 = \frac{1}{6}, \quad B_3 = 0, \quad B_4 = \frac{-1}{30} \quad \ldots$$

Lemma 8.23. The Bernoulli's number is rational, and $B_{2k+3} = 0$ for any $k \in \mathbb{N}$.

Proof. Obeserve that

$$\frac{ze^z}{e^z - 1} = \frac{-z}{e^{-z} - 1}$$

which allows us to write

$$z = \frac{ze^z}{e^z - 1} - \frac{z}{e^z - 1} = \sum_{n \geq 0} ((-1)^n - 1) \frac{B_n}{n!} \cdot z^n$$

at the voisinage of zero, hence by comparing the coefficients we can conclude that $B_0 = 1$ and $B_n = 0$ for any odd integer $n \geq 3$. To prove the rationalty, by the properties of analytical function we know

$$B_n = n! \frac{d^n}{dz^n} \left(\frac{z}{e^z - 1} \right) \bigg|_{z=0}$$

the derivatives can be proved to be a rational number by induction on n, hence the result holds true. □

Proposition 8.24. Let k be a positive integer, then $\zeta(-2k) = 0$ and

$$\zeta(-n) = -\frac{B_{n+1}}{n+1}$$

by the functional equation a corollary is that

$$\zeta(2k) = (-1)^{k+1} \frac{(2\pi)^{2k} B_{2k}}{2(2k)!}$$

Proof. The first result is immediate by the functional equation (3) by setting $s = 1 + 2k$, then $\Gamma(s)$ and $\zeta(s)$ are both finite, while the consine term vanishes, hence $\zeta(-2k) = 0$.

For the second result, a classical formula about cotangent function is

$$\pi z \cot(\pi z) = 1 + \sum_{n=1}^{\infty} \frac{2z^2}{z^2 - n^2}, \qquad z \in \mathbb{C} \backslash \mathbb{Z}$$

and we can expand the formula by geometric series as following

$$\begin{aligned}\pi z\cot(\pi z) &= 1 + 2z^2 \sum_{n\geq 1} \frac{-1}{n^2} \cdot \frac{1}{1 - z^2/n^2} \\ &= 1 - 2z^2 \sum_{n\geq 1} \frac{1}{n^2} \sum_{k\geq 0} \left(\frac{z^2}{n^2}\right)^k \\ &= 1 - 2\sum_{k\geq 1} \zeta(2k) z^{2k}\end{aligned}$$

here the formula holds true for $|z| < 1$, and the last equality holds by Fubini's theorem, so we get a power series expansion at zeros. To finish the proof, we write the trigonometric function in terms of exponential function, then we can calculate

$$z\cot(z) = \frac{2iz}{e^{2iz} - 1} + iz = 1 + \sum_{k\geq 1} (-1)^k \frac{2^{2k}}{(2k)!} B_{2k} \cdot z^{2k}$$

which allows us to get another power series expansion at zeros, if we replace z by πz and compare the two expansions, then we can conclude the first formula. □

Remark 8.14. The absance of values on positive odd integers is a mystery, if we set $s = 1 + 2k$ then $\cos(\pi s/2) = 0$, which prevents us to get any information from the functional equation, although $\zeta(-2k)$ and $\zeta(2k+1)$ are lie on two sides, before the work of Apéry in 1978, we even can not determine whether $\zeta(3)$ is rational or not.

Remark 8.15. Another obeservation about zeros is via the completed zeta function

$$Z(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)$$

as we know in theorem 8.6, Z is finite at $s = -2n$, while $s \mapsto \Gamma(s/2)$ has a simple pole at $s = -2n$, so the unique possible is that ζ has a simple zero at $s = -2n$ to cancel the pole, and we call such zeros from the Gamma factor the **trivial zeros** of zeta function.

Theorem 8.7 (Hamadard,1896)**.** ζ does not vanish on the line $\mathrm{Re}(s) = 1$.

Proof. □

Corollary 8.25. The non-trival zeros of zeta function lie in the **critical strip** $\{s \in \mathbb{C} : 0 < \Re(s) < 1\}$.

8.3.3 Von Mangoldt function

A remarkable arithmetic function in the study of Riemann and Tchebychev is the Von Mangoldt function, which is defined as following:

$$\Lambda(n) := \begin{cases} \log p & \text{if } n = p^k \\ 0 & \text{otherwise} \end{cases}$$

Notice that $\Lambda(1) = 0$ and it is only related to the prime numbers, and we view it in the context of arithemetic functions, it is not hard to prove the following properties:

Proposition 8.26. By dirichlet convolution,

$$\log n = (\Lambda * \mathbf{1})(n) = \sum_{d|n} \Lambda(d)$$

and by mobius inversion formula

$$\Lambda(n) = (\log *\mu)(n) = \sum_{d|n} \mu(d) \log(n/d)$$

This function appears in the Riemann's paper first time although it is not named as such.

Lemma 8.27 (Riemann,1859)**.** For any $\Re(s) > 1$, we have

$$\log \zeta(s) = \sum_{n\geq 2} \frac{\Lambda(n)}{\log(n)} \frac{1}{n^s}$$

and

$$-\frac{\zeta(s)}{\zeta(s)} = \sum_{n\geq 2} \frac{\Lambda(n)}{n^s}, \quad \mathrm{Re}(s) > 1$$

Proof. Take the logarithm to zeta function of the Euler product expansion, we have

$$\log \zeta(s) = -\sum_{p} \log(1 - p^{-s})$$

notice that $\log(1-x)$ has a power series expansion if $|x| < 1$, and here $|p^{-s}| = |p|^{-\mathrm{Re}(s)} < 1$, so we can write

$$\log \zeta(s) = \sum_{p} \sum_{k\geq 1} \frac{1}{k} p^{-ks}$$

Notice that 1 is not a prime by convention, and Λ only supports on the prime powers, so the sum can be rewritten as

$$\sum_{n=p^k} \frac{\Lambda(n)}{\log n} \frac{1}{n^s}$$

since

$$\frac{\Lambda(p^k)}{\log p^k} = \frac{\log p}{k \log p}$$

$\square$

Tchebychev consider a type of summation, which is closed to the average of the von mangoldt function, that is

$$\theta(x) := \sum_{p\leq x} \log p$$

and we call it **the first Tchebychev function**, and **the second Tchebychev function** is defined by

$$\psi(x) := \sum_{n \le x} \Lambda(n)$$

they are equivalent in the sense of asymptotic behavior, and he estimated the growth of θ, which can be translated as following:

Lemma 8.28 (Tchebychev, 1850)**.** $\psi(x) \asymp x$, that means there exists two positive constants c_1, c_2 such that for any x large enough, we have

$$c_1 x \le \psi(x) \le c_2 x$$

Proof. □

Corollary 8.29. ψ and θ have the same asymptotic behavior.

8.3.4 Prime number theorem

Fix prime counting function

$$\pi(x) = \sum_{p \le x} 1$$

Claim 8.30. As $x \to \infty$, the following asymptotic behavior are equivalent:

(a) $\pi(x)/x \sim 1/\log x$

(b) $\psi(x) \sim x$

(c) $\psi_1(x) \sim x^2/2$

and

8.4 Dirichlet L-function

8.4.1 Analytical continuation

Definition 8.7. The dirichlet L-function associated to a dirichlet character χ mod N is defined by the series

$$L(s, \chi) = \sum_{n \ge 1} \frac{\chi(n)}{n^s}$$

about the analytic properties, some quick descriptions are as following:

1. $s \mapsto L(s, \chi)$ defines a holomorphic function on the half-plane $\Re(s) > 1$, and the Euler product formula holds as well one the half-plane

$$L(s, \chi) = \prod_{p \text{ prime}} \frac{1}{1 - \chi(p)p^{-s}}$$

   the prove is similar to the case of zeta function.

2. If χ is a character with conductor q of the form $\chi = \psi_q \cdot \chi_0$, where χ_0 is the trival character mod N and ψ_q is a primitive character mod q, then we have the factorization

$$L(s, \chi) = L(s, \psi_q) \prod_{p|N} (1 - \psi_q(p)p^{-s})$$

   which allows us to reduce the analytic properties of $L(s, \chi)$ to the primitive case.

3. Riemann zeta function can be viewed as a L-function associated to the character mod 1, and for any trival character χ_0 mod N, we have

$$L(s, \chi_0) = \sum_{n \geq 1, (n,N)=1} \frac{1}{n^s} = \zeta(s) \prod_{p|N} (1 - p^{-s})$$

   so if χ_0 is trival, then $L(s, \chi_0)$ has a meomorhic continuation to the whole complex plane with only a simple pole at $s = 1$ with residue $\prod_{p|N}(1 - p^{-1})$.

To do the analytical continuation, we start from the integral we use in the Riemann zeta function

$$\pi^{-\frac{s}{2}} \Gamma(\frac{s}{2}) \frac{1}{n^s} = \int_0^\infty e^{-\pi n^2 t} t^{\frac{s}{2}-1} dt \qquad \Re(s) > 1$$

to make use of the Gauss sum of the character, so it is natural to modify the integral a little bit **with a asuumption that the character is non trivial and primitive χ mod q**

$$(q/\pi)^{s/2} \Gamma(\frac{s}{2}) \frac{\chi(n)}{n^s} = \frac{1}{\sqrt{q}} \int_0^\infty [\chi(n) e^{-\pi n^2 t/p}] t^{\frac{s}{2}-1} dt \tag{4}$$

In the case of $\chi(-1) = 1$, we can sum equation (4) over n to get

$$(q/\pi)^{s/2} \Gamma(\frac{s}{2}) L(s, \chi) = \frac{1}{2\sqrt{q}} \int_0^\infty \psi(t, \chi)\, t^{\frac{s}{2}-1} dt \tag{5}$$

with

$$\psi(t, \chi) = \sum_{n \in \mathbb{Z}} \chi(n) e^{-\pi n^2 t/q}$$

In the case of $\chi(-1) = -1$, then $\psi(t, \chi) = 0$ such that representation (5) will make no sense in this case, so we should modify the equation (4) a little bit (introduce a translation $s \mapsto s + 1$) to get

$$(q/\pi)^{(s+1)/2} \Gamma(\frac{s+1}{2}) \frac{\chi(n)}{n^s} = \frac{1}{2\sqrt{q}} \int_0^\infty \psi(t, \chi)\, t^{\frac{s-1}{2}} dt$$

with

$$\psi(t,\chi)=\sum_{n\in\mathbb{Z}} n\chi(n)e^{-\pi n^2 t/q}$$

above discussion can be summarized as following:

Lemma 8.31. On the half-plane $\Re(s)>1$, the integral repsentation of a L-function $L(s,\chi)$ associated to a non-trivial primitive character χ mod q is given by

$$(q/\pi)^{\frac{s+\mathfrak{v}}{2}}\Gamma(\frac{s+\mathfrak{v}}{2})L(s,\chi)=\frac{1}{2\sqrt{q}}\int_0^\infty \psi(t,\chi)t^{\frac{s+\mathfrak{v}}{2}-1}dt$$

with $\mathfrak{v}=\frac{1-\chi(-1)}{2}$ and auxiliary function

$$\psi(t,\chi)=\sum_{n\in\mathbb{Z}} n^{\mathfrak{v}}\chi(n)e^{-\pi n^2 t/q}$$

So we should know the functional equation of ψ firstly.

Lemma 8.32. Let $t>0$, then ψ satisfies the equation

$$\tau(\bar{\chi})\psi(t,\chi)=(i/t)^{\mathfrak{v}}\sqrt{q/t}\cdot\psi(\bar{\chi},1/t)$$

where $\tau(\chi)=G(1,\chi)$ as the Gauss sum of χ.

Proof. We consider the $f(x)=e^{-\pi x^2 t}$ for some $t>0$, and then we can calculate

$$\hat{f}(s)=\frac{1}{\sqrt{t}}f(\frac{s}{t})=\frac{1}{\sqrt{t}}e^{-\pi s^2/t}$$

hence by poisson's formula we have

$$\sum_{n\in\mathbb{Z}} e^{-\pi(n+a)^2 t}=\frac{1}{\sqrt{t}}\sum_{n\in\mathbb{Z}} e^{-\pi n^2/t}e^{2\pi i n a} \tag{6}$$

now we consider χ is a arithemetic function on $\mathbb{Z}$ with period q, then we can write

$$\chi(n)=\frac{\tau(\chi)}{q}\sum_{m=1}^{q}\bar{\chi}(m)e^{2\pi i m n/q}$$

by the primitive character $\overline{\tau(\chi)}=\chi(-1)\tau(\bar{\chi})$, we have

$$\tau(\bar{\chi})\chi(n)=\sum_{m=1}^{q}\bar{\chi}(m)e^{2\pi i m n/q}$$

hence we can calculate

$$\begin{aligned}\tau(\bar{\chi})\psi(t,\chi)&=\sum_{n\in\mathbb{Z}} n^{\mathfrak{v}}\tau(\bar{\chi})\chi(n)e^{-\pi n^2 t/q}\\&=\sum_{m=1}^{q}\bar{\chi}(m)\sum_{n\in\mathbb{Z}} n^{\mathfrak{v}}e^{-\pi n^2 t/q}e^{2\pi i n m/q}\end{aligned}$$

when $\chi(-1) = 1$, i.e. $\mathfrak{v} = 0$, we can apply the Poisson's formula (6) to get

$$\begin{aligned}
\tau(\bar{\chi})\psi(t,\chi) &= \sqrt{q/t} \cdot \sum_{m=1}^{q} \bar{\chi}(m) \cdot \sum_{n\in\mathbb{Z}} e^{-\pi(qn+m)^2/qt} \\
&= \sqrt{q/t} \cdot \sum_{l\in\mathbb{Z}} \bar{\chi}(l) e^{-\pi l^2/t} \\
&= \sqrt{q/t} \cdot \psi(\bar{\chi}, 1/t)
\end{aligned}$$

when $\chi(-1) = -1$, i.e. $\mathfrak{v} = 1$, we can modify the poisson's formula (6) by taking derivative of $f(x)$, then we have

$$it^{3/2} \sum_{n\in\mathbb{Z}} (n+a) e^{-\pi(n+a)^2 t} = \sum_{n\in\mathbb{Z}} n e^{-\pi n^2/t} e^{2\pi i n a}$$

then apply it we can get

$$\begin{aligned}
\tau(\bar{\chi})\psi(t,\chi) &= \sum_{m=1}^{q} \bar{\chi}(m) \cdot i(q/t)^{3/2} \sum_{n\in\mathbb{Z}} (n + \frac{m}{q}) e^{-\pi(qn+m)^2/qt} \\
&= (i/t)\sqrt{q/t} \cdot \sum_{l\in\mathbb{Z}} l \cdot \bar{\chi}(l) e^{-\pi l^2/t} \\
&= (i/t)\sqrt{q/t} \cdot \psi(\bar{\chi}, 1/t)
\end{aligned}$$

□

Finally we can finish the analytical continuation as we do in Riemann zeta function

Theorem 8.8. For any L-function associated to a non-trivial primitive character χ mod q, the completed L-function is defined by

$$\Lambda(s,\chi) = (q/\pi)^{\frac{s+\mathfrak{v}}{2}} \Gamma(\frac{s+\mathfrak{v}}{2}) L(s,\chi)$$

then $s \mapsto \Lambda(s,\chi)$ has a **holomorphic continuation** to the whole complex plane, and it satisfies the functional equation

$$\Lambda(1-s,\bar{\chi}) = \frac{i^{\mathfrak{v}}\sqrt{q}}{\tau(\chi)} \Lambda(s,\chi)$$

Proof. when $\chi(-1) = 1$, i.e. $\mathfrak{v} = 0$, we have

$$\begin{aligned}
2\sqrt{q}\Lambda(s,\chi) &= \int_0^1 \psi(t,\chi) t^{\frac{s}{2}-1} dt + \int_1^\infty \psi(t,\chi) t^{\frac{s}{2}-1} dt \\
&= \int_1^\infty \psi(1/u,\chi) u^{-\frac{s}{2}-1} du + \int_1^\infty \psi(t,\chi) t^{\frac{s}{2}-1} dt \\
&= \frac{\tau(\chi)}{\sqrt{q}} \int_1^\infty \psi(u,\bar{\chi}) u^{\frac{-s-1}{2}} du + \int_1^\infty \psi(t,\chi) t^{\frac{s}{2}-1} dt \quad (a)
\end{aligned}$$

notice that in this case we have $\overline{\tau(\chi)} = \tau(\bar{\chi})$, so we can write

$$\begin{aligned} 2\sqrt{q}\Lambda(s,\chi) &= \frac{\tau(\chi)}{\sqrt{q}}[\int_1^\infty \psi(u,\bar{\chi})u^{\frac{-s-1}{2}}du + \frac{\tau(\bar{\chi})}{\sqrt{q}}\int_1^\infty \psi(t,\bar{\chi})t^{\frac{-s-1}{2}-1}dt] \\ &= \frac{\tau(\chi)}{\sqrt{q}} \cdot 2\sqrt{q}\Lambda(1-s,\bar{\chi}) \end{aligned}$$

Finally, when $\chi(-1) = -1$, i.e. $\mathfrak{v} = 1$, we have

$$\begin{aligned} 2\sqrt{q}\Lambda(s,\chi) &= \int_0^1 \psi(t,\chi)t^{\frac{s-1}{2}}dt + \int_1^\infty \psi(t,\chi)t^{\frac{s-1}{2}}dt \\ &= \int_1^\infty \psi(1/u,\chi)u^{-\frac{s+3}{2}}du + \int_1^\infty \psi(t,\chi)t^{\frac{s-1}{2}}dt \\ &= \frac{\tau(\chi)}{i\sqrt{q}}\int_1^\infty \psi(u,\bar{\chi})u^{-\frac{s}{2}}du + \int_1^\infty \psi(t,\chi)t^{\frac{s-1}{2}}dt \quad (b) \end{aligned}$$

and notice that in this case we have $\overline{\tau(\chi)} = -\tau(\bar{\chi})$, so we can write

$$\begin{aligned} 2\sqrt{q}\Lambda(s,\chi) &= \frac{\tau(\chi)}{i\sqrt{q}}[\int_1^\infty \psi(u,\bar{\chi})u^{-\frac{s}{2}}du + \frac{\tau(\bar{\chi})}{i\sqrt{q}}\int_1^\infty \psi(t,\bar{\chi})t^{\frac{s-1}{2}}dt] \\ &= \frac{\tau(\chi)}{i\sqrt{q}} \cdot 2\sqrt{q}\Lambda(1-s,\bar{\chi}) \end{aligned}$$

□

Remark 8.16. Notice that in above proof, (a) and (b) shows that the functions$s \mapsto 2\sqrt{q}\Lambda(s,\chi)$ defines an entire function, since the integrand nicely converges uniformly on $(1,\infty)$, so it indicates the holomoprhic continuation of Λ.

8.4.2 The arithmetic progressions

8.5 Dedekind L-function